

MATH2931 Higher Linear Models



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STATISTICAL LEARNING

There are two major goals for modeling data: 1) to accurately predict some future quantity of interest, given some observed data, and 2) to discover unusual or interesting patterns in the data.

1.1 Supervised and Unsupervised Learning

Given an input or *feature* vector $\mathbf{x}$, one of the main goals of machine learning is to predict an output or *response* variable y . For example, $\mathbf{x}$ could be a digitized signature and y a binary variable that indicates whether the signature is genuine or false. Another example is where $\mathbf{x}$ represents the weight and smoking habits of an expecting mother and y the birth weight of the baby. The data science attempt at this prediction is encoded in a mathematical function g , called the *prediction function*, which takes as an input $\mathbf{x}$ and outputs a guess $g(\mathbf{x})$ for y (denoted by $\hat{y}$, for example). In a sense, g encompasses all the information about the relationship between the variables $\mathbf{x}$ and y , excluding the effects of chance and randomness in nature.

In *regression* problems, the response variable y can take any real value.

In contrast, when y can only lie in a finite set, say $y \in \{0, \dots, c-1\}$, then predicting y is conceptually the same as classifying the input $\mathbf{x}$ into one of c categories, and so prediction becomes a *classification* problem.

We can measure the accuracy of a prediction $\hat{y}$ with respect to a given response y by using some *loss function* $\text{Loss}(y, \hat{y})$. In a regression setting the usual choice is the squared-error loss $(y - \hat{y})^2$. In the case of classification, the zero-one (also written 0-1) loss function $\text{Loss}(y, \hat{y}) = \mathbb{1}\{y \neq \hat{y}\}$ is often used, which incurs a loss of 1 whenever the predicted class $\hat{y}$ is not equal to the class y . Later on in this book, we will encounter various other useful loss functions, such as the cross-entropy and hinge loss functions.

It is unlikely that any mathematical function g will be able to make accurate predictions for all possible pairs $(\mathbf{x}, y)$ one may encounter in Nature. One reason for this is that, even with the same input $\mathbf{x}$, the output y may be different, depending on chance circumstances or randomness. For this reason, we adopt a probabilistic

FEATURE
RESPONSE

PREDICTION
FUNCTION

REGRESSION

CLASSIFICA-
TION

LOSS
FUNCTION

RISK

approach and assume that each pair $(\mathbf{x}, y)$ is the outcome of a random pair (X, Y) that has some joint probability density $f(\mathbf{x}, y)$. We then assess the predictive performance via the expected loss, usually called the *risk*, for g :

$$\ell(g) = \mathbb{E} \text{Loss}(Y, g(X)). \quad (1.1)$$

REGRESSION
FUNCTION

As already mentioned, for regression the most widely-used loss function is the squared-error loss. In this setting, the optimal prediction function g^* is often called the *regression function*. The following theorem specifies its exact form.

Theorem 1.1: Optimal Prediction Function for Squared-Error Loss

For the squared-error loss $\text{Loss}(y, \hat{y}) = (y - \hat{y})^2$, the optimal prediction function g^* is equal to the conditional expectation of Y given $X = \mathbf{x}$:

$$g^*(\mathbf{x}) = \mathbb{E}[Y | X = \mathbf{x}].$$

Proof: Let $g^*(\mathbf{x}) = \mathbb{E}[Y | X = \mathbf{x}]$. For any function g , the squared-error risk satisfies

$$\begin{aligned} \mathbb{E}(Y - g(X))^2 &= \mathbb{E}[(Y - g^*(X) + g^*(X) - g(X))^2] \\ &= \mathbb{E}(Y - g^*(X))^2 + 2\mathbb{E}[(Y - g^*(X))(g^*(X) - g(X))] + \mathbb{E}(g^*(X) - g(X))^2 \\ &\geq \mathbb{E}(Y - g^*(X))^2 + 2\mathbb{E}[(Y - g^*(X))(g^*(X) - g(X))] \\ &= \mathbb{E}(Y - g^*(X))^2 + 2\mathbb{E}\{(g^*(X) - g(X))\mathbb{E}[Y - g^*(X) | X]\}. \end{aligned}$$

106

In the last equation we used the tower property. By the definition of the conditional expectation, we have $\mathbb{E}[Y - g^*(X) | X] = 0$. It follows that $\mathbb{E}(Y - g(X))^2 \geq \mathbb{E}(Y - g^*(X))^2$, showing that g^* yields the smallest squared-error risk. $\square$

One consequence of Theorem 1.1 is that, conditional on $X = \mathbf{x}$, the (random) response Y can be written as

$$Y = g^*(\mathbf{x}) + \varepsilon(\mathbf{x}), \quad (1.2)$$

where $\varepsilon(\mathbf{x})$ can be viewed as the random deviation of the response from its conditional mean at $\mathbf{x}$. This random deviation satisfies $\mathbb{E} \varepsilon(\mathbf{x}) = 0$. Further, the conditional variance of the response Y at $\mathbf{x}$ can be written as $\mathbb{V}ar \varepsilon(\mathbf{x}) = v^2(\mathbf{x})$ for some unknown positive function v . Note that, in general, the probability distribution of $\varepsilon(\mathbf{x})$ is unspecified.

TRAINING
SET

Since, the optimal prediction function g^* depends on the typically unknown joint distribution of (X, Y) , it is not available in practice. Instead, all that we have available is a finite number of (usually) independent realizations from the joint density $f(\mathbf{x}, y)$. We denote this sample by $\mathcal{T} = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ and call it the *training set* ($\mathcal{T}$ is a mnemonic for training) with n examples. It will be important to distinguish between a random training set $\mathcal{T}$ and its (deterministic) outcome $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$. We will use the notation τ for the latter. We will also add the subscript n in τ_n when we wish to emphasize the size of the training set.

Our goal is thus to “learn” the unknown g^* using the n examples in the training set $\mathcal{T}$. Let us denote by $g_{\mathcal{T}}$ the best (by some criterion) approximation for g^* that we can construct from $\mathcal{T}$. Note that $g_{\mathcal{T}}$ is a random function. A particular outcome is denoted by g_{τ} . It is often useful to think of a teacher–learner metaphor, whereby the function $g_{\mathcal{T}}$ is a *learner* who learns the unknown functional relationship $g^* : \mathbf{x} \mapsto y$ from the training data $\mathcal{T}$. We can imagine a “teacher” who provides n examples of the true relationship between the output Y_i and the input X_i for $i = 1, \dots, n$, and thus “trains” the learner $g_{\mathcal{T}}$ to predict the output of a new input X , for which the correct output Y is not provided by the teacher (is unknown).

LEARNER

The above setting is called *supervised learning*, because one tries to learn the functional relationship between the feature vector $\mathbf{x}$ and response y in the presence of a teacher who provides n examples. It is common to speak of “explaining” or predicting y on the basis of $\mathbf{x}$, where $\mathbf{x}$ is a vector of *explanatory variables*.

SUPERVISED
LEARNING

An example of supervised learning is email spam detection. The goal is to train the learner $g_{\mathcal{T}}$ to accurately predict whether any future email, as represented by the feature vector $\mathbf{x}$, is spam or not. The training data consists of the feature vectors of a number of different email examples as well as the corresponding labels (spam or not spam). For instance, a feature vector could consist of the number of times sales-pitch words like “free”, “sale”, or “miss out” occur within a given email.

EXPLANAT-
ORY
VARIABLES

As seen from the above discussion, most questions of interest in supervised learning can be answered if we know the conditional pdf $f(y|\mathbf{x})$, because we can then in principle work out the function value $g^*(\mathbf{x})$.

In contrast, *unsupervised learning* makes no distinction between response and explanatory variables, and the objective is simply to learn the structure of the unknown distribution of the data. In other words, we need to learn $f(\mathbf{x})$. In this case the guess $g(\mathbf{x})$ is an approximation of $f(\mathbf{x})$ and the risk is of the form

UNSUPER-
VISED
LEARNING

$$\ell(g) = \mathbb{E} \text{Loss}(f(\mathbf{X}), g(\mathbf{X})).$$

An example of unsupervised learning is when we wish to analyze the purchasing behaviors of the customers of a grocery shop that has a total of, say, a hundred items on sale. A feature vector here could be a binary vector $\mathbf{x} \in \{0, 1\}^{100}$ representing the items bought by a customer on a visit to the shop (a 1 in the k -th position if a customer bought item $k \in \{1, \dots, 100\}$ and a 0 otherwise). Based on a training set $\tau = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, we wish to find any interesting or unusual purchasing patterns. In general, it is difficult to know if an unsupervised learner is doing a good job, because there is no teacher to provide examples of accurate predictions.

1.2 Training and Test Loss

Given an arbitrary prediction function g , it is typically not possible to compute its risk $\ell(g)$ in (1.1). However, using the training sample $\mathcal{T}$, we can approximate $\ell(g)$ via the

empirical (sample average) risk

$$\ell_{\mathcal{T}}(g) = \frac{1}{n} \sum_{i=1}^n \text{Loss}(Y_i, g(\mathbf{X}_i)), \quad (1.3)$$

TRAINING
LOSS

which we call the *training loss*. The training loss is thus an unbiased estimator of the risk (the expected loss) for a prediction function g , based on the training data.

To approximate the optimal prediction function g^* (the minimizer of the risk $\ell(g)$) we first select a suitable collection of approximating functions $\mathcal{G}$ and then take our *learner* to be the function in $\mathcal{G}$ that minimizes the training loss; that is,

$$g_{\mathcal{T}}^{\mathcal{G}} = \operatorname{argmin}_{g \in \mathcal{G}} \ell_{\mathcal{T}}(g). \quad (1.4)$$

For example, the simplest and most useful $\mathcal{G}$, which is studied in detail in this course, is the set of *linear* functions of $\mathbf{x}$; that is, the set of all functions $g : \mathbf{x} \mapsto \boldsymbol{\beta}^{\top} \mathbf{x}$ for some real-valued vector $\boldsymbol{\beta}$.

We suppress the superscript $\mathcal{G}$ when it is clear that the linear class of function is used. Note that minimizing the training loss over all possible functions g (rather than over all $g \in \mathcal{G}$) does not lead to a meaningful optimization problem, as any function g for which $g(\mathbf{X}_i) = Y_i$ for all i gives minimal training loss. In particular, for a squared-error loss, the training loss will be 0. Unfortunately, such functions have a poor ability to predict new (that is, independent from $\mathcal{T}$) pairs of data. This poor generalization performance is called *overfitting*.

OVERFITTING



By choosing g a function that predicts the training data exactly (and is, for example, 0 otherwise), the squared-error training loss is zero. Minimizing the training loss is not the ultimate goal!

GENERALIZA-
TION RISK

The prediction accuracy of new pairs of data is measured by the *generalization risk* of the learner. For a *fixed* training set τ it is defined as

$$\ell(g_{\tau}^{\mathcal{G}}) = \mathbb{E} \text{Loss}(Y, g_{\tau}^{\mathcal{G}}(\mathbf{X})), \quad (1.5)$$

where $(\mathbf{X}, Y)$ is distributed according to $f(\mathbf{x}, y)$. In the discrete case the generalization risk is therefore: $\ell(g_{\tau}^{\mathcal{G}}) = \sum_{\mathbf{x}, y} \text{Loss}(y, g_{\tau}^{\mathcal{G}}(\mathbf{x})) f(\mathbf{x}, y)$ (replace the sum with an integral for the continuous case).

For a *random* training set $\mathcal{T}$, the generalization risk is thus a random variable that depends on $\mathcal{T}$ (and $\mathcal{G}$). If we average the generalization risk over all possible instances of $\mathcal{T}$, we obtain the *expected generalization risk*:

EXPECTED
GENERALIZA-
TION RISK

$$\mathbb{E} \ell(g_{\mathcal{T}}^{\mathcal{G}}) = \mathbb{E} \text{Loss}(Y, g_{\mathcal{T}}^{\mathcal{G}}(\mathbf{X})), \quad (1.6)$$

where $(\mathbf{X}, Y)$ in the expectation above is independent of $\mathcal{T}$. In the discrete case, we have $\mathbb{E} \ell(g_{\mathcal{T}}^{\mathcal{G}}) = \sum_{\mathbf{x}, y, \mathbf{x}_1, y_1, \dots, \mathbf{x}_n, y_n} \text{Loss}(y, g_{\mathcal{T}}^{\mathcal{G}}(\mathbf{x})) f(\mathbf{x}, y) f(\mathbf{x}_1, y_1) \cdots f(\mathbf{x}_n, y_n)$.

For any outcome τ of the training data, we can estimate the generalization risk without bias by taking the sample average

$$\ell_{\mathcal{T}'}(g_{\tau}^{\mathcal{G}}) := \frac{1}{n'} \sum_{i=1}^{n'} \text{Loss}(Y'_i, g_{\tau}^{\mathcal{G}}(X'_i)), \quad (1.7)$$

where $\{(X'_1, Y'_1), \dots, (X'_{n'}, Y'_{n'})\} =: \mathcal{T}'$ is a so-called *test sample*. The test sample is completely separate from $\mathcal{T}$, but is drawn in the same way as $\mathcal{T}$; that is, via independent draws from $f(\mathbf{x}, y)$, for some sample size n' . We call the estimator (1.7) the *test loss*. For a random training set $\mathcal{T}$ we can define $\ell_{\mathcal{T}'}(g_{\tau}^{\mathcal{G}})$ similarly. It is then crucial to assume that $\mathcal{T}$ is independent of $\mathcal{T}'$. Table 1.1 summarizes the main definitions and notation for supervised learning.

TEST SAMPLE

TEST LOSS

To compare the predictive performance of various learners in the function class $\mathcal{G}$, as measured by the test loss, we can use the *same* fixed training set τ and test set τ' for all learners. When there is an abundance of data, the “overall” data set is usually (randomly) divided into a training and test set, as depicted in Figure 1.1. We then

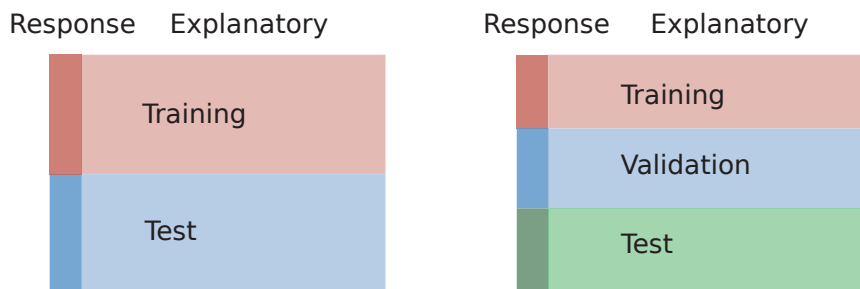


Figure 1.1: Statistical learning algorithms often require the data to be divided into training and test data. If the latter is used for model selection, a third set is needed for testing the performance of the selected model.

use the training data to construct various learners $g_{\tau}^{\mathcal{G}_1}, g_{\tau}^{\mathcal{G}_2}, \dots$, and use the test data to select the best (with the smallest test loss) among these learners. In this context the test set is called the *validation set*. Once the best learner has been chosen, a third “test” set can be used to assess the predictive performance of the best learner. The training, validation, and test sets can again be obtained from the overall data set via a random allocation. When the overall data set is of modest size, it is customary to perform the validation phase (model selection) on the training set only, using cross-validation. This is the topic of Section 1.2.1.

VALIDATION SET

We next consider a concrete example that illustrates the concepts introduced so far.



14

■ **Example 1.1 (Polynomial Regression)** In what follows, it will appear that we have arbitrarily replaced the symbols $\mathbf{x}, g, \mathcal{G}$ with $u, h, \mathcal{H}$, respectively. The reason for this switch of notation will become clear at the end of the example.

The data (depicted as dots) in Figure 1.2 are $n = 100$ points $(u_i, y_i), i = 1, \dots, n$ drawn from iid random points $(U_i, Y_i), i = 1, \dots, n$, where the $\{U_i\}$ are uniformly distributed on the interval $(0, 1)$ and, given $U_i = u_i$, the random variable Y_i has a normal distribution with expectation $10 - 140u_i + 400u_i^2 - 250u_i^3$ and variance $\ell^* = 25$. This is an example of a *polynomial regression model*. Using a squared-error loss, the optimal prediction function $h^*(u) = \mathbb{E}[Y | U = u]$ is thus

$$h^*(u) = 10 - 140u + 400u^2 - 250u^3,$$

which is depicted by the dashed curve in Figure 1.2.

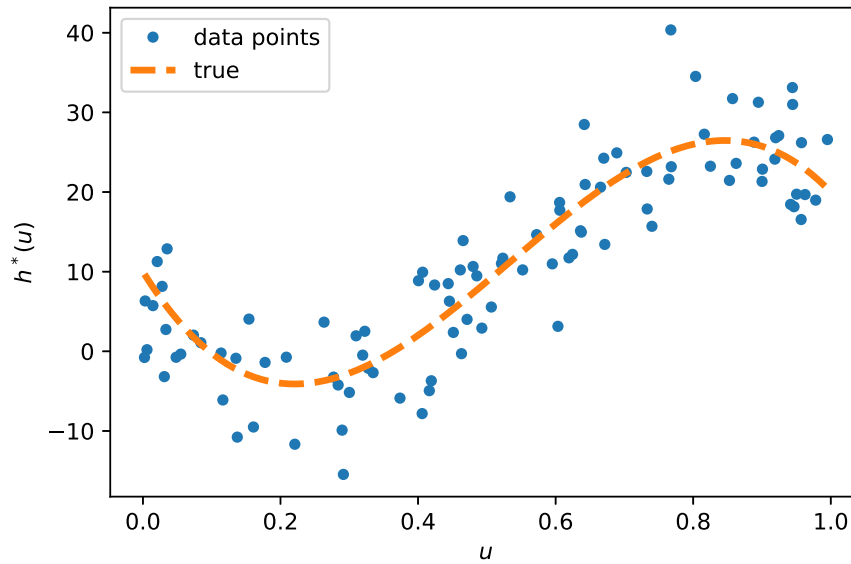


Figure 1.2: Training data and the optimal polynomial prediction function h^* .

To obtain a good estimate of $h^*(u)$ based on the training set $\tau = \{(u_i, y_i), i = 1, \dots, n\}$, we minimize the outcome of the training loss (1.3):

$$\ell_\tau(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h(u_i))^2, \quad (1.8)$$

over a suitable set $\mathcal{H}$ of candidate functions. Let us take the set $\mathcal{H}_p$ of polynomial functions in u of order $p - 1$:

$$h(u) := \beta_1 + \beta_2 u + \beta_3 u^2 + \dots + \beta_p u^{p-1} \quad (1.9)$$

for $p = 1, 2, \dots$ and parameter vector $\boldsymbol{\beta} = [\beta_1, \beta_2, \dots, \beta_p]^\top$. This function class contains the best possible $h^*(u) = \mathbb{E}[Y | U = u]$ for $p \geq 4$. Note that optimization over $\mathcal{H}_p$ is a parametric optimization problem, in that we need to find the best $\boldsymbol{\beta}$. Optimization of (1.8) over $\mathcal{H}_p$ is not straightforward, unless we notice that (1.9) is

a *linear* function in β . In particular, if we map each feature u to a feature vector $\mathbf{x} = [1, u, u^2, \dots, u^{p-1}]^\top$, then the right-hand side of (1.9) can be written as the function

$$g(\mathbf{x}) = \mathbf{x}^\top \beta,$$

which is linear in $\mathbf{x}$ (as well as β). The optimal $h^*(u)$ in $\mathcal{H}_p$ for $p \geq 4$ then corresponds to the function $g^*(\mathbf{x}) = \mathbf{x}^\top \beta^*$ in the set $\mathcal{G}_p$ of linear functions from $\mathbb{R}^p$ to $\mathbb{R}$, where $\beta^* = [10, -140, 400, -250, 0, \dots, 0]^\top$. Thus, instead of working with the set $\mathcal{H}_p$ of polynomial functions we may prefer to work with the set $\mathcal{G}_p$ of linear functions. This brings us to a very important idea in statistical learning:



Expand the feature space to obtain a *linear* prediction function.

Let us now reformulate the learning problem in terms of the new explanatory (feature) variables $\mathbf{x}_i = [1, u_i, u_i^2, \dots, u_i^{p-1}]^\top$, $i = 1, \dots, n$. It will be convenient to arrange these feature vectors into a matrix $\mathbf{X}$ with rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_n^\top$:

$$\mathbf{X} = \begin{bmatrix} 1 & u_1 & u_1^2 & \cdots & u_1^{p-1} \\ 1 & u_2 & u_2^2 & \cdots & u_2^{p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & u_n & u_n^2 & \cdots & u_n^{p-1} \end{bmatrix}. \quad (1.10)$$

Collecting the responses $\{y_i\}$ into a column vector $\mathbf{y}$, the training loss (1.3) can now be written compactly as

$$\frac{1}{n} \|\mathbf{y} - \mathbf{X}\beta\|^2. \quad (1.11)$$

To find the optimal learner (1.4) in the class $\mathcal{G}_p$ we need to find the minimizer of (1.11):

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\beta\|^2, \quad (1.12)$$

which is called the *ordinary least-squares* solution. As is illustrated in Figure 1.3, to find $\hat{\beta}$, we choose $\mathbf{X}\hat{\beta}$ to be equal to the orthogonal projection of $\mathbf{y}$ onto the linear space spanned by the columns of the matrix $\mathbf{X}$; that is, $\mathbf{X}\hat{\beta} = \mathbf{P}\mathbf{y}$, where $\mathbf{P}$ is the *projection matrix*.

ORDINARY
LEAST-
SQUARES

PROJECTION
MATRIX

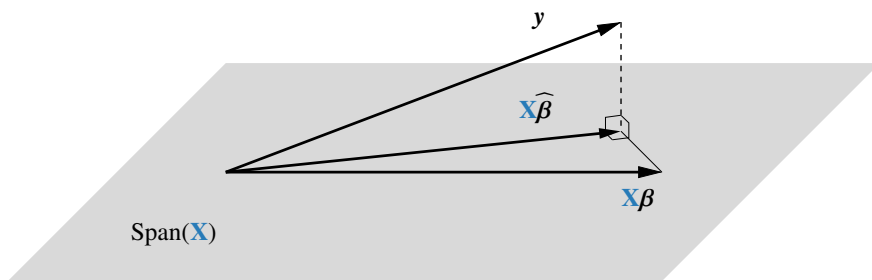


Figure 1.3: $\mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the linear space spanned by the columns of the matrix $\mathbf{X}$.

77

According to Theorem A.4, the projection matrix is given by

$$\mathbf{P} = \mathbf{X}\mathbf{X}^+, \quad (1.13)$$

72

PSEUDO-
INVERSE

68

where the $p \times n$ matrix $\mathbf{X}^+$ in (1.13) is the *pseudo-inverse* of $\mathbf{X}$. If $\mathbf{X}$ happens to be of *full column rank* (so that none of the columns can be expressed as a linear combination of the other columns), then $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$.

In any case, from $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{P}\mathbf{y}$ and $\mathbf{P}\mathbf{X} = \mathbf{X}$, we can see that $\hat{\boldsymbol{\beta}}$ satisfies the *normal equations*:

NORMAL
EQUATIONS

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{P} \mathbf{y} = (\mathbf{P} \mathbf{X})^\top \mathbf{y} = \mathbf{X}^\top \mathbf{y}. \quad (1.14)$$

This is a set of linear equations, which can be solved very fast and whose solution can be written explicitly as:

$$\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}. \quad (1.15)$$

Figure 1.4 shows the trained learners for various values of p :

$$h_{\tau}^{\mathcal{H}_p}(u) = g_{\tau}^{\mathcal{G}_p}(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$$

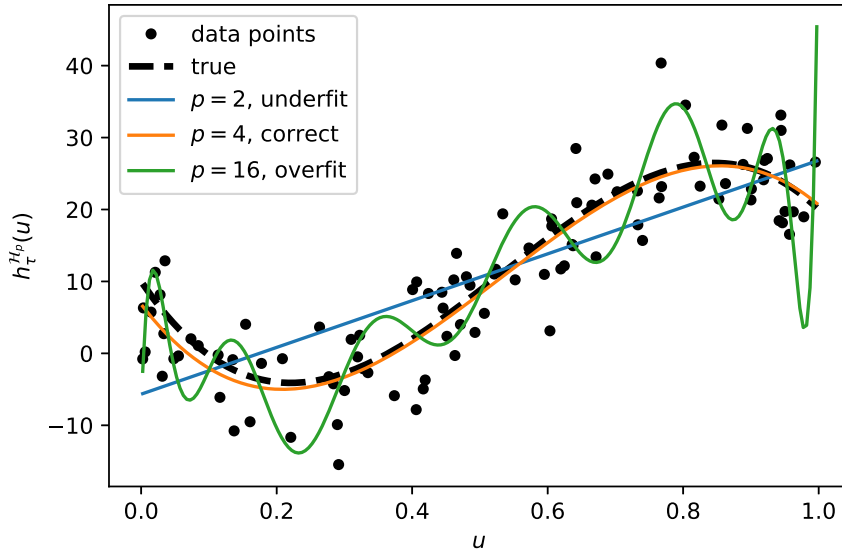


Figure 1.4: Training data with fitted curves for $p = 2, 4$, and 16 . The true cubic polynomial curve for $p = 4$ is also plotted (dashed line).

We see that for $p = 16$ the fitted curve lies closer to the data points, but is further away from the dashed true polynomial curve, indicating that we overfit. The choice $p = 4$ (the true cubic polynomial) is much better than $p = 16$, or indeed $p = 2$ (straight line).

Each function class $\mathcal{G}_p$ gives a different learner $g_{\tau}^{\mathcal{G}_p}$, $p = 1, 2, \dots$. To assess which is better, we should not simply take the one that gives the smallest training loss. We can always get a *zero* training loss by taking $p = n$, because for any set of n points there exists a polynomial of degree $n - 1$ that interpolates all points!

Instead, we assess the predictive performance of the learners using the test loss (1.7), computed from a test data set. If we collect all n' test feature vectors in a matrix $\mathbf{X}'$ and the corresponding test responses in a vector $\mathbf{y}'$, then, similar to (1.11), the test loss can be written compactly as

$$\ell_{\tau'}(g_{\tau}^{\mathcal{G}_p}) = \frac{1}{n'} \|\mathbf{y}' - \mathbf{X}'\widehat{\boldsymbol{\beta}}\|^2,$$

where $\widehat{\boldsymbol{\beta}}$ is given by (1.15), using the training data.

Figure 1.5 shows a plot of the test loss against the number of parameters in the vector $\boldsymbol{\beta}$; that is, p . The graph has a characteristic “bath-tub” shape and is at its lowest for $p = 4$, correctly identifying the polynomial order 3 for the true model. Note that the test loss, as an estimate for the generalization risk (1.7), becomes numerically unreliable after $p = 16$ (the graph goes down, where it should go up). The reader may check that the graph for the training loss exhibits a similar numerical instability for large p , and in fact fails to numerically decrease to 0 for large p , contrary to what it should do in theory. The numerical problems arise from the fact that for large p the columns of the

(Vandermonde) matrix $\mathbf{X}$ are of vastly different magnitudes and so floating point errors quickly become very large.

Finally, observe that the lower bound for the test loss is here around 21, which corresponds to an estimate of the minimal (squared-error) risk $\ell^* = 25$.

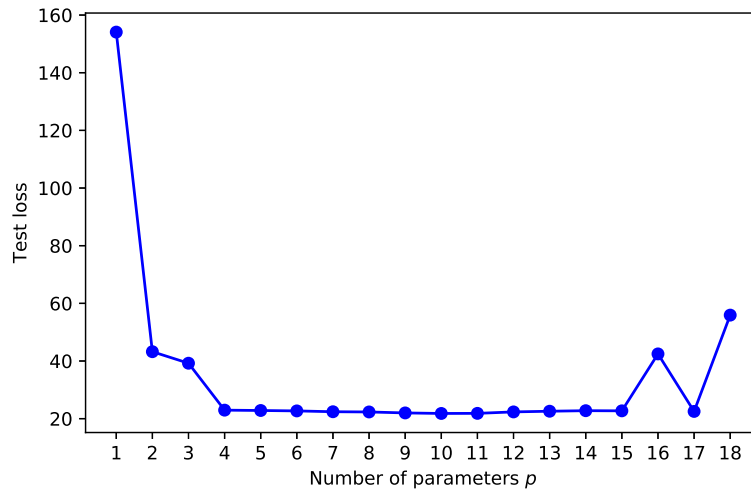


Figure 1.5: Test loss as function of the number of parameters p of the model.

This script shows how the training data were generated and plotted in Python:

polyreg1.py

```
import numpy as np
from numpy.random import rand , randn
from numpy.linalg import norm , solve
import matplotlib.pyplot as plt
def generate_data(beta , sig, n):
    u = np.random.rand(n, 1)
    y = (u ** np.arange(0, 4)) @ beta + sig * np.random.randn(n, 1)
    return u, y

np.random.seed(12)
beta = np.array([[10, -140, 400, -250]]).T
n = 100
sig = 5
u, y = generate_data(beta , sig, n)
xx = np.arange(np.min(u), np.max(u)+5e-3, 5e-3)
yy = np.polyval(np.flip(beta), xx)
plt.plot(u, y, '.', markersize=8)
plt.plot(xx, yy, '--', linewidth=3)
plt.xlabel(r'$u$')
plt.ylabel(r'$h^{(4)}(u)$')
plt.legend(['data points', 'true'])
```

```
plt.show()
```

The following code, which imports the code above, fits polynomial models with $p = 1, \dots, K = 18$ parameters to the training data and plots a selection of fitted curves, as shown in Figure 1.4.

polyreg2.py

```
from polyreg1 import *

max_p = 18
p_range = np.arange(1, max_p + 1, 1)
X = np.ones((n, 1))
betahat, trainloss = {}, {}

for p in p_range: # p is the number of parameters
    if p > 1:
        X = np.hstack((X, u**(p-1))) # add column to matrix

        betahat[p] = solve(X.T @ X, X.T @ y)
        trainloss[p] = (norm(y - X @ betahat[p])**2/n)

p = [2, 4, 16] # select three curves

#replot the points and true line and store in the list "plots"
plots = [plt.plot(u, y, 'k.', markersize=8)[0],
         plt.plot(xx, yy, 'k--', linewidth=3)[0]]
# add the three curves
for i in p:
    yy = np.polyval(np.flip(betahat[i]), xx)
    plots.append(plt.plot(xx, yy)[0])

plt.xlabel(r'$u$')
plt.ylabel(r'$h^{\mathcal{H}_p}_{\tau}(u)$')
plt.legend(plots, ('data points', 'true', '$p=2$, underfit',
                  '$p=4$, correct', '$p=16$, overfit'))
plt.savefig('polyfitpy.pdf', format='pdf')
plt.show()
```

The last code snippet which imports the previous code, generates the test data and plots the graph of the test loss, as shown in Figure 1.5.

polyreg3.py

```
from polyreg2 import *

# generate test data
u_test, y_test = generate_data(beta, sig, n)
```

```

MSE = []
X_test = np.ones((n, 1))

for p in p_range:
    if p > 1:
        X_test = np.hstack((X_test, u_test**(p-1)))

    y_hat = X_test @ betahat[p] # predictions
    MSE.append(np.sum((y_test - y_hat)**2/n))

plt.plot(p_range, MSE, 'b', p_range, MSE, 'bo')
plt.xticks(ticks=p_range)
plt.xlabel('Number of parameters $p$')
plt.ylabel('Test loss')

```

1.2.1 Cross-Validation

In general, for complex function classes $\mathcal{G}$, it is very difficult to derive simple formulas of the approximation and statistical errors, let alone for the generalization risk or expected generalization risk. As we saw, when there is an abundance of data, the easiest way to assess the generalization risk for a given training set τ is to obtain a test set τ' and evaluate the test loss (1.7). When a sufficiently large test set is not available but computational resources are cheap, one can instead gain direct knowledge of the expected generalization risk via a computationally intensive method called *cross-validation*.

The idea is to make multiple identical copies of the data set, and to partition each copy into different training and test sets, as illustrated in Figure 1.6. Here, there are four copies of the data set (consisting of response and explanatory variables). Each copy is divided into a test set (colored blue) and training set (colored pink). For each of these sets, we estimate the model parameters using only training data and then predict the responses for the test set. The average loss between the predicted and observed responses is then a measure for the predictive power of the model.

In particular, suppose we partition a data set $\mathcal{T}$ of size n into K folds $C_1, \dots, C_K$ of sizes $n_1, \dots, n_K$ (hence, $n_1 + \dots + n_K = n$). Typically $n_k \approx n/K$, $k = 1, \dots, K$.

Let ℓ_{C_k} be the test loss when using C_k as test data and all remaining data, denoted $\mathcal{T}_{-k}$, as training data. Each ℓ_{C_k} is an unbiased estimator of the generalization risk for training set $\mathcal{T}_{-k}$; that is, for $\ell(g_{\mathcal{T}_{-k}})$.

7

CROSS-
VALIDATION

FOLDS

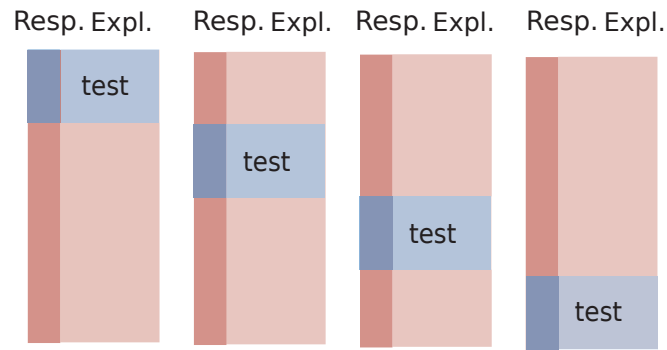


Figure 1.6: An illustration of four-fold cross-validation, representing four copies of the same data set. The data in each copy is partitioned into a training set (pink) and a test set (blue). The darker columns represent the response variable and the lighter ones the explanatory variables.

The K -fold cross-validation loss is the weighted average of these risk estimators:

**K -FOLD
CROSS-
VALIDATION**

$$\begin{aligned} \text{CV}_K &= \sum_{k=1}^K \frac{n_k}{n} \ell_{C_k}(g_{\mathcal{T}_k}) \\ &= \frac{1}{n} \sum_{k=1}^K \sum_{i \in C_k} \text{Loss}(g_{\mathcal{T}_k}(\mathbf{x}_i), y_i) \\ &= \frac{1}{n} \sum_{i=1}^n \text{Loss}(g_{\mathcal{T}_{\kappa(i)}}(\mathbf{x}_i), y_i), \end{aligned}$$

where the function $\kappa : \{1, \dots, n\} \mapsto \{1, \dots, K\}$ indicates to which of the K folds each of the n observations belongs. In other words, $\kappa(i)$ indicates which fold contains the i -th observation. In the special case of $K = n$, we have $\kappa(i) = i$. As the average is taken over varying training sets $\{\mathcal{T}_k\}$, it estimates the expected generalization risk $\mathbb{E} \ell(g_{\mathcal{T}})$, rather than the generalization risk $\ell(g_{\mathcal{T}})$ for the particular training set $\mathcal{T}$.

■ **Example 1.2 (Polynomial Regression (cont.))** For the polynomial regression example, we can calculate a K -fold cross-validation loss with a nonrandom partitioning of the training set using the following code, which imports the previous code for the polynomial regression example. We omit the full plotting code.

polyregCV.py

```
from polyreg3 import *

K_vals = [5, 10, 100] # number of folds
cv = np.zeros((len(K_vals), max_p)) # cv loss
X = np.ones((n, 1))
```

```

for p in p_range:
    if p > 1:
        X = np.hstack((X, u**(p-1)))
    j = 0
    for K in K_vals:
        loss = []
        for k in range(1, K+1):
            # integer indices of test samples
            test_ind = ((n/K)*(k-1) + np.arange(1,n/K+1)-1).astype('int')
            train_ind = np.setdiff1d(np.arange(n), test_ind)

            X_train, y_train = X[train_ind, :], y[train_ind, :]
            X_test, y_test = X[test_ind, :], y[test_ind]

            # fit model and evaluate test loss
            betahat = solve(X_train.T @ X_train, X_train.T @ y_train)
            loss.append(norm(y_test - X_test @ betahat) ** 2)

        cv[j, p-1] = sum(loss)/n
        j += 1

# basic plotting
plt.plot(p_range, cv[0, :], 'k-.')
plt.plot(p_range, cv[1, :], 'r')
plt.plot(p_range, cv[2, :], 'b--')
plt.show()

```

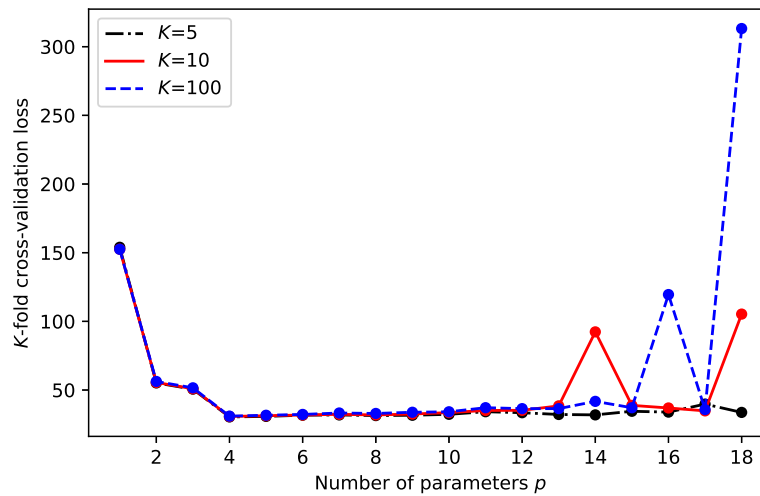


Figure 1.7: K -fold cross-validation for the polynomial regression example.

Figure 1.7 shows the cross-validation loss for $K \in \{5, 10, 100\}$. The case $K = 100$

corresponds to the *leave-one-out cross-validation*, which can be computed more efficiently using the formula in Theorem 2.1. ■

LEAVE-ONE-
OUT
CROSS-
VALIDATION



29

1.3 Linear Models

In a *supervised* setting, where the data is represented by a vector $\mathbf{x}$ of explanatory variables and a response y , the general model is that $(\mathbf{x}, y)$ is an outcome of $(\mathbf{X}, Y) \sim f$ for some unknown f . And for a training sequence $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$ the default model assumption is that $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n) \sim_{\text{iid}} f$. As explained in Section 1.1, the analysis primarily involves the conditional pdf $f(y|\mathbf{x})$ and in particular (when using the squared-error loss) the conditional expectation $g^*(\mathbf{x}) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}]$. The resulting representation (1.2) allows us to then write the response at $\mathbf{X} = \mathbf{x}$ as a function of the feature $\mathbf{x}$ plus an error term: $Y = g^*(\mathbf{x}) + \varepsilon(\mathbf{x})$.

This leads to the simplest and most important model for supervised learning, where we choose a *linear* class $\mathcal{G}$ of prediction or guess functions and assume that it is rich enough to contain the true g^* . If we further assume that, conditional on $\mathbf{X} = \mathbf{x}$, the error term ε does not depend on $\mathbf{x}$, that is, $\mathbb{E} \varepsilon = 0$ and $\text{Var} \varepsilon = \sigma^2$, then we obtain the following model.

Definition 1.1: Linear Model

In a *linear model* the response Y depends on a p -dimensional explanatory variable $\mathbf{x} = [x_1, \dots, x_p]^\top$ via the linear relationship

$$Y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon, \quad (1.16)$$

where $\mathbb{E} \varepsilon = 0$ and $\text{Var} \varepsilon = \sigma^2$.

LINEAR
MODEL

Note that (1.16) is a model for a single pair $(\mathbf{x}, Y)$. The model for the training set $\{(\mathbf{x}_i, Y_i)\}$ is simply that each Y_i satisfies (1.16) (with $\mathbf{x} = \mathbf{x}_i$) and that the $\{Y_i\}$ are independent. Gathering all responses in the vector $\mathbf{Y} = [Y_1, \dots, Y_n]^\top$, we can write

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (1.17)$$

where $\boldsymbol{\varepsilon} = [\varepsilon_1, \dots, \varepsilon_n]^\top$ is a vector of iid copies of ε and $\mathbf{X}$ is the so-called *model matrix*, with rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_n^\top$. Linear models are fundamental building blocks of statistical learning algorithms.

MODEL
MATRIX

■ **Example 1.3 (Polynomial Regression (cont.))** For our running Example 1.1, we see that the data is described by a linear model of the form (1.17), with model matrix $\mathbf{X}$ given in (1.10). ■



7

1.4 Multivariate Normal Models

A standard model for numerical observations $x_1, \dots, x_n$ (forming, e.g., a column in a spreadsheet or data frame) is that they are the outcomes of iid normal random variables

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

It is helpful to view a normally distributed random variable as a simple transformation of a standard normal random variable. To wit, if Z has a standard normal distribution, then $X = \mu + \sigma Z$ has a $\mathcal{N}(\mu, \sigma^2)$ distribution. The generalization to n dimensions is discussed in Appendix B.7. We summarize the main points: Let $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. The pdf of $\mathbf{Z} = [Z_1, \dots, Z_n]^\top$ (that is, the joint pdf of $Z_1, \dots, Z_n$) is given by

$$f_{\mathbf{Z}}(\mathbf{z}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z_i^2} = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\mathbf{z}^\top \mathbf{z}}, \quad \mathbf{z} \in \mathbb{R}^n. \quad (1.18)$$

We write $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ and say that $\mathbf{Z}$ has a standard normal distribution in $\mathbb{R}^n$. Let

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{B} \mathbf{Z} \quad (1.19)$$

for some $m \times n$ matrix $\mathbf{B}$ and m -dimensional vector $\boldsymbol{\mu}$. Then $\mathbf{X}$ has expectation vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{B}^\top$; see (B.20) and (B.21). This leads to the following definition.

Definition 1.2: Multivariate Normal Distribution

An m -dimensional random vector $\mathbf{X}$ that can be written in the form (1.19) for some m -dimensional vector $\boldsymbol{\mu}$ and $m \times n$ matrix $\mathbf{B}$, with $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$, is said to have a *multivariate normal* or *multivariate Gaussian* distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{B}^\top$. We write $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

The m -dimensional density of a multivariate normal distribution has a very similar form to the density of the one-dimensional normal distribution and is given in the next theorem. We leave the proof as an exercise;

Theorem 1.2: Density of a Multivariate Random Vector

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where the $m \times m$ covariance matrix $\boldsymbol{\Sigma}$ is invertible. Then $\mathbf{X}$ has pdf

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^m |\boldsymbol{\Sigma}|}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}, \quad \mathbf{x} \in \mathbb{R}^m. \quad (1.20)$$

109

107

MULTIVARI-
ATE NORMAL

Figure 1.8 shows the pdfs of two bivariate (that is, two-dimensional) normal distributions. In both cases the mean vector is $\boldsymbol{\mu} = [0, 0]^\top$ and the variances (the diagonal elements of Σ) are 1. The correlation coefficients (or, equivalently here, the covariances) are respectively $\varrho = 0$ and $\varrho = 0.8$.

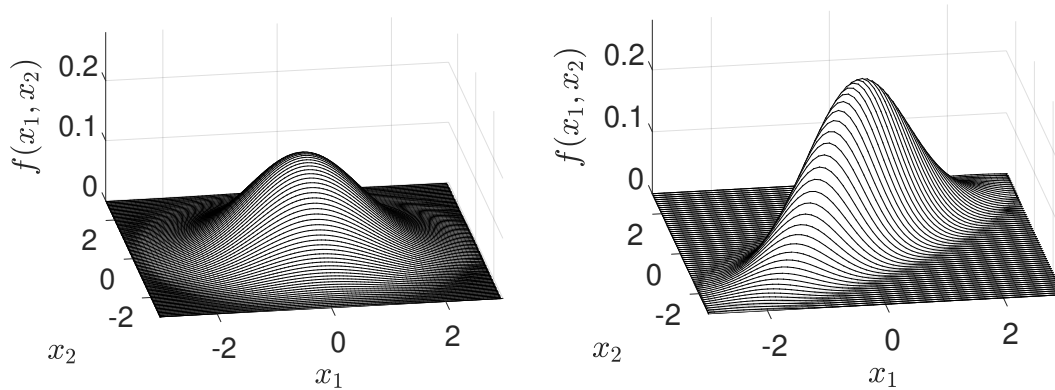


Figure 1.8: Pdfs of bivariate normal distributions with means zero, variances 1, and correlation coefficients 0 (left) and 0.8 (right).

The main reason why the multivariate normal distribution plays an important role in data science and machine learning is that it satisfies the following properties, the details and proofs of which can be found in Appendix B.7:

1. Affine combinations are normal.
2. Marginal distributions are normal.
3. Conditional distributions are normal.



109

1.5 Normal Linear Models

Normal linear models combine the simplicity of the linear model with the tractability of the Gaussian distribution. They are the principal model for traditional statistics, and include the classic linear regression and analysis of variance models.

Definition 1.3: Normal Linear Model

In a *normal linear model* the response Y depends on a p -dimensional explanatory variable $\mathbf{x} = [x_1, \dots, x_p]^\top$, via the linear relationship

$$Y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon, \quad (1.21)$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

NORMAL
LINEAR
MODEL

Thus, a normal linear model is a linear model (in the sense of Definition 1.1) with normal error terms. Similar to (1.17), the corresponding normal linear model for the whole training set $\{(\mathbf{x}_i, Y_i)\}$ has the form

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (1.22)$$

where $\mathbf{X}$ is the model matrix comprised of rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_n^\top$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. Consequently, $\mathbf{Y}$ can be written as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \sigma \mathbf{Z}$, where $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$, so that $\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$. It follows from (1.20) that its joint density is given by

$$g(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2, \mathbf{X}) = (2\pi\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}. \quad (1.23)$$

Estimation of the parameter $\boldsymbol{\beta}$ can be performed via the least-squares method, as discussed in Example 1.1. An estimate can also be obtained via the maximum likelihood method. This simply means finding the parameters σ^2 and $\boldsymbol{\beta}$ that maximize the likelihood of the outcome $\mathbf{y}$, given by the right-hand side of (1.23). It is clear that for every value of σ^2 the likelihood is maximal when $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ is minimal. As a consequence, the maximum likelihood estimate for $\boldsymbol{\beta}$ is the same as the least-squares estimate (1.15). We leave it as an exercise to show that the maximum likelihood estimate of σ^2 is equal to

$$\widehat{\sigma^2} = \frac{\|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2}{n}, \quad (1.24)$$

where $\widehat{\boldsymbol{\beta}}$ is the maximum likelihood estimate (least squares estimate in this case) of $\boldsymbol{\beta}$.

Table 1.1: Summary of definitions for supervised learning.

$\mathbf{x}$	Fixed explanatory (feature) vector.
$\mathbf{X}$	Random explanatory (feature) vector.
y	Fixed (real-valued) response.
Y	Random response.
$f(\mathbf{x}, y)$	Joint pdf of $\mathbf{X}$ and Y , evaluated at $(\mathbf{x}, y)$.
$f(y \mathbf{x})$	Conditional pdf of Y given $\mathbf{X} = \mathbf{x}$, evaluated at y .
τ or τ_n	Fixed training data $\{(\mathbf{x}_i, y_i), i = 1, \dots, n\}$.
$\mathcal{T}$ or $\mathcal{T}_n$	Random training data $\{(\mathbf{X}_i, Y_i), i = 1, \dots, n\}$.
$\mathbf{X}$	Matrix of explanatory variables, with n rows $\mathbf{x}_i^\top, i = 1, \dots, n$ and $\dim(\mathbf{x})$ feature columns; one of the features may be the constant 1.
$\mathbf{y}$	Vector of response variables $(y_1, \dots, y_n)^\top$.
g	Prediction (guess) function.
$\text{Loss}(y, \hat{y})$	Loss incurred when predicting response y with $\hat{y}$.
$\ell(g)$	Risk for prediction function g ; that is, $\mathbb{E} \text{Loss}(Y, g(\mathbf{X}))$.
g^*	Optimal prediction function; that is, $\arg\min_g \ell(g)$.
$g^{\mathcal{G}}$	Optimal prediction function in function class $\mathcal{G}$; that is, $\arg\min_{g \in \mathcal{G}} \ell(g)$.
$\ell_\tau(g)$	Training loss for prediction function g ; that is, the sample average estimate of $\ell(g)$ based on a fixed training sample τ .
$\ell_{\mathcal{T}}(g)$	The same as $\ell_\tau(g)$, but now for a random training sample $\mathcal{T}$.
$g_\tau^{\mathcal{G}}$ or g_τ	The <i>learner</i> : $\arg\min_{g \in \mathcal{G}} \ell_\tau(g)$. That is, the optimal prediction function based on a fixed training set τ and function class $\mathcal{G}$. We suppress the superscript $\mathcal{G}$ if the function class is implicit.
$g_{\mathcal{T}}^{\mathcal{G}}$ or $g_{\mathcal{T}}$	The learner, where we have replaced τ with a random training set $\mathcal{T}$.

LINEAR REGRESSION

2.1 Introduction

Francis Galton observed in an article in 1889 that the heights of adult offspring are, on the whole, more “average” than the heights of their parents. Galton interpreted this as a degenerative phenomenon, using the term “regression” to indicate this “return to mediocrity”. Nowadays, *regression* refers to a broad class of supervised learning techniques where the aim is to predict a quantitative response (output) variable y via a function $g(\mathbf{x})$ of an explanatory (input) vector $\mathbf{x} = [x_1, \dots, x_p]^\top$, consisting of p features, each of which can be continuous or discrete. For instance, regression could be used to predict the birth weight of a baby (the response variable) from the weight of the mother, her socio-economic status, and her smoking habits (the explanatory variables).

REGRESSION

Let us recapitulate the framework of supervised learning established in Chapter 1. The aim is to find a prediction function g that best guesses¹ what the random output Y will be for a random input vector $\mathbf{X}$. The joint pdf $f(\mathbf{x}, y)$ of $\mathbf{X}$ and Y is unknown, but a training set $\tau = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is available, which is thought of as the outcome of a random training set $\mathcal{T} = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$ of iid copies of $(\mathbf{X}, Y)$. Once we have selected a loss function $\text{Loss}(y, \hat{y})$, such as the *squared-error loss*

3

SQUARED-
ERROR LOSS

$$\text{Loss}(y, \hat{y}) = (y - \hat{y})^2, \quad (2.1)$$

then the “best” prediction function g is defined as the one that minimizes the *risk* $\ell(g) = \mathbb{E} \text{Loss}(Y, g(\mathbf{X}))$. We saw in Section 1.1 that for the squared-error loss this optimal prediction function is the conditional expectation

RISK

$$g^*(\mathbf{x}) = \mathbb{E}[Y | \mathbf{X} = \mathbf{x}].$$

As the squared-error loss is the most widely-used loss function for regression, we will adopt this loss function in most of this chapter.

¹Recall the mnemonic use of “g” for “guess”

The optimal prediction function g^* has to be learned from the training set τ by minimizing the training loss

$$\ell_\tau(g) = \frac{1}{n} \sum_{i=1}^n (y_i - g(\mathbf{x}_i))^2 \quad (2.2)$$

over a suitable class of functions $\mathcal{G}$. Note that in the above definition, the training set τ is assumed to be fixed. For a random training set $\mathcal{T}$, we will write the training loss as $\ell_{\mathcal{T}}(g)$. The function $g_{\mathcal{T}}^{\mathcal{G}}$ that minimizes the training loss is the function we use for prediction — the so-called *learner*. When the function class $\mathcal{G}$ is linear, we drop the superscript in the notation.

LEARNER

4

As we already saw in (1.2), conditional on $\mathbf{X} = \mathbf{x}$, the response Y can be written as

$$Y = g^*(\mathbf{x}) + \varepsilon(\mathbf{x}),$$

where $\mathbb{E} \varepsilon(\mathbf{x}) = 0$. This motivates a standard modeling assumption in supervised learning, in which the responses $Y_1, \dots, Y_n$, conditional on the explanatory variables $\mathbf{X}_1 = \mathbf{x}_1, \dots, \mathbf{X}_n = \mathbf{x}_n$, are assumed to be of the form

$$Y_i = g(\mathbf{x}_i) + \varepsilon_i, \quad i = 1, \dots, n,$$

where the $\{\varepsilon_i\}$ are independent with $\mathbb{E} \varepsilon_i = 0$ and $\text{Var} \varepsilon_i = \sigma^2$ for some function $g \in \mathcal{G}$ and variance σ^2 . The above model is usually further specified by assuming that g is completely known up to an unknown parameter vector; that is,

$$Y_i = g(\mathbf{x}_i | \boldsymbol{\beta}) + \varepsilon_i, \quad i = 1, \dots, n. \quad (2.3)$$

While the model (2.3) is described *conditional* on the explanatory variables, it will be convenient to make one further model simplification, and view (2.3) as if the $\{\mathbf{x}_i\}$ were *fixed*, while the $\{Y_i\}$ are random.



For the remainder of this chapter, we assume that the training feature vectors $\{\mathbf{x}_i\}$ are fixed and only the responses are random; that is, $\mathcal{T} = \{(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)\}$.

The advantage of the model (2.3) is that the problem of estimating the *function* g from the training data is reduced to the (much simpler) problem of estimating the *parameter vector* $\boldsymbol{\beta}$. An obvious disadvantage is that functions of the form $g(\cdot | \boldsymbol{\beta})$ may not accurately approximate the true unknown g^* . The remainder of this chapter deals with the analysis of models of the form (2.3). In the important case where the function $g(\cdot | \boldsymbol{\beta})$ is *linear*, the analysis proceeds through the class of linear models. If, in addition, the error terms $\{\varepsilon_i\}$ are assumed to be *Gaussian*, this analysis can be carried out using the rich theory of normal linear models.

2.2 Linear Regression

The most basic regression model involves a linear relationship between the response and a single explanatory variable. In particular, we have measurements $(x_1, y_1), \dots, (x_n, y_n)$ that lie approximately on a straight line, as in Figure 2.1.

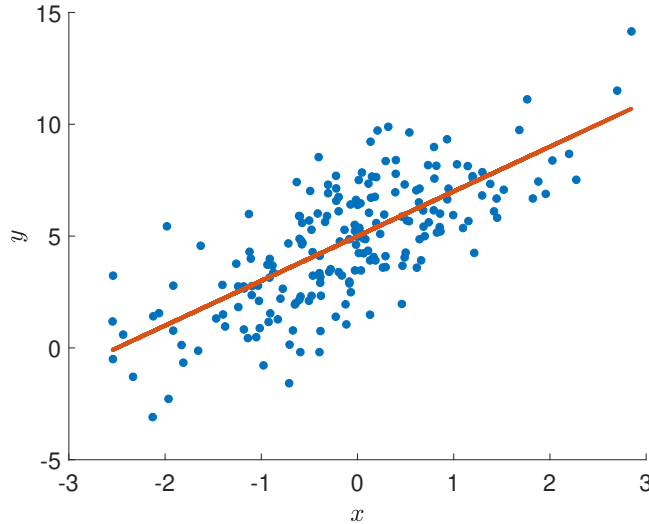


Figure 2.1: Data from a simple linear regression model.

Following the general scheme captured in (2.3), a simple model for these data is that the $\{x_i\}$ are fixed and variables $\{Y_i\}$ are random such that

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (2.4)$$

for certain *unknown* parameters β_0 and β_1 . The $\{\varepsilon_i\}$ are assumed to be independent with expectation 0 and unknown variance σ^2 . The unknown line

$$y = \underbrace{\beta_0 + \beta_1 x}_{g(x|\boldsymbol{\beta})} \quad (2.5)$$

is called the *regression line*. Thus, we view the responses as random variables that would lie exactly on the regression line, were it not for some “disturbance” or “error” term represented by the $\{\varepsilon_i\}$. The extent of the disturbance is modeled by the parameter σ^2 . The model in (2.4) is called *simple linear regression*. This model can easily be extended to incorporate more than one explanatory variable, as follows.

REGRESSION
LINE

SIMPLE
LINEAR
REGRESSION
MODEL

Definition 2.1: Multiple Linear Regression Model

In a *multiple linear regression model* the response Y depends on a d -dimensional explanatory vector $\mathbf{x} = [x_1, \dots, x_d]^\top$, via the linear relationship

$$Y = \beta_0 + \beta_1 x_1 + \dots + \beta_d x_d + \varepsilon, \quad (2.6)$$

where $\mathbb{E} \varepsilon = 0$ and $\text{Var} \varepsilon = \sigma^2$.

Thus, the data lie approximately on a d -dimensional affine hyperplane

$$y = \underbrace{\beta_0 + \beta_1 x_1 + \dots + \beta_d x_d}_{g(\mathbf{x} | \boldsymbol{\beta})},$$

where we define $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_d]^\top$. The function $g(\mathbf{x} | \boldsymbol{\beta})$ is linear in $\boldsymbol{\beta}$, but not linear in the feature vector $\mathbf{x}$, due to the constant β_0 . However, *augmenting* the feature space with the constant 1, the mapping $[1, \mathbf{x}^\top]^\top \mapsto g(\mathbf{x} | \boldsymbol{\beta}) := [1, \mathbf{x}^\top] \boldsymbol{\beta}$ becomes linear in the feature space and so (2.6) becomes a *linear model* (see Section 1.1). Most software packages for regression include 1 as a feature by default.

Note that in (2.6) we only specified the model for a single pair $(\mathbf{x}, Y)$. The model for the training set $\mathcal{T} = \{(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)\}$ is simply that each Y_i satisfies (2.6) (with $\mathbf{x} = \mathbf{x}_i$) and that the $\{Y_i\}$ are independent. Setting $\mathbf{Y} = [Y_1, \dots, Y_n]^\top$, we can write the multiple linear regression model for the training data compactly as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (2.7)$$

where $\boldsymbol{\varepsilon} = [\varepsilon_1, \dots, \varepsilon_n]^\top$ is a vector of iid copies of ε and $\mathbf{X}$ is the *model matrix* given by

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1d} \\ 1 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{x}_1^\top \\ 1 & \mathbf{x}_2^\top \\ \vdots & \vdots \\ 1 & \mathbf{x}_n^\top \end{bmatrix}.$$

■ **Example 2.1 (Multiple Linear Regression Model)** Figure 2.2 depicts a realization of the multiple linear regression model

$$Y_i = x_{i1} + x_{i2} + \varepsilon_i, \quad i = 1, \dots, 100,$$

where $\varepsilon_1, \dots, \varepsilon_{100} \sim_{\text{iid}} \mathcal{N}(0, 1/16)$. The fixed feature vectors (vectors of explanatory variables) $\mathbf{x}_i = [x_{i1}, x_{i2}]^\top$, $i = 1, \dots, 100$ lie in the unit square.

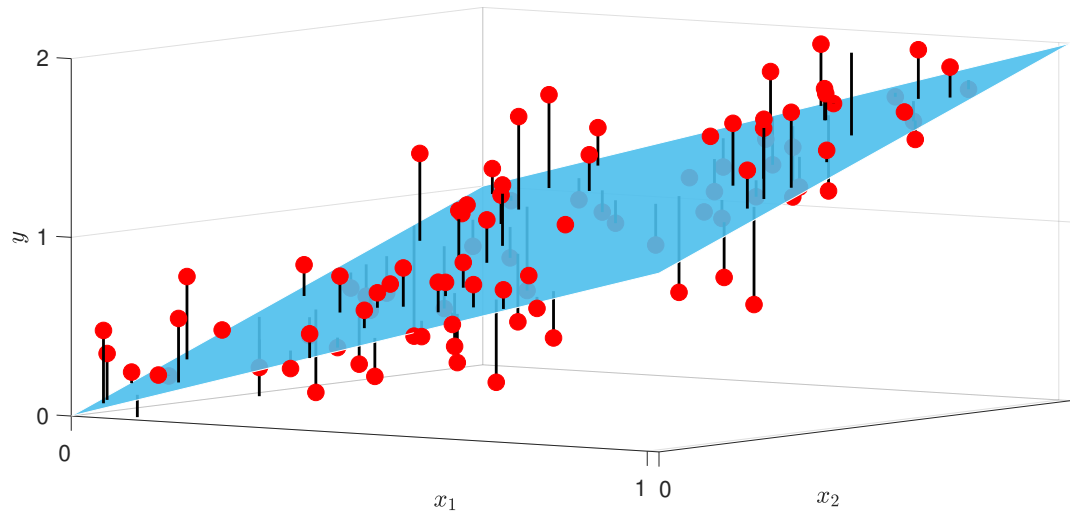


Figure 2.2: Data from a multiple linear regression model.



2.3 Analysis via Linear Models

Analysis of data from a linear regression model is greatly simplified through the linear model representation (2.7). In this section we present the main ideas for parameter estimation and model selection for a general linear model of the form

$$Y = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (2.8)$$

where $\mathbf{X}$ is an $n \times p$ matrix, $\boldsymbol{\beta} = [\beta_1, \dots, \beta_p]^\top$ a vector of p parameters, and $\boldsymbol{\varepsilon} = [\varepsilon_1, \dots, \varepsilon_n]^\top$ an n -dimensional vector of independent error terms, with $\mathbb{E} \varepsilon_i = 0$ and $\text{Var} \varepsilon_i = \sigma^2$, $i = 1, \dots, n$. Note that the model matrix $\mathbf{X}$ is assumed to be fixed, and Y and $\boldsymbol{\varepsilon}$ are random. A specific outcome of Y is denoted by y (in accordance with the notation in Section 1.5).



19



Note that the multiple linear regression model in (2.7) was defined using a different parameterization; in particular, there we used $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_d]^\top$. So, when applying the results in the present section to such models, be aware that $p = d + 1$. Also, in this section a feature vector $\mathbf{x}$ includes the constant 1, so that $\mathbf{X}^\top = [\mathbf{x}_1, \dots, \mathbf{x}_n]$.

2.3.1 Parameter Estimation

The linear model $Y = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ contains two unknown parameters, $\boldsymbol{\beta}$ and σ^2 , which have to be estimated from the training data τ . To estimate $\boldsymbol{\beta}$, we can repeat exactly the same reasoning used in our recurring polynomial regression Example 1.1 as follows. For a linear prediction function $g(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}$, the (squared-error) training loss can be written as

$$\ell_\tau(g) = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2,$$

and the optimal learner g_τ minimizes this quantity, leading to the least-squares estimate $\hat{\boldsymbol{\beta}}$, which satisfies the normal equations

$$\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}. \quad (2.9)$$

The corresponding training loss can be taken as an estimate of σ^2 ; that is,

$$\widehat{\sigma^2} = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2. \quad (2.10)$$

To justify the latter, note that σ^2 is the second moment of the model errors $\varepsilon_i, i = 1, \dots, n$, in (2.8) and could be estimated via the method of moments (see Section B.9.1) using the sample average $n^{-1} \sum_i \varepsilon_i^2 = \|\boldsymbol{\varepsilon}\|^2/n = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2/n$, if $\boldsymbol{\beta}$ were known. By replacing $\boldsymbol{\beta}$ with its estimator, we arrive at (2.10). Note that no distributional properties of the $\{\varepsilon_i\}$ were used other than $\mathbb{E} \varepsilon_i = 0$ and $\text{Var} \varepsilon_i = \sigma^2, i = 1, \dots, n$. The vector $\mathbf{e} := \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ is called the vector of *residuals* and approximates the (unknown) vector of model errors $\boldsymbol{\varepsilon}$. The quantity $\|\mathbf{e}\|^2 = \sum_{i=1}^n e_i^2$ is called the *residual sum of squares* (RSS). Dividing the RSS by $n - p$ gives an unbiased estimate of σ^2 , which we call the estimated *residual squared error* (RSE).

In terms of the notation given in the summary Table 1.1 for supervised learning, we thus have:

1. The (observed) training data is $\tau = \{\mathbf{X}, \mathbf{y}\}$.
2. The function class $\mathcal{G}$ is the class of linear functions of $\mathbf{x}$; that is $\mathcal{G} = \{g(\cdot | \boldsymbol{\beta}) : \mathbf{x} \mapsto \mathbf{x}^\top \boldsymbol{\beta}, \boldsymbol{\beta} \in \mathbb{R}^p\}$.
3. The (squared-error) training loss is $\ell_\tau(g(\cdot | \boldsymbol{\beta})) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2/n$.
4. The learner g_τ is given by $g_\tau(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$, where $\hat{\boldsymbol{\beta}} = \text{argmin}_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$.
5. The minimal training loss is $\ell_\tau(g_\tau) = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2/n = \widehat{\sigma^2}$.

7

118

RESIDUALS

RESIDUAL
SUM OF
SQUARESRESIDUAL
SQUARED
ERROR

21

2.3.2 Cross-Validation and Predictive Residual Sum of Squares

We start by considering the n -fold cross-validation, also called *leave-one-out cross-validation*, for the linear model (2.8). We partition the data into n data sets, leaving out precisely one observation per data set, which we then predict based on the $n - 1$ remaining observations; see Section 1.2.1 for the general case. Let $\hat{y}_{-i}$ denote the prediction for the i -th observation using all the data except y_i . The error in the prediction, $y_i - \hat{y}_{-i}$, is called a *predicted residual* — in contrast to an ordinary residual, $e_i = y_i - \hat{y}_i$, which is the difference between an observation and its fitted value $\hat{y}_i = g_\tau(\mathbf{x}_i)$ obtained using the whole sample. In this way, we obtain the collection of predicted residuals $\{y_i - \hat{y}_{-i}\}_{i=1}^n$ and summarize them through the *predicted residual sum of squares (PRESS)*:

$$\text{PRESS} = \sum_{i=1}^n (y_i - \hat{y}_{-i})^2.$$

Dividing the PRESS by n gives an estimate of the expected generalization risk.

In general, computing the PRESS is computationally intensive as it involves training and predicting n separate times. For linear models, however, the predicted residuals can be calculated quickly using only the ordinary residuals and the projection matrix $\mathbf{P} = \mathbf{X}\mathbf{X}^+$ onto the linear space spanned by the columns of the model matrix $\mathbf{X}$ (see (1.13)). The i -th diagonal element $\mathbf{P}_{ii}$ of the projection matrix is called the i -th *leverage*, and it can be shown that $0 \leq \mathbf{P}_{ii} \leq 1$.

Theorem 2.1: PRESS for Linear Models

Consider the linear model (2.8), where the $n \times p$ model matrix $\mathbf{X}$ is of full rank. Given an outcome $\mathbf{y} = [y_1, \dots, y_n]^\top$ of $\mathbf{Y}$, the fitted values can be obtained as $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$, where $\mathbf{P} = \mathbf{X}\mathbf{X}^+ = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is the projection matrix. If the leverage value $p_i := \mathbf{P}_{ii} \neq 1$ for all $i = 1, \dots, n$, then the predicted residual sum of squares can be written as

$$\text{PRESS} = \sum_{i=1}^n \left(\frac{e_i}{1 - p_i} \right)^2,$$

where $e_i = y_i - \hat{y}_i = y_i - (\mathbf{X}\hat{\boldsymbol{\beta}})_i$ is the i -th residual.

Proof: It suffices to show that the i -th predicted residual can be written as $y_i - \hat{y}_{-i} = e_i / (1 - p_i)$. Let $\mathbf{X}_{-i}$ denote the model matrix $\mathbf{X}$ with the i -th row, $\mathbf{x}_i^\top$, removed, and define $\mathbf{y}_{-i}$ similarly. Then, the least-squares estimate for $\boldsymbol{\beta}$ using all but the i -th observation is $\hat{\boldsymbol{\beta}}_{-i} = (\mathbf{X}_{-i}^\top \mathbf{X}_{-i})^{-1} \mathbf{X}_{-i}^\top \mathbf{y}_{-i}$. Writing $\mathbf{X}^\top \mathbf{X} = \mathbf{X}_{-i}^\top \mathbf{X}_{-i} + \mathbf{x}_i \mathbf{x}_i^\top$, we have by the Sherman–Morrison formula

$$(\mathbf{X}_{-i}^\top \mathbf{X}_{-i})^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1}}{1 - \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i},$$

14

LEAVE-ONE-OUT
CROSS-VALIDATION

PREDICTED
RESIDUAL

PRESS

27

10

LEVERAGE

87

where $\mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i = p_i < 1$. Also, $\mathbf{X}_{-i}^\top \mathbf{y}_{-i} = \mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i$. Combining all these identities, we have

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{-i} &= (\mathbf{X}_{-i}^\top \mathbf{X}_{-i})^{-1} \mathbf{X}_{-i}^\top \mathbf{y}_{-i} \\ &= \left((\mathbf{X}^\top \mathbf{X})^{-1} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1}}{1 - p_i} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) \\ &= \hat{\boldsymbol{\beta}} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}}{1 - p_i} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i y_i - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i p_i y_i}{1 - p_i} \\ &= \hat{\boldsymbol{\beta}} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}}{1 - p_i} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i y_i}{1 - p_i} \\ &= \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i (y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}})}{1 - p_i} = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - p_i}. \end{aligned}$$

It follows that the predicted value for the i -th observation is given by

$$\hat{y}_{-i} = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_{-i} = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}} - \frac{\mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - p_i} = \hat{y}_i - \frac{p_i e_i}{1 - p_i}.$$

Hence, $y_i - \hat{y}_{-i} = e_i + p_i e_i / (1 - p_i) = e_i / (1 - p_i)$. $\square$

Since the PRESS formula in Theorem 2.1 is equal to n times the n -fold cross-validation, it is natural to consider if a similar formula exists for K -fold cross-validation.

14

Using the notation from Section 1.2.1, let $C_k \equiv \{\mathbf{X}_k, \mathbf{y}_k\}$ be the test set in the K -th fold and $\hat{\boldsymbol{\beta}}_{\mathcal{T}_k}$ be the estimate of $\boldsymbol{\beta}$ using all the remaining data (that is, $\mathcal{T}_k \equiv \mathcal{T} \setminus C_k$) as training set. Note that with this notation, we can write

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \vdots \\ \mathbf{X}_K \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_K \end{bmatrix},$$

87

where both matrix $\mathbf{X}_k$ and vector $\mathbf{y}_k$ have n_k rows. Then, using the Woodbury identity and similar computations to the one in the proof of Theorem 2.1, we can show that

$$\hat{\boldsymbol{\beta}}_{\mathcal{T}_k} = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_k^\top (\mathbf{I}_{n_k} - \mathbf{X}_k (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_k^\top)^{-1} (\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}}).$$

Additionally, the predictive residuals $\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}}_{\mathcal{T}_k}$ can be expressed in terms of the ordinary residuals:

$$\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}}_{\mathcal{T}_k} = (\mathbf{I}_{n_k} - \mathbf{X}_k (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_k^\top)^{-1} (\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}}).$$

Therefore, the formula for the K -fold cross-validation simplifies as follows:

$$\text{CV}_K = \sum_{k=1}^K \frac{n_k}{n} \frac{\|\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}}_{\mathcal{T}_k}\|^2}{n_k} = \frac{1}{n} \sum_{k=1}^K \|(\mathbf{I}_{n_k} - \mathbf{P}_{n_k})^{-1} (\mathbf{y}_k - \mathbf{X}_k \hat{\boldsymbol{\beta}})\|^2, \quad (2.11)$$

where $\mathbf{P}_{n_k}$ is the $n_k \times n_k$ diagonal sub-block of $\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. For example, for $K = 2$ we have

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_{n_1} & * \\ * & \mathbf{P}_{n_2} \end{bmatrix},$$

and for $K = n$ each $\mathbf{P}_{n_k}$ is simply the k -th diagonal element of $\mathbf{P}$, in agreement with the PRESS formula in Theorem 2.1.

■ **Example 2.2 (Polynomial Regression (cont.))** We return to Example 1.1, where we estimated the generalization risk for various polynomial prediction functions using independent validation data. Instead, let us estimate the expected generalization risk via cross-validation (thus using only the training set) and apply Theorem 2.1 to compute the PRESS.



7



29

polyregress.py

```
import numpy as np
import matplotlib.pyplot as plt

def generate_data(beta, sig, n):
    u = np.random.rand(n, 1)
    y = u ** np.arange(0, 4) @ beta.reshape(4,1) + (
        sig * np.random.randn(n, 1))
    return u, y

np.random.seed(12)
beta = np.array([[10.0, -140, 400, -250]]).T;
sig=5; n = 10**2;
u,y = generate_data(beta,sig,n)

X = np.ones((n, 1))
K = 12 #maximum number of parameters
press = np.zeros(K+1)
for k in range(1,K):
    if k > 1:
        X = np.hstack((X, u**(k-1))) # add column to matrix
    P = X @ np.linalg.pinv(X) # projection matrix
    e = y - P @ y

    press[k] = np.sum((e/(1-np.diag(P).reshape(n,1)))**2)

plt.plot(press[1:K]/n)
```

The PRESS values divided by $n = 100$ for the constant, linear, quadratic, cubic, and quartic order polynomial regression models are, respectively, 152.487, 56.249, 51.606, 30.999, and 31.634. Hence, the cubic polynomial regression model has the lowest PRESS, indicating that it has the best predictive performance. ■

2.3.3 In-Sample Risk

Given a fixed data set τ with associated response vector $\mathbf{y}$ and $n \times p$ matrix of explanatory variables $\mathbf{X}$, the in-sample risk of a prediction function g is defined as

$$\ell_{\text{in}}(g) := \mathbb{E}_{\mathbf{X}} \text{Loss}(Y, g(\mathbf{X})), \quad (2.12)$$

where $\mathbb{E}_{\mathbf{X}}$ signifies that the expectation is taken under a different probability model, in which $\mathbf{X}$ takes the values $\mathbf{x}_1, \dots, \mathbf{x}_n$ with equal probability, and given $\mathbf{X} = \mathbf{x}_i$ the random variable Y is drawn from the conditional pdf $f(y | \mathbf{x}_i)$.

For a fixed training set τ , we can compare the training loss of the learner with the in-sample risk. Their difference,

$$\text{op}_{\tau} = \ell_{\text{in}}(g_{\tau}) - \ell_{\tau}(g_{\tau}),$$

is called the *optimism* (of the training loss), because it measures how much the training loss underestimates (is optimistic about) the unknown in-sample risk. Mathematically, it is simpler to work with the *expected optimism*:

$$\mathbb{E}[\text{op}_{\tau} | \mathbf{X}_1 = \mathbf{x}_1, \dots, \mathbf{X}_n = \mathbf{x}_n] =: \mathbb{E}_{\mathbf{X}} \text{op}_{\tau},$$

where the expectation is taken over a random training set $\mathcal{T}$, conditional on $\mathbf{X}_i = \mathbf{x}_i$, $i = 1, \dots, n$. For ease of notation, we have abbreviated the expected optimism to $\mathbb{E}_{\mathbf{X}} \text{op}_{\tau}$, where $\mathbb{E}_{\mathbf{X}}$ denotes the expectation operator conditional on $\mathbf{X}_i = \mathbf{x}_i$, $i = 1, \dots, n$. As in Example 1.1, the feature vectors are stored as the rows of an $n \times p$ matrix $\mathbf{X}$. It turns out that the expected optimism for various loss functions can be expressed in terms of the (conditional) covariance between the observed and predicted response.

Theorem 2.2: Expected Optimism

For the squared-error loss the expected optimism is

$$\mathbb{E}_{\mathbf{X}} \text{op}_{\tau} = \frac{2}{n} \sum_{i=1}^n \text{Cov}_{\mathbf{X}}(g_{\tau}(\mathbf{x}_i), Y_i). \quad (2.13)$$

Proof: In what follows, all expectations are taken conditional on $\mathbf{X}_1 = \mathbf{x}_1, \dots, \mathbf{X}_n = \mathbf{x}_n$. Let Y_i be the response for $\mathbf{x}_i$ and let $\widehat{Y}_i = g_{\tau}(\mathbf{x}_i)$ be the predicted value. Note that the latter depends on $Y_1, \dots, Y_n$. Also, let Y'_i be an independent copy of Y_i for the *same* $\mathbf{x}_i$. In particular, Y'_i has the same distribution as Y_i and is statistically independent of all $\{Y_j\}$, including Y_i , and therefore is also independent of $\widehat{Y}_i$. We have

$$\begin{aligned} \mathbb{E}_{\mathbf{X}} \text{op}_{\tau} &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathbf{X}} \left[(Y'_i - \widehat{Y}_i)^2 - (Y_i - \widehat{Y}_i)^2 \right] = \frac{2}{n} \sum_{i=1}^n \mathbb{E}_{\mathbf{X}} \left[(Y_i - Y'_i) \widehat{Y}_i \right] \\ &= \frac{2}{n} \sum_{i=1}^n \left(\mathbb{E}_{\mathbf{X}} [Y_i \widehat{Y}_i] - \mathbb{E}_{\mathbf{X}} Y_i \mathbb{E}_{\mathbf{X}} \widehat{Y}_i \right) = \frac{2}{n} \sum_{i=1}^n \text{Cov}_{\mathbf{X}}(\widehat{Y}_i, Y_i). \end{aligned}$$

□

In summary, the expected optimism indicates how much, on average, the training loss deviates from the expected in-sample risk. Since the covariance of independent random variables is zero, the expected optimism is zero if the learner $g_{\mathcal{T}}$ is statistically independent from the responses $Y_1, \dots, Y_n$.

If the conditional variance of the error $Y - g^*(X)$ given $X = \mathbf{x}$ does not depend on $\mathbf{x}$, then the expected in-sample risk of a learner $g_{\mathcal{T}}$, averaged over all training sets, has a simple expression:

Theorem 2.3: Expected In-Sample Risk for Linear Models

Let $\mathbf{X}$ be the model matrix for a linear model, of dimension $n \times p$. If $\text{Var}[Y - g^*(X) | X = \mathbf{x}] =: v^2$ does not depend on $\mathbf{x}$, then the expected in-sample risk (with respect to the squared-error loss) for a random learner $g_{\mathcal{T}}$ is given by

$$\mathbb{E}_{\mathbf{X}} \ell_{\text{in}}(g_{\mathcal{T}}) = \mathbb{E}_{\mathbf{X}} \ell_{\mathcal{T}}(g_{\mathcal{T}}) + \frac{2\ell^* p}{n}, \quad (2.14)$$

where ℓ^* is the irreducible risk.

Proof: Using the formula (1.15) for the least-squares estimator $\hat{\beta}$, the expected optimism (2.13) is

$$\begin{aligned} \frac{2}{n} \sum_{i=1}^n \text{Cov}_{\mathbf{X}}(\mathbf{x}_i^{\top} \hat{\beta}, Y_i) &= \frac{2}{n} \text{tr}(\text{Cov}_{\mathbf{X}}(\mathbf{X} \hat{\beta}, Y)) = \frac{2}{n} \text{tr}(\text{Cov}_{\mathbf{X}}(\mathbf{X} \mathbf{X}^+ Y, Y)) \\ &= \frac{2 \text{tr}(\mathbf{X} \mathbf{X}^+ \text{Cov}_{\mathbf{X}}(Y, Y))}{n} = \frac{2\ell^* \text{tr}(\mathbf{X} \mathbf{X}^+)}{n} = \frac{2\ell^* p}{n}. \end{aligned}$$

In the last equation we used the cyclic property of the trace (Theorem A.1): $\text{tr}(\mathbf{X} \mathbf{X}^+) = \text{tr}(\mathbf{X}^+ \mathbf{X}) = \text{tr}(\mathbf{I}_p)$, assuming that $\text{rank}(\mathbf{X}) = p$. □

69

Therefore, an estimate for the in-sample risk is:

$$\hat{\ell}_{\text{in}}(g_{\mathcal{T}}) = \ell_{\mathcal{T}}(g_{\mathcal{T}}) + 2\ell^* p/n, \quad (2.15)$$

Equation (2.14) is the basis of the following model comparison heuristic: Estimate the irreducible risk $\ell^* = v^2$ via $\hat{v}^2$, using a model with relatively high complexity. Then choose the linear model with the lowest value of

$$\|\mathbf{y} - \mathbf{X} \hat{\beta}\|^2 + 2\hat{v}^2 p. \quad (2.16)$$

Figure 2.3 shows that this estimate is very close to the test loss from Figure 1.5 for the polynomial regression in Example 1.1.

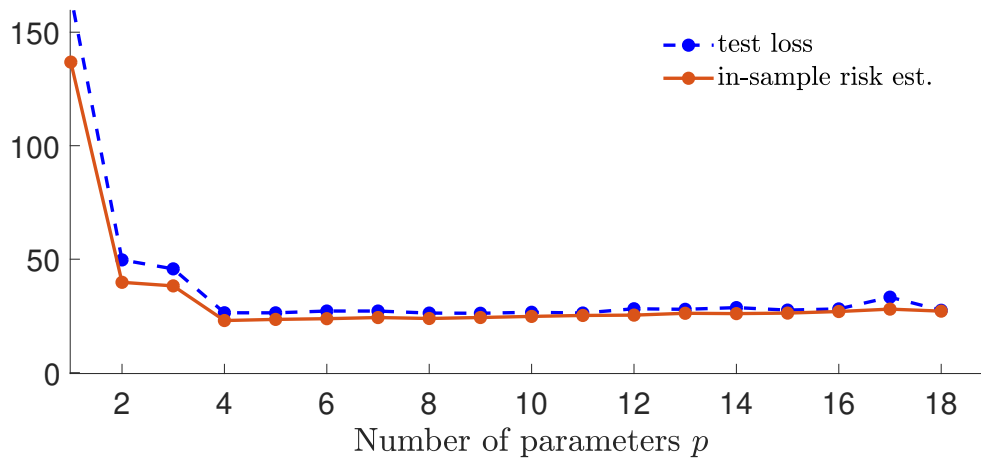


Figure 2.3: In-sample risk estimate $\widehat{\ell}_{\text{in}}(g_{\tau})$ as a function of the number of parameters p of the model. The test loss is superimposed as a blue dashed curve.

2.3.4 Categorical Features

Suppose that the data is given in the form of a spreadsheet or data frame with n rows and $p + 1$ columns, where the first element of row i is the response variable y_i , and the remaining p elements form the vector of explanatory variables $\mathbf{x}_i^{\top}$. When all the explanatory variables (features, predictors) are *quantitative*, then the model matrix $\mathbf{X}$ can be directly read off from the data frame as the $n \times p$ matrix with rows $\mathbf{x}_i^{\top}, i = 1, \dots, n$.

However, when some explanatory variables are *qualitative* (categorical), such a one-to-one correspondence between data frame and model matrix no longer holds. The solution is to include *indicator* or *dummy* variables.

Linear models with continuous responses and categorical explanatory variables often arise in *factorial experiments*. These are controlled statistical experiments in which the aim is to assess how a response variable is affected by one or more *factors* tested at several *levels*. A typical example is an agricultural experiment where one wishes to investigate how the yield of a food crop depends on factors such as location, pesticide, and fertilizer.

■ **Example 2.3 (Crop Yield)** The data in Table 2.1 lists the yield of a food crop for four different crop treatments (e.g., strengths of fertilizer) on four different blocks (plots).

FACTORIAL
EXPERI-
MENTS
FACTORS
LEVELS

Table 2.1: Crop yield for different treatments and blocks.

Block	Treatment			
	1	2	3	4
1	9.2988	9.4978	9.7604	10.1025
2	8.2111	8.3387	8.5018	8.1942
3	9.0688	9.1284	9.3484	9.5086
4	8.2552	7.8999	8.4859	8.9485

The corresponding data frame, given in Table 2.2, has 16 rows and 3 columns: one column for the crop yield (the response variable), one column for the Block, with levels 1, 2, 3, 4, and one column for the Treatment, also with levels 1, 2, 3, 4. The values 1, 2, 3, and 4 have no quantitative meaning (it does not make sense to take their average, for example) — they merely identify the category of the treatment or block.

Table 2.2: Crop yield data organized as a data frame in standard format.

Yield	Treatment	Block
9.2988	1	1
8.2111	1	2
9.0688	1	3
8.2552	1	4
9.4978	2	1
8.3387	2	2
$\vdots$	$\vdots$	$\vdots$
9.5086	4	3
8.9485	4	4



In general, suppose there are r factor (categorical) variables $u_1, \dots, u_r$, where the j -th factor has p_j mutually exclusive levels, denoted by $1, \dots, p_j$. In order to include these categorical variables in a linear model, a common approach is to introduce an *indicator feature* $x_{jk} = \mathbb{1}\{u_j = k\}$ for each factor j at level k . Thus, $x_{jk} = 1$ if the value of factor j is k and 0 otherwise. Since $\sum_k \mathbb{1}\{u_j = k\} = 1$, it suffices to consider only $p_j - 1$ of these indicator features for each factor j (this prevents the model matrix from being rank deficient). For a single response Y , the feature vector $\mathbf{x}^\top$ is thus a row vector of binary variables that indicates which levels were observed for each factor. The model assumption is that Y depends in a linear way on the indicator features, apart

INDICATOR
FEATURE

from an error term. That is,

$$Y = \beta_0 + \sum_{j=1}^r \sum_{k=2}^{p_j} \beta_{jk} \underbrace{\mathbb{1}\{u_j = k\}}_{x_{jk}} + \varepsilon,$$

where we have omitted one indicator feature (corresponding to level 1) for each factor j . For independent responses $Y_1, \dots, Y_n$ corresponding to the factors $\{u_{1j}\}_{j=1}^{p_1}, \dots, \{u_{rj}\}_{j=1}^{p_r}$, let $x_{ijk} = \mathbb{1}\{u_{ij} = k\}$. Then, the linear model for the data becomes

$$Y_i = \beta_0 + \sum_{j=1}^r \sum_{k=2}^{p_j} \beta_{jk} x_{ijk} + \varepsilon_i, \quad (2.17)$$

where the $\{\varepsilon_i\}$ are independent with expectation 0 and some variance σ^2 . By gathering the β_0 and $\{\beta_{jk}\}$ into a vector $\boldsymbol{\beta}$, and the $\{x_{ijk}\}$ into a matrix $\mathbf{X}$, we have again a linear model of the form (2.8). The model matrix $\mathbf{X}$ has n rows and $1 + \sum_{j=1}^r (p_j - 1)$ columns. Using the above convention that the β_{j1} parameters are subsumed in the parameter β_0 (corresponding to the “constant” feature), we can interpret β_0 as a baseline response when using the explanatory vector $\mathbf{x}^\top$ for which $x_{j1} = 1$ for all factors $j = 1, \dots, r$. The other parameters $\{\beta_{jk}\}$ can be viewed as *incremental effects* relative to this baseline effect. For example, β_{12} describes by how much the response is expected to change if level 2 is used instead of level 1 for factor 1.

INCRE-
MENTAL
EFFECTS

■ **Example 2.4 (Single-Factor ANOVA Model)** The simplest ANOVA model is the single-factor one, where the response is modeled as:

$$Y_i = \beta_0 + \beta_{12} \mathbb{1}\{u_i = 2\} + \dots + \beta_{1p} \mathbb{1}\{u_i = p\} + \varepsilon_i,$$

and for $p = 4$ can be arranged as:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{1}_{n_1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{1}_{n_2} & \mathbf{1}_{n_2} & \mathbf{0} & \mathbf{0} \\ \mathbf{1}_{n_3} & \mathbf{0} & \mathbf{1}_{n_3} & \mathbf{0} \\ \mathbf{1}_{n_p} & \mathbf{0} & \mathbf{0} & \mathbf{1}_{n_p} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_{12} \\ \beta_{13} \\ \beta_{14} \end{bmatrix} + \boldsymbol{\varepsilon},$$

where $\mathbf{Y}$ of length $n = n_1 + \dots + n_p$ is organized as

$$\underbrace{Y_1, \dots, Y_{n_1}}_{\text{level 1}}, \underbrace{Y_{n_1+1}, \dots, Y_{n_1+n_2}}_{\text{level 2}}, \dots, \underbrace{Y_{n-n_p+1}, \dots, Y_{n_p}}_{\text{level } p}.$$

■

■ **Example 2.5 (Crop Yield (cont.))** In Example 2.3, the linear model (2.17) has eight parameters: $\beta_0, \beta_{12}, \beta_{13}, \beta_{14}, \beta_{22}, \beta_{23}, \beta_{24}$, and σ^2 . The model matrix $\mathbf{X}$ depends on how the crop yields are organized in a vector $\mathbf{y}$ and on the ordering of the

factors. Let us order y column-wise from Table 2.1, as in $y = [9.2988, 8.2111, 9.0688, 8.2552, 9.4978, \dots, 8.9485]^\top$, and let Treatment be Factor 1 and Block be Factor 2. Then we can write (2.17) as

$$Y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & C \\ 1 & 1 & 0 & 0 & C \\ 1 & 0 & 1 & 0 & C \\ 1 & 0 & 0 & 1 & C \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_{12} \\ \beta_{13} \\ \beta_{14} \\ \beta_{22} \\ \beta_{23} \\ \beta_{24} \end{bmatrix}}_{\boldsymbol{\beta}} + \boldsymbol{\varepsilon}, \quad \text{where} \quad \mathbf{C} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and with $\mathbf{1} = [1, 1, 1, 1]^\top$ and $\mathbf{0} = [0, 0, 0, 0]^\top$. Estimation of $\hat{\boldsymbol{\beta}}$ and σ^2 , model selection, and prediction can now be carried out in the usual manner for linear models.

■

In the context of factorial experiments, the model matrix is often called the *design matrix*, as it specifies the design of the experiment; e.g., how many replications are taken for each combination of factor levels. The model (2.17) can be extended by adding products of indicator variables as new features. Such features are called *interaction terms*.

DESIGN
MATRIX

INTERACTION

2.3.5 Nested Models

Let $\mathbf{X}$ be a $n \times p$ model matrix of the form $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, where $\mathbf{X}_1$ and $\mathbf{X}_2$ are model matrices of dimension $n \times k$ and $n \times (p - k)$, respectively. The linear models $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$ and $\mathbf{Y} = \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ are said to be *nested within* the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. This simply means that certain features in $\mathbf{X}$ are ignored in each of the first two models. Note that $\boldsymbol{\beta}$, $\boldsymbol{\beta}_1$, and $\boldsymbol{\beta}_2$ are parameter vectors of dimension p , k , and $p - k$, respectively. In what follows, we assume that $n \geq p$ and that all model matrices are full-rank.

NESTED
MODELS

Suppose we wish to assess whether to use the full model matrix $\mathbf{X}$ or the reduced model matrix $\mathbf{X}_1$. Let $\hat{\boldsymbol{\beta}}$ be the estimate of $\boldsymbol{\beta}$ under the full model (that is, obtained via (2.9)), and let $\hat{\boldsymbol{\beta}}_1$ denote the estimate of $\boldsymbol{\beta}_1$ for the reduced model. Let $\mathbf{Y}^{(2)} = \mathbf{X}\hat{\boldsymbol{\beta}}$ be the projection of $\mathbf{Y}$ onto the space $\text{Span}(\mathbf{X})$ spanned by the columns of $\mathbf{X}$; and let $\mathbf{Y}^{(1)} = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ be the projection of $\mathbf{Y}$ onto the space $\text{Span}(\mathbf{X}_1)$ spanned by the columns of $\mathbf{X}_1$ only; see Figure 2.4. In order to decide whether the features in $\mathbf{X}_2$ are needed, we may compare the estimated error terms of the two models, as calculated by (2.10); that is, by the residual sum of squares divided by the number of observations n . If the outcome of this comparison is that there is little difference between the model error for the full and reduced model, then it is appropriate to adopt the reduced model, as it has fewer parameters than the full model, while explaining the data just as well. The comparison is thus between the squared norms $\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2$ and $\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2$.

Because of the nested nature of the linear models, $\text{Span}(\mathbf{X}_1)$ is a subspace of $\text{Span}(\mathbf{X})$ and, consequently, the orthogonal projection of $\mathbf{Y}^{(2)}$ onto $\text{Span}(\mathbf{X}_1)$ is the same as the orthogonal projection of $\mathbf{Y}$ onto $\text{Span}(\mathbf{X}_1)$; that is, $\mathbf{Y}^{(1)}$. By Pythagoras' theorem, we thus have the decomposition $\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2 + \|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2 = \|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2$. This is also illustrated in Figure 2.4.

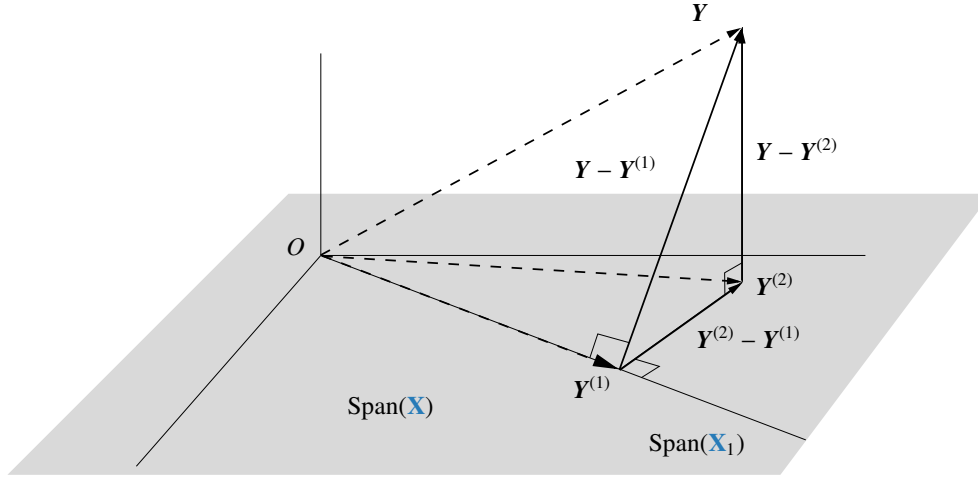


Figure 2.4: The residual sum of squares for the full model corresponds to $\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2$ and for the reduced model it is $\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2$. By Pythagoras's theorem, the difference is $\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2$.

The above decomposition can be generalized to more than two model matrices. Suppose that the model matrix can be decomposed into d submatrices: $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_d]$, where the matrix $\mathbf{X}_i$ has p_i columns and n rows, $i = 1, \dots, d$. Thus, the number of columns² in the full model matrix is $p = p_1 + \dots + p_d$. This creates an increasing sequence of “nested” model matrices: $\mathbf{X}_1, [\mathbf{X}_1, \mathbf{X}_2], \dots, [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_d]$, from (say) the baseline normal model matrix $\mathbf{X}_1 = \mathbf{1}$ to the full model matrix $\mathbf{X}$. Think of each model matrix corresponding to specific variables in the model.

We follow a similar projection procedure as in Figure 2.4: First project $\mathbf{Y}$ onto $\text{Span}(\mathbf{X})$ to yield the vector $\mathbf{Y}^{(d)}$, then project $\mathbf{Y}^{(d)}$ onto $\text{Span}([\mathbf{X}_1, \dots, \mathbf{X}_{d-1}])$ to obtain $\mathbf{Y}^{(d-1)}$, and so on, until $\mathbf{Y}^{(2)}$ is projected onto $\text{Span}(\mathbf{X}_1)$ to yield $\mathbf{Y}^{(1)} = \bar{Y}\mathbf{1}$ (in the case that $\mathbf{X}_1 = \mathbf{1}$).

By applying Pythagoras' theorem, the total sum of squares can be decomposed as

$$\underbrace{\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2}_{\text{df}=n-p_1} = \underbrace{\|\mathbf{Y} - \mathbf{Y}^{(d)}\|^2}_{\text{df}=n-p} + \underbrace{\|\mathbf{Y}^{(d)} - \mathbf{Y}^{(d-1)}\|^2}_{\text{df}=p_d} + \dots + \underbrace{\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2}_{\text{df}=p_2}. \quad (2.18)$$

Software packages typically report the sums of squares as well as the corresponding *degrees of freedom* (df): $n - p, p_d, \dots, p_2$.

²As always, we assume the columns are linearly independent.

2.3.6 Coefficient of Determination

To assess how a linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ compares to the default model $\mathbf{Y} = \beta_0 \mathbf{1} + \boldsymbol{\varepsilon}$, we can compare the variance of the original data, estimated via $\sum_i (Y_i - \bar{Y})^2/n = \|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2/n$, with the variance of the fitted data; estimated via $\sum_i (\hat{Y}_i - \bar{Y})^2/n = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2/n$, where $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$. The sum $\sum_i (Y_i - \bar{Y})^2/n = \|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2$ is sometimes called the *total sum of squares* (TSS), and the quantity

TOTAL SUM
OF SQUARES

$$R^2 = \frac{\|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2}{\|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2} \quad (2.19)$$

is called the *coefficient of determination* of the linear model. In the notation of Figure 2.4, $\hat{\mathbf{Y}} = \mathbf{Y}^{(2)}$ and $\bar{Y}\mathbf{1} = \mathbf{Y}^{(1)}$, so that

COEFFICIENT
OF DETERM-
INATION

$$R^2 = \frac{\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2}{\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2} = \frac{\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2 - \|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2}{\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2} = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}.$$

Note that R^2 lies between 0 and 1. An R^2 value close to 1 indicates that a large proportion of the variance in the data has been explained by the model.

Many software packages also give the *adjusted coefficient of determination*, or simply the adjusted R^2 , defined by

ADJUSTED
COEFFICIENT
OF DETERM-
INATION

$$R_{\text{adjusted}}^2 = 1 - (1 - R^2) \frac{n-1}{n-p}.$$

The regular R^2 is always *non-decreasing* in the number of parameters (tutorial Exercise), but this may not indicate better predictive power. The adjusted R^2 compensates for this increase by decreasing the regular R^2 as the number of variables increases. This heuristic adjustment can make it easier to compare the quality of two competing models.

2.4 Inference for Normal Linear Models

So far we have not assumed any distribution for the random vector of errors $\boldsymbol{\varepsilon} = [\varepsilon_1, \dots, \varepsilon_n]^\top$ in a linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. When the error terms $\{\varepsilon_i\}$ are assumed to be normally distributed (that is, $\{\varepsilon_i\} \sim_{\text{iid}} \mathcal{N}(0, \sigma^2)$), whole new avenues open up for inference on linear models. In Section 1.5 we already saw that for such *normal linear models*, estimation of $\boldsymbol{\beta}$ and σ^2 can be carried out via maximum likelihood methods, yielding the same estimators from (2.9) and (2.10).



19

The following theorem lists the properties of these estimators. In particular, it shows that $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2 n/(n-p)$ are independent and unbiased estimators of $\boldsymbol{\beta}$ and σ^2 , respectively.

Theorem 2.4: Properties of the Estimators for a Normal Linear Model

Consider the linear model $Y = \mathbf{X}\beta + \varepsilon$, with $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, where β is a p -dimensional vector of parameters and σ^2 a dispersion parameter. The following results hold.

1. The maximum likelihood estimators $\hat{\beta}$ and $\hat{\sigma}^2$ are independent.
2. $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^+)$.
3. $n\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$, where $p = \text{rank}(\mathbf{X})$.

72 *Proof:* Using the pseudo-inverse (Definition A.2), we can write the random vector $\hat{\beta}$ as $\mathbf{X}^+ \mathbf{Y}$, which is a linear transformation of a normal random vector. Consequently, $\hat{\beta}$ has a multivariate normal distribution; see Theorem B.6. The mean vector and covariance matrix follow from the same theorem:

$$\mathbb{E} \hat{\beta} = \mathbf{X}^+ \mathbb{E} Y = \mathbf{X}^+ \mathbf{X} \beta = \beta$$

and

$$\text{Cov}(\hat{\beta}) = \mathbf{X}^+ \sigma^2 \mathbf{I}_n (\mathbf{X}^+)^{\top} = \sigma^2 (\mathbf{X}^{\top} \mathbf{X})^+.$$

114 To show that $\hat{\beta}$ and $\hat{\sigma}^2$ are independent, define $Y^{(2)} = \mathbf{X}\hat{\beta}$. Note that Y/σ has a $\mathcal{N}(\mu, \mathbf{I}_n)$ distribution, with expectation vector $\mu = \mathbf{X}\beta/\sigma$. A direct application of Theorem B.10 now shows that $(Y - Y^{(2)})/\sigma$ is independent of $Y^{(2)}/\sigma$. Since $\hat{\beta} = \mathbf{X}^+ \mathbf{X} \hat{\beta} = \mathbf{X}^+ Y^{(2)}$ and $\hat{\sigma}^2 = \|Y - Y^{(2)}\|^2/n$, it follows that $\hat{\sigma}^2$ is independent of $\hat{\beta}$. Finally, by the same theorem, the random variable $\|Y - Y^{(2)}\|^2/\sigma^2$ has a χ_{n-p}^2 distribution, as $Y^{(2)}$ has the same expectation vector as Y . $\square$

As a corollary, we see that each estimator $\hat{\beta}_i$ of β_i has a normal distribution with expectation β_i and variance $\sigma^2 \mathbf{u}_i^{\top} \mathbf{X}^+ (\mathbf{X}^+)^{\top} \mathbf{u}_i = \sigma^2 \|\mathbf{u}_i^{\top} \mathbf{X}^+\|^2$, where $\mathbf{u}_i = [0, \dots, 0, 1, 0, \dots, 0]^{\top}$ is the i -th unit vector; in other words, the variance is $\sigma^2 [(\mathbf{X}^{\top} \mathbf{X})^+]_{ii}$.

122 It is of interest to test whether certain regression parameters β_i are 0 or not, since if $\beta_i = 0$, the i -th explanatory variable has no direct effect on the expected response and so could be removed from the model. A standard procedure is to conduct a hypothesis test (see Section B.11 for a review of hypothesis testing) to test the null hypothesis $H_0 : \beta_i = 0$ against the alternative $H_1 : \beta_i \neq 0$, using the test statistic

$$T = \frac{\hat{\beta}_i / \|\mathbf{u}_i^{\top} \mathbf{X}^+\|}{\sqrt{\text{RSE}}}, \quad (2.20)$$

where RSE is the residual squared error; that is $\text{RSE} = \text{RSS}/(n-p)$. This test statistic has a t_{n-p} distribution under H_0 . To see this, write $T = Z/\sqrt{V/(n-p)}$, with

$$Z = \frac{\hat{\beta}_i}{\sigma \|\mathbf{u}_i^{\top} \mathbf{X}^+\|} \quad \text{and} \quad V = n\hat{\sigma}^2/\sigma^2.$$

Then, by Theorem 2.4, $Z \sim \mathcal{N}(0, 1)$ under H_0 , $V \sim \chi_{n-p}^2$, and Z and V are independent. The result now follows directly from Corollary B.1.



115

2.4.1 Comparing Two Normal Linear Models

Suppose we have the following normal linear model for data $\mathbf{Y} = [Y_1, \dots, Y_n]^\top$:

$$\mathbf{Y} = \underbrace{\mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2}_{\mathbf{X}\boldsymbol{\beta}} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (2.21)$$

where $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are unknown vectors of dimension k and $p - k$, respectively; and $\mathbf{X}_1$ and $\mathbf{X}_2$ are full-rank model matrices of dimensions $n \times k$ and $n \times (p - k)$, respectively. Above we implicitly defined $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ and $\boldsymbol{\beta}^\top = [\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top]$. Suppose we wish to test the hypothesis $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$ against $H_1 : \boldsymbol{\beta}_2 \neq \mathbf{0}$. Following Section 2.3.5, the idea is to compare the residual sum of squares for both models, expressed as $\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2$ and $\|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2$. Using Pythagoras' theorem we saw that $\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2 - \|\mathbf{Y} - \mathbf{Y}^{(1)}\|^2 = \|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2$, and so it makes sense to base the decision whether to retain or reject H_0 on the basis of the quotient of $\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2$ and $\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2$. This leads to the following test statistics.

Theorem 2.5: Test Statistic for Comparing Two Normal Linear Models

For the model (2.21), let $\mathbf{Y}^{(2)}$ and $\mathbf{Y}^{(1)}$ be the projections of $\mathbf{Y}$ onto the space spanned by the p columns of $\mathbf{X}$ and the k columns of $\mathbf{X}_1$, respectively. Then under $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$ the test statistic

$$T = \frac{\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2 / (p - k)}{\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2 / (n - p)} \quad (2.22)$$

has an $F(p - k, n - p)$ distribution.

Proof: Define $\mathbf{X} := \mathbf{Y}/\sigma$ with expectation $\boldsymbol{\mu} := \mathbf{X}\boldsymbol{\beta}/\sigma$, and $\mathbf{X}_j := \mathbf{Y}^{(j)}/\sigma$ with expectation $\boldsymbol{\mu}_j$, $j = k, p$. Note that $\boldsymbol{\mu}_p = \boldsymbol{\mu}$ and, under H_0 , $\boldsymbol{\mu}_k = \boldsymbol{\mu}_p$. We can directly apply Theorem B.10 to find that $\|\mathbf{Y} - \mathbf{Y}^{(2)}\|^2/\sigma^2 = \|\mathbf{X} - \mathbf{X}_p\|^2 \sim \chi_{n-p}^2$ and, under H_0 , $\|\mathbf{Y}^{(2)} - \mathbf{Y}^{(1)}\|^2/\sigma^2 = \|\mathbf{X}_p - \mathbf{X}_k\|^2 \sim \chi_{p-k}^2$. Moreover, these random variables are independent of each other. The proof is completed by applying Theorem B.11. $\square$



114

Note that H_0 is rejected for large values of T . The testing procedure thus proceeds as follows:

1. Compute the outcome, t say, of the test statistic T in (2.22).
2. Evaluate the P-value $\mathbb{P}(T \geq t)$, with $T \sim F(p - k, n - p)$.

3. Reject H_0 if this P-value is too small, say less than 0.05.

41

For nested models $[\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k]$, $k = 1, 2, \dots, d$, as in Section 2.3.5, the F test statistic in Theorem 2.5 can now be used to test whether certain $\mathbf{X}_i$ are needed or not. In particular, software packages will report the outcomes of

$$F_i = \frac{\|\mathbf{Y}^{(i)} - \mathbf{Y}^{(i-1)}\|^2 / p_i}{\|\mathbf{Y} - \mathbf{Y}^{(d)}\|^2 / (n - p)}, \quad (2.23)$$

in the order $i = 2, 3, \dots, d$. Under the null hypothesis that $\mathbf{Y}^{(i)}$ and $\mathbf{Y}^{(i-1)}$ have the same expectation (that is, adding $\mathbf{X}_i$ to $\mathbf{X}_{i-1}$ has no additional effect on reducing the approximation error), the test statistic F_i has an $F(p_i, n - p)$ distribution, and the corresponding P-values quantify the strength of the decision to include an additional variable in the model or not. This procedure is called *analysis of variance* (ANOVA).

ANALYSIS OF
VARIANCE

Note that the output of an ANOVA table depends on the order in which the variables are considered.

■ **Example 2.6 (Crop Yield (cont.))** We continue Examples 2.3 and 2.5. Decompose the linear model as

$$\mathbf{Y} = \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}_{\mathbf{X}_1} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}}_{\beta_1} + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{X}_2} \underbrace{\begin{bmatrix} \beta_{12} \\ \beta_{13} \\ \beta_{14} \end{bmatrix}}_{\beta_2} + \underbrace{\begin{bmatrix} \mathbf{C} \\ \mathbf{C} \\ \mathbf{C} \\ \mathbf{C} \end{bmatrix}}_{\mathbf{X}_3} \underbrace{\begin{bmatrix} \beta_{22} \\ \beta_{23} \\ \beta_{24} \end{bmatrix}}_{\beta_3} + \boldsymbol{\varepsilon}.$$

Is the crop yield dependent on treatment levels as well as blocks? We first test whether we can remove Block as a factor in the model against it playing a significant role in explaining the crop yields. Specifically, we test $\beta_3 = \mathbf{0}$ versus $\beta_3 \neq \mathbf{0}$ using Theorem 2.5. Now the vector $\mathbf{Y}^{(2)}$ is the projection of $\mathbf{Y}$ onto the $(p = 7)$ -dimensional space spanned by the columns of $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3]$; and $\mathbf{Y}^{(1)}$ is the projection of $\mathbf{Y}$ onto the $(k = 4)$ -dimensional space spanned by the columns of $\mathbf{X}_{12} := [\mathbf{X}_1, \mathbf{X}_2]$. The test statistic, T_{12} say, under H_0 has an $F(3, 9)$ distribution.

For the outcome of the test statistic T_{12} and the corresponding P-value, we find $t_{12} = 34.9998$, which gives a P-value 2.73×10^{-5} . This shows that the block effects are extremely important for explaining the data.

Using the extended model (including the block effects), we can test whether $\beta_2 = \mathbf{0}$ or not; that is, whether the treatments have a significant effect on the crop yield in the presence of the Block factor. This is done in the last six lines of the code below. The outcome of the test statistic is 4.4878, with a P-value of 0.0346. By including the block effects, we effectively reduce the uncertainty in the model and are able to more

accurately assess the effects of the treatments, to conclude that the treatment seems to have an effect on the crop yield. A closer look at the data shows that within each block (row) the crop yield roughly increases with the treatment level.

crop.py

```
import numpy as np
from scipy.stats import f
from numpy.linalg import lstsq, norm

yy = np.array([9.2988, 9.4978, 9.7604, 10.1025,
               8.2111, 8.3387, 8.5018, 8.1942,
               9.0688, 9.1284, 9.3484, 9.5086,
               8.2552, 7.8999, 8.4859, 8.9485]).reshape(4,4).T

nrow, ncol = yy.shape[0], yy.shape[1]
n = nrow * ncol
y = yy.reshape(16,)
X_1 = np.ones((n,1))

KM = np.kron(np.eye(ncol), np.ones((nrow,1)))
KM[:,0]
X_2 = KM[:,1:ncol]
IM = np.eye(nrow)
C = IM[:,1:nrow]

X_3 = np.vstack((C, C))
X_3 = np.vstack((X_3, C))
X_3 = np.vstack((X_3, C))

X = np.hstack((X_1, X_2))
X = np.hstack((X, X_3))

p = X.shape[1] #number of parameters in full model
betahat = lstsq(X, y, rcond=None)[0] #estimate under the full
model

ym = X @ betahat

X_12 = np.hstack((X_1, X_2)) #omitting the block effect
k = X_12.shape[1] #number of parameters in reduced model
betahat_12 = lstsq(X_12, y, rcond=None)[0]
y_12 = X_12 @ betahat_12
T_12 = (n-p)/(p-k) * (norm(y-y_12)**2 - norm(y-ym)**2) / norm(y-ym)**2
pval_12 = 1 - f.cdf(T_12, p-k, n-p)

X_13 = np.hstack((X_1, X_3)) #omitting the treatment effect
k = X_13.shape[1] #number of parameters in reduced model
betahat_13 = lstsq(X_13, y, rcond=None)[0]
y_13 = X_13 @ betahat_13
```

```
T_13=(n-p)/(p-k)*(norm(y-y_13)**2 - norm(y-ym)**2)/norm(y-ym)**2
pval_13 = 1 - f.cdf(T_13,p-k,n-p)
```

2.4.2 Confidence and Prediction Intervals

As in all supervised learning settings, linear regression is most useful when we wish to predict how a new response variable will behave on the basis of a new explanatory vector $\mathbf{x}$. For example, it may be difficult to measure the response variable, but by knowing the estimated regression line and the value for $\mathbf{x}$, we will have a reasonably good idea what Y or the expected value of Y is going to be.

Thus, consider a new $\mathbf{x}$ and let $Y \sim \mathcal{N}(\mathbf{x}^\top \boldsymbol{\beta}, \sigma^2)$, with $\boldsymbol{\beta}$ and σ^2 unknown. First we are going to look at the *expected* value of Y , that is $\mathbb{E}Y = \mathbf{x}^\top \boldsymbol{\beta}$. Since $\boldsymbol{\beta}$ is unknown, we do not know $\mathbb{E}Y$ either. However, we can estimate it via the estimator $\hat{Y} = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$, where $\hat{\boldsymbol{\beta}} \sim \mathcal{N}(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^+)$, by Theorem 2.4. Being linear in the components of $\boldsymbol{\beta}$, $\hat{Y}$ therefore has a normal distribution with expectation $\mathbf{x}^\top \boldsymbol{\beta}$ and variance $\sigma^2 \|\mathbf{x}^\top \mathbf{X}^+\|^2$. Let $Z \sim \mathcal{N}(0, 1)$ be the standardized version of $\hat{Y}$ and $V = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / \sigma^2 \sim \chi_{n-p}^2$. Then the random variable

$$T := \frac{(\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta}) / \|\mathbf{x}^\top \mathbf{X}^+\|}{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\| / \sqrt{(n-p)}} = \frac{Z}{\sqrt{V/(n-p)}} \quad (2.24)$$

115 has, by Corollary B.1, a t_{n-p} distribution. After rearranging the identity $\mathbb{P}(|T| \leq t_{n-p;1-\alpha/2}) = 1 - \alpha$, where $t_{n-p;1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the t_{n-p} distribution, we arrive at the stochastic *confidence interval*

$$\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm t_{n-p;1-\alpha/2} \sqrt{\text{RSE}} \|\mathbf{x}^\top \mathbf{X}^+\|, \quad (2.25)$$

where we have identified $\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / (n-p)$ with RSE. This confidence interval quantifies the uncertainty in the learner (regression surface).

PREDICTION INTERVAL A *prediction interval* for a new response Y is different from a confidence interval for $\mathbb{E}Y$. Here the idea is to construct an interval such that Y lies in this interval with a certain guaranteed probability. Note that now we have *two* sources of variation:

1. $Y \sim \mathcal{N}(\mathbf{x}^\top \boldsymbol{\beta}, \sigma^2)$ itself is a random variable.
2. Estimating $\mathbf{x}^\top \boldsymbol{\beta}$ via $\hat{Y}$ brings another source of variation.

We can construct a $(1 - \alpha)$ prediction interval, by finding two random bounds such that the random variable Y lies between these bounds with probability $1 - \alpha$. We can reason as follows. Firstly, note that $Y \sim \mathcal{N}(\mathbf{x}^\top \boldsymbol{\beta}, \sigma^2)$ and $\hat{Y} \sim \mathcal{N}(\mathbf{x}^\top \boldsymbol{\beta}, \sigma^2 \|\mathbf{x}^\top \mathbf{X}^+\|^2)$ are independent. It follows that $Y - \hat{Y}$ has a normal distribution with expectation 0 and variance

$$\sigma^2(1 + \|\mathbf{x}^\top \mathbf{X}^+\|^2). \quad (2.26)$$

Secondly, letting $Z \sim \mathcal{N}(0, 1)$ be the standardized version of $Y - \widehat{Y}$, and repeating the steps used for the construction of the confidence interval (2.25), we arrive at the prediction interval

$$\mathbf{x}^\top \widehat{\boldsymbol{\beta}} \pm t_{n-p; 1-\alpha/2} \sqrt{\text{RSE}} \sqrt{1 + \|\mathbf{x}^\top \mathbf{X}^+\|^2}. \quad (2.27)$$

This prediction interval captures the uncertainty from an as-yet-unobserved response as well as the uncertainty in the parameters of the regression model itself.

■ **Example 2.7 (Confidence Limits in Simple Linear Regression)** The following program draws $n = 100$ samples from a simple linear regression model with parameters $\boldsymbol{\beta} = [6, 13]^\top$ and $\sigma = 2$, where the x -coordinates are evenly spaced on the interval $[0, 1]$. The parameters are estimated in the third block of the code. Estimates for $\boldsymbol{\beta}$ and σ are $[6.03, 13.09]^\top$ and $\widehat{\sigma} = 1.60$, respectively. The program then proceeds by calculating the 95% numeric confidence and prediction intervals for various values of the explanatory variable. Figure 2.5 shows the results.

confpred.py

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import t
from numpy.linalg import inv, lstsq, norm
np.random.seed(123)

n = 100
x = np.linspace(0.01, 1, 100).reshape(n, 1)
# parameters
beta = np.array([6, 13])
sigma = 2
Xmat = np.hstack((np.ones((n, 1)), x)) # design matrix
y = Xmat @ beta + sigma*np.random.randn(n)

# solve the normal equations
betahat = lstsq(Xmat, y, rcond=None)[0]
# estimate for sigma
sqMSE = norm(y - Xmat @ betahat)/np.sqrt(n-2)

tquant = t.ppf(0.975, n-2) # 0.975 quantile
ucl = np.zeros(n) # upper conf. limits
lcl = np.zeros(n) # lower conf. limits
upl = np.zeros(n)
lpl = np.zeros(n)
rl = np.zeros(n) # (true) regression line
u = 0

for i in range(n):
    u = u + 1/n;
```

```

xvec = np.array([1,u])
sqc = np.sqrt(xvec.T @ inv(Xmat.T @ Xmat) @ xvec)
sqp = np.sqrt(1 + xvec.T @ inv(Xmat.T @ Xmat) @ xvec)
rl[i] = xvec.T @ beta;
ucl[i] = xvec.T @ betahat + tquant*sqMSE*sqc;
lcl[i] = xvec.T @ betahat - tquant*sqMSE*sqc;
upl[i] = xvec.T @ betahat + tquant*sqMSE*sqp;
lpl[i] = xvec.T @ betahat - tquant*sqMSE*sqp;

plt.plot(x,y, '.')
plt.plot(x,rl,'b')
plt.plot(x,ucl,'k:')
plt.plot(x,lcl,'k:')
plt.plot(x,upl,'r--')
plt.plot(x,lpl,'r--')

```

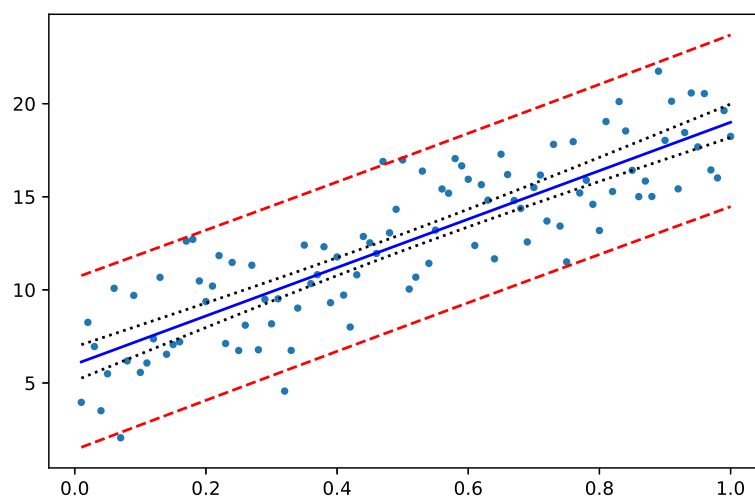


Figure 2.5: The true regression line (blue, solid) and the upper and lower 95% prediction curves (red, dashed) and confidence curves (dotted). ■

REGULARIZED LINEAR REGRESSION

3.1 Ridge Linear Regression

Recall that the minimizer of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$, called the *ordinary least squares (OLS)* estimate of $\boldsymbol{\beta}$, is easily written down using the pseudoinverse:

$$\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}.$$

Suppose that¹

$$\mathbf{X} = \sum_{i=1}^p s_i \mathbf{u}_i \mathbf{v}_i^\top = \mathbf{U} \mathbf{S} \mathbf{V}^\top, \quad (3.1)$$

is the full SVD of $\mathbf{X}$, where $\mathbf{V}$ and $\mathbf{U}$ are square, but $\mathbf{S} \in \mathbb{R}^{n \times p}$ is (possibly) rectangular. Then, $\mathbf{X}^+ = \sum_{i:s_i \neq 0} \frac{1}{s_i} \mathbf{v}_i \mathbf{u}_i^\top$ and one of the defining properties of $\mathbf{X}^+$ is the following limit.

Theorem 3.1: Pseudoinverse as a Limit

The pseudoinverse of $\mathbf{X} \in \mathbb{R}^{n \times p}$ satisfies

$$\mathbf{X}^+ = \lim_{\lambda \downarrow 0} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \lim_{\lambda \downarrow 0} \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}.$$

Proof: The SVD in (3.1) implies that

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top &= (\mathbf{V} \mathbf{S}^\top \mathbf{S} \mathbf{V}^\top + \lambda \mathbf{I}_p)^{-1} \mathbf{V} \mathbf{S}^\top \mathbf{U}^\top \\ &= (\mathbf{V} [\mathbf{S}^\top \mathbf{S} + \lambda \mathbf{I}_p] \mathbf{V}^\top)^{-1} \mathbf{V} \mathbf{S}^\top \mathbf{U}^\top \\ &= \mathbf{V} (\mathbf{S}^\top \mathbf{S} + \lambda \mathbf{I}_p)^{-1} \mathbf{S}^\top \mathbf{U}^\top = \sum_{i:s_i \neq 0} \frac{s_i}{s_i^2 + \lambda} \mathbf{v}_i \mathbf{u}_i^\top. \end{aligned}$$

¹Here $\mathbf{v}_k$ is the k -th column of $\mathbf{X}$ and $\mathbf{v}_k$ is the k -th column of $\mathbf{V}$.

10
ORDINARY
LEAST
SQUARES
(OLS)

92

Therefore, $\lim_{\lambda \downarrow 0} \sum_{i: s_i \neq 0} \frac{s_i}{s_i^2 + \lambda} \mathbf{v}_i \mathbf{u}_i^\top = \sum_{i: s_i \neq 0} \frac{1}{s_i} \mathbf{v}_i \mathbf{u}_i^\top = \mathbf{X}^+$. This establishes the first limit. The second limit follows from a similar calculation. $\square$

The limiting property suggests that $\mathbf{X}^+ \approx (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top$ for small $\lambda > 0$ and that, instead of the OLS estimate $\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}$, we may consider the estimate

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{y} \quad (3.2)$$

RIDGE
ESTIMATE

for some small $\lambda > 0$. The $\hat{\boldsymbol{\beta}}_\lambda$ thus computed is called the *ridge estimate* of the coefficient $\boldsymbol{\beta}$ with *regularization parameter* $\lambda > 0$.

Just like the OLS, which is the minimizer of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$, the ridge estimate is also a minimizer:

$$\hat{\boldsymbol{\beta}}_\lambda = \underset{\mathbf{b}}{\operatorname{argmin}} \{ \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|^2 \}.$$

In other words, the ridge estimate minimizes the sum of squares with the addition of the penalty term

$$\lambda \|\mathbf{b}\|^2 = \lambda \sum_{i=1}^p b_i^2.$$

This can be most easily seen by noting that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|^2 = \|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\mathbf{b}\|^2$, where

$$\tilde{\mathbf{y}} := \begin{bmatrix} \mathbf{y} \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^{n+p}, \quad \tilde{\mathbf{X}} := \begin{bmatrix} \mathbf{X} \\ \sqrt{\lambda} \mathbf{I}_p \end{bmatrix},$$

and then verifying that $\tilde{\mathbf{X}}^+ \tilde{\mathbf{y}}$ equals the right-hand side of (3.2).

■ Example 3.1 (Ridge Estimate via Coordinate Descent) *Mention Gauss-Seidel* ■

The penalty is smaller when λ is smaller and, as a matter of fact, $\lim_{\lambda \downarrow 0} \hat{\boldsymbol{\beta}}_\lambda$ is the OLS estimate. In other words, as the regularization parameter λ becomes smaller and smaller, the ridge estimate becomes indistinguishable from the OLS estimate. In contrast, as λ tends to infinity, the ridge estimate approaches the zero vector. The phenomenon, whereby as λ ranges from zero to infinity, the ridge estimate $\mathbf{b}_\lambda$ ranges from the OLS estimate to the zero estimate, is called *shrinkage*. The following result summarizes these observations.

SHRINKAGE

Theorem 3.2: Shrinkage & Ridge Estimate

Let $\hat{\boldsymbol{\beta}}_\lambda$ be the ridge estimate (3.2), then

$$\lim_{\lambda \uparrow \infty} \hat{\boldsymbol{\beta}}_\lambda = \mathbf{0} \quad \text{and} \quad \lim_{\lambda \downarrow 0} \hat{\boldsymbol{\beta}}_\lambda = \hat{\boldsymbol{\beta}}.$$

Proof: Using the SVD in (3.1), we obtain $\widehat{\beta}_\lambda = \sum_{i=1}^p \frac{s_i}{s_i^2 + \lambda} (\mathbf{u}_i^\top \mathbf{y}) \mathbf{v}_i$. It follows that $\lim_{\lambda \uparrow \infty} \widehat{\beta}_\lambda \rightarrow \mathbf{0}$ and:

$$\lim_{\lambda \downarrow 0} \widehat{\beta}_\lambda = \lim_{\lambda \downarrow 0} \sum_{i:s_i \neq 0} \frac{s_i}{s_i^2 + \lambda} \mathbf{v}_i (\mathbf{u}_i^\top \mathbf{y}) = \left[\sum_{i:s_i \neq 0} \frac{1}{s_i} \mathbf{v}_i \mathbf{u}_i^\top \right] \mathbf{y} = \mathbf{X}^+ \mathbf{y}.$$

□

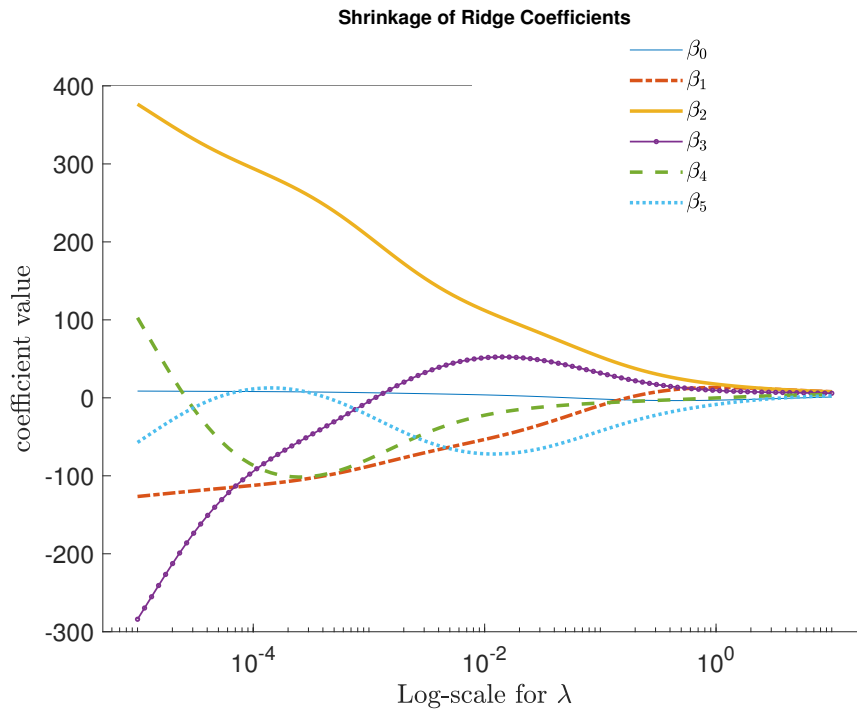


Figure 3.1: Ridge coefficient values as a function of the ridge parameter λ . The coefficients are smoothly shrunk to zero as $\lambda \uparrow \infty$, but are not zero for any finite λ .

■ **Example 3.2 (Shrinkage for Polynomial Regression)** We revisit the polynomial regression dataset from Example 1.1. Figure 3.1 shows the estimated coefficients as a function of λ when we fit a polynomial of degree five (that is, $p = 6$). When λ is small we can clearly see that the estimated values converge to $\mathbf{X}^+ \mathbf{y}$ and when λ becomes larger and larger, all these coefficients are smoothly shrunk to zero (there are not corners or kinks in the paths). The smooth nature of this shrinkage does not aid the selection of features. ■

We now know how to compute the ridge estimator. However, we still do not know what value to use for λ , and why we should use a value other than 0. To find out why a suitably chosen $\lambda > 0$ is beneficial, we need to derive the statistical properties of the ridge learner.

3.2 Statistical Properties of Ridge Regression

To discuss any statistical properties, we need to make some basic assumptions about the linear regression. Here we assume that the response variable Y obeys the usual *linear model*, in Definition 1.1. This means that $g^*(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}$ is the best possible prediction function at $\mathbf{x}$ with irreducible risk $\ell^* = \sigma^2$. The class of linear learners dependent on λ is here denoted as $\mathcal{G}_\lambda$.

3.2.1 Risk Properties of Ridge Learner

To understand how the regularization parameter λ influences predictions made using the ridge learner $g_{\mathcal{T}}^{\lambda}(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}}_\lambda$ we can study the effect of λ on its bias and variance. Using the SVD (3.1) and the observations

$$\mathbb{E}_{\mathbf{X}}[\mathbf{U}^\top \mathbf{Y}] = \mathbf{S} \mathbf{V}^\top \boldsymbol{\beta} \quad \text{and} \quad \mathbb{V}_{\mathbf{X}}[\mathbf{U}^\top \mathbf{Y}] = \sigma^2 \mathbf{I}_n,$$

the magnitude of the squared bias of the learner $\mathbf{X} \hat{\boldsymbol{\beta}}_\lambda$ is (see Exercise):

$$\|\mathbf{X} \mathbb{E}_{\mathbf{X}}[\hat{\boldsymbol{\beta}}_\lambda] - \boldsymbol{\beta}\|^2 = \sum_i \frac{\lambda^2 s_i^2 (\mathbf{v}_i^\top \boldsymbol{\beta})^2}{(s_i^2 + \lambda)^2}, \quad (3.3)$$

and its variance is:

$$\mathbb{V}_{\mathbf{X}}[\mathbf{X} \hat{\boldsymbol{\beta}}_\lambda] = \sum_i \frac{\sigma^2 s_i^4}{(s_i^2 + \lambda)^2} \mathbf{u}_i \mathbf{u}_i^\top. \quad (3.4)$$

The bias and variance computations help us obtain an explicit formula for the *expected in-sample risk* of the ridge learner.

Theorem 3.3: Expected In-sample Risk

The expected in-sample risk of the ridge learner can be written as

$$\mathbb{E}_{\mathbf{X}} \ell_{\text{in}}(g_{\mathcal{T}}^{\lambda}) = \sigma^2 + \frac{\lambda^2}{n} \sum_i \frac{s_i^2 (\mathbf{v}_i^\top \boldsymbol{\beta})^2}{(s_i^2 + \lambda)^2} + \frac{\sigma^2}{n} \sum_i \frac{s_i^4}{(s_i^2 + \lambda)^2}.$$

Proof: In the usual notation, where Y' is a (test) response drawn from $f(y | \mathbf{x})$ and independent of the training set $\mathcal{T}$, we have

$$\begin{aligned}\mathbb{E}_{\mathbf{X}} \left(Y' - g_{\mathcal{T}}^{\mathcal{G}_\lambda}(\mathbf{x}) \right)^2 &= \mathbb{E}_{\mathbf{X}} \left(Y' - \mathbf{x}^\top \boldsymbol{\beta} - \mathbf{x}^\top (\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}) \right)^2 \\ &= \sigma^2 + \mathbb{E}_{\mathbf{X}} \left(\mathbf{x}^\top (\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}) \right)^2.\end{aligned}$$

Since $\mathcal{G}_\lambda$ is a linear class of learners, the mean squared error $\mathbb{E}_{\mathbf{X}} \left(\mathbf{x}^\top (\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}) \right)^2$ can be decomposed into a bias and variance term:

$$\mathbb{E}_{\mathbf{X}} \left(\mathbf{x}^\top (\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}) \right)^2 = \underbrace{\left(\mathbf{x}^\top \mathbb{E}_{\mathbf{X}} [\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}] \right)^2}_{\text{pointwise squared bias}} + \underbrace{\mathbb{V}\text{ar}_{\mathbf{X}} [\mathbf{x}^\top \widehat{\boldsymbol{\beta}}_\lambda]}_{\text{pointwise variance}}.$$

Combining the last two equations then yields the formula for $\mathbb{E}_{\mathbf{X}} \ell_{\text{in}}(g_{\mathcal{T}}^{\mathcal{G}_\lambda})$:

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathbf{X}} (Y'_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_\lambda)^2 &= \sigma^2 + \frac{1}{n} \sum_{i=1}^n \left(\mathbf{x}_i^\top \mathbb{E}_{\mathbf{X}} [\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}] \right)^2 + \frac{1}{n} \sum_{i=1}^n \mathbb{V}\text{ar}_{\mathbf{X}} [\mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_\lambda] \\ &= \sigma^2 + \frac{\|\mathbf{X} \mathbb{E}_{\mathbf{X}} [\widehat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}]\|^2}{n} + \frac{\text{tr}(\mathbb{V}\text{ar}_{\mathbf{X}} [\mathbf{X} \widehat{\boldsymbol{\beta}}_\lambda])}{n}.\end{aligned}$$

Finally, applying (3.3) and (3.4) in the last line above yield the desired formula. $\square$

Note how the eigenvalues, $\{\sigma^2 s_i^4 / (s_i^2 + \lambda)^2\}$, of the covariance matrix in (3.4) are shrunk towards zero as λ increases. In contrast, the squared bias (3.3) is an increasing function of λ . Thus, it is reasonable to expect that there exists a λ that will provide an optimal tradeoff between the bias and the variance. The following result shows that when λ is chosen judiciously, the ridge learner will always outperform the OLS learner in terms of expected in-sample risk.

Theorem 3.4: Ridge & Ordinary-Least-Squares Learner

Let $\widehat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{Y}$ be the OLS estimator and $\mathbf{Y}'$ be test data, and define $\boldsymbol{\Omega} := (\mathbf{X}^\top \mathbf{X} / n)^+$. If

$$\lambda^* = \frac{\sigma^2 \text{tr}(\boldsymbol{\Omega})}{\boldsymbol{\beta}^\top \boldsymbol{\Omega} \boldsymbol{\beta} + 3\sigma^2 \text{tr}(\boldsymbol{\Omega}^2) / n},$$

then the ridge learner has smaller expected in-sample risk than the OLS learner:

$$\frac{1}{n} \mathbb{E}_{\mathbf{X}} \|\mathbf{Y}' - \mathbf{X} \widehat{\boldsymbol{\beta}}\|^2 \geq \frac{1}{n} \mathbb{E}_{\mathbf{X}} \|\mathbf{Y}' - \mathbf{X} \widehat{\boldsymbol{\beta}}_{\lambda^*}\|^2 + \frac{\sigma^2}{n^2} \lambda^* \text{tr}(\boldsymbol{\Omega}).$$

Proof: First, note the inequalities for $x \geq 0$:

$$\sum_{k=0}^{2n+1} (-1)^k (k+1)x^k \leq (1+x)^{-2} \leq \sum_{k=0}^{2n} (-1)^k (k+1)x^k, \quad n = 0, 1, 2, \dots \quad (3.5)$$

Second, the expected in-sample risk for the OLS learner can be obtained from either Theorem 2.3, or from the limit of the formula in Theorem 3.3: $\lim_{\lambda \downarrow 0} \mathbb{E}_{\mathbf{X}} \ell_{\text{in}} \left(g_{\mathcal{T}}^{\mathcal{G}_{\lambda}} \right) = \sigma^2 + \frac{\sigma^2}{n} \sum_{i:s_i \neq 0} s_i^4 / s_i^4 = \sigma^2 + \frac{\sigma^2}{n} \text{rank}(\mathbf{X})$. Finally, we apply the inequalities (3.5) to:

$$\begin{aligned} \mathbb{E}_{\mathbf{X}} \ell_{\text{in}} \left(g_{\mathcal{T}}^{\mathcal{G}_{\lambda}} \right) &= \sigma^2 + \frac{\lambda^2}{n} \sum_{i:s_i \neq 0} \frac{(\mathbf{v}_i^{\top} \boldsymbol{\beta})^2 / s_i^2}{(1 + \lambda / s_i^2)^2} + \frac{\sigma^2}{n} \sum_{i:s_i \neq 0} \frac{1}{(1 + \lambda / s_i^2)^2} \\ &\leq \sigma^2 + \frac{\lambda^2}{n} \sum_{i:s_i \neq 0} \frac{(\mathbf{v}_i^{\top} \boldsymbol{\beta})^2}{s_i^2} + \frac{\sigma^2}{n} \sum_{i:s_i \neq 0} 1 - \frac{2\lambda}{s_i^2} + \frac{3\lambda^2}{s_i^4} \\ &= \lambda^2 \times \frac{\boldsymbol{\beta}^{\top} \boldsymbol{\Omega} \boldsymbol{\beta} + 3\sigma^2 \text{tr}(\boldsymbol{\Omega}^2) / n}{n^2} - 2\lambda \times \frac{\sigma^2 \text{tr}(\boldsymbol{\Omega})}{n^2} + \sigma^2 + \frac{\sigma^2}{n} \text{rank}(\mathbf{X}). \end{aligned}$$

Since the last expression is a quadratic minimized at λ^* , we obtain the desired inequality. Note that using (3.5) we also have

$$0 \leq \mathbb{E}_{\mathbf{X}} \ell_{\text{in}} \left(g_{\mathcal{T}}^{\mathcal{G}} \right) - \mathbb{E}_{\mathbf{X}} \ell_{\text{in}} \left(g_{\mathcal{T}}^{\mathcal{G}_{\lambda^*}} \right) - \frac{\sigma^2}{n^2} \lambda^* \text{tr}(\boldsymbol{\Omega}) \leq \frac{2}{n^3} [\lambda^*]^3 \left[\boldsymbol{\beta}^{\top} \boldsymbol{\Omega}^2 \boldsymbol{\beta} + \frac{2\sigma^2}{n} \text{tr}(\boldsymbol{\Omega}^3) \right].$$

□

3.2.2 Plug-in Estimate of λ^* for $n > p$

While Theorem 3.4 provides a formula for the regularization parameter, which is guaranteed to reduce the expected in-sample risk, the formula involves the (typically) unknown parameters σ^2 and $\boldsymbol{\beta}$. How we estimate these unknown parameters depends on whether $n > p$, or $n \leq p$.

When $n > p$, then a simple solution is to replace the unknown σ^2 with its unbiased estimator:

$$\hat{\sigma}^2 = \frac{\|(\mathbf{I} - \mathbf{P}_{\mathbf{X}}) \mathbf{Y}\|^2}{\text{tr}(\mathbf{I} - \mathbf{P}_{\mathbf{X}})} = \frac{\|\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}\|^2}{n - p},$$

and replace $\boldsymbol{\beta}^{\top} \boldsymbol{\Omega} \boldsymbol{\beta}$ with its unbiased estimator ($\mathbf{A} := \mathbf{X} \mathbf{X}^{\top} / n$):

$$n \sum_i \frac{(\mathbf{u}_i^{\top} \mathbf{Y})^2}{s_i^4} - \hat{\sigma}^2 n \sum_i \frac{1}{s_i^4} = \|\mathbf{A}^+ \mathbf{Y}\|^2 / n - \hat{\sigma}^2 \text{tr}(\boldsymbol{\Omega}^2) / n.$$

This yields the direct *plug-in estimator* of the regularization parameter λ^* :

$$\lambda_{\text{dpi}}^* = \frac{n \hat{\sigma}^2 \text{tr}(\boldsymbol{\Omega})}{\|\mathbf{A}^+ \mathbf{Y}\|^2 + 2\hat{\sigma}^2 \text{tr}(\boldsymbol{\Omega}^2)}. \quad (3.6)$$

■ Example 3.3 (Polynomial Regression) ■

3.3 Lasso Linear Regression

Consider a regression setting in which $\mathbf{X}$ does not include the intercept feature corresponding to β_0 . Thus, the model is

$$Y = \mathbf{1}\beta_0 + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\boldsymbol{\beta} \in \mathbb{R}^p$ (rather than using $d = p - 1$). Assuming that we do not wish to penalize the intercept β_0 , the ridge regression estimator then solves

$$\min_{b_0, \mathbf{b}} \|\mathbf{y} - \mathbf{1}b_0 - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|^2.$$

The usual computations show that $b_0 = \mathbf{1}^+(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{1}^\top(\mathbf{y} - \mathbf{X}\mathbf{b})/n$, so that eliminating b_0 from the optimization is equivalent to applying the centering matrix

$$\mathbf{C} = \mathbf{I} - \mathbf{1}\mathbf{1}^\top/n$$

to both $\mathbf{X}$ and $\mathbf{y}$. In other words, by centering the data we eliminate b_0 from the optimization and can focus on the regularization, free from the intercept:

$$\min_{\mathbf{b}} \|\mathbf{C}\mathbf{y} - \mathbf{C}\mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|^2.$$

Another type of regularization is the *lasso* regression, where lasso stands for *least absolute value shrinkage and selection operator*. Roughly, this type of regularization replaces the $\|\mathbf{b}\|^2$ penalty with the $\sum_i |b_i|$ penalty. This is the L^1 -norm penalty, which we can write as $\|\mathbf{b}\|_1 = \sum_i |b_i|$.

LASSO

After centering the data

$$\mathbf{y} \leftarrow \mathbf{y} - \mathbf{1}\bar{y}, \quad \mathbf{X} \leftarrow \mathbf{X} - \mathbf{1}\mathbf{1}^\top\mathbf{X}/n,$$

the lasso estimator is defined as:²

$$\hat{\boldsymbol{\beta}}_\lambda = \operatorname{argmin}_{\mathbf{b}} \frac{\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2}{2n} + \frac{\lambda}{\sqrt{n}} \sum_i \|\mathbf{v}_i\| \times |b_i|. \quad (3.7)$$

It is common to standardize the variances of the features in $\mathbf{X}/\sqrt{n}$ to be unity. In other words, the columns $\mathbf{v}_1, \dots, \mathbf{v}_p$ of $\mathbf{X}$ are all rescaled in such a way that $\|\mathbf{v}_i\|^2 = n$ for all i . This standardization can be accomplished by working with the matrix $\mathbf{X}\boldsymbol{\Upsilon}$, rather than $\mathbf{X}$, where $\boldsymbol{\Upsilon}$ is the rescaling matrix

$$\boldsymbol{\Upsilon} := \operatorname{diag}(\sqrt{n}/\|\mathbf{v}_1\|, \dots, \sqrt{n}/\|\mathbf{v}_p\|).$$

²Here we assume the estimator is unique.

The solution to (3.7) (without standardization of the features) is equivalent to the solution

$$\hat{\beta}_\lambda = \Upsilon \times \underset{b}{\operatorname{argmin}} \frac{\|y - [\mathbf{X}\Upsilon]b\|^2}{2n} + \lambda \sum_i |b_i|. \quad (3.8)$$

As we shall see, an advantage of this type of regularization is that many of the estimates in $\hat{\beta}_\lambda$ are estimated as zeros, making it very easy to tell which features are important and which are not.

■ **Example 3.4 (Shrinkage and Feature Selection for Polynomial Regression)** We revisit the polynomial regression dataset from Example 1.1. Figure 3.2 shows the values of the coefficients (with the exception of the intercept β_0) as a function of λ . Compared to Figure 3.1 the paths traced by the values of the coefficients are not smooth and have kinks and corners. Some of these kinks correspond to a coefficient hitting zero. This property aids in the selection of features — features with coefficient equal to zero are excluded from the regression. Again, as λ converges to zero, the coefficient estimates converge to $\mathbf{X}^+y$. However, for a large enough λ , all the coefficients drop to zero.

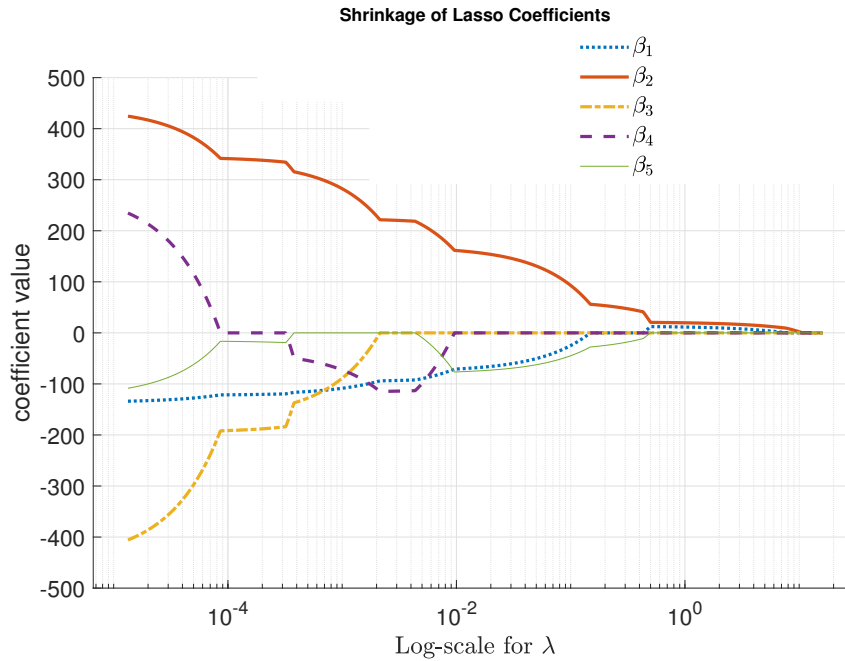


Figure 3.2: Lasso coefficient values as a function of the ridge parameter λ . The lasso coefficients are shrunk to zero abruptly as λ becomes larger and larger. The existence of zero coefficients for finite λ aids the selection of features.



Lasso regression can be used to select which features are relevant (without the need for complex F-tests, ANOVA, etc.) and works even when $p > n$.

3.3.1 Computing Lasso Estimator

To understand how the feature selection in Lasso works, we need to study the properties of the minimizer (3.7). Just as with the ridge regression, we set the gradient of

$$\frac{\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2}{2n} + \frac{\lambda}{\sqrt{n}} \sum_i \|\mathbf{v}_i\| \times |b_i|$$

with respect to $\mathbf{b}$ to zero and then solve the system of equations. The only issue is computing the derivative of $|x|$, which is not as simple as that of x^2 . We may guess that when $x > 0$, then $d|x|/dx = 1$ and when $x < 0$, then $d|x|/dx = -1$. This is, in fact, correct. Thus, the only issue remains with determining the derivative value at $x = 0$. The difficulty here is that the derivative from the left:

$$\lim_{\varepsilon \downarrow 0, x=0} \frac{|x| - |x - \varepsilon|}{\varepsilon} = -1$$

does not equal the derivative from the right:

$$\lim_{\varepsilon \downarrow 0, x=0} \frac{|x + \varepsilon| - |x|}{\varepsilon} = 1.$$

So the usual derivative (which requires that these two limits agree) does not exist at $x = 0$. This calls for a generalization of the derivative called *subderivative*.

SUBDERIVATIVE

A subderivative of a convex function f in the neighborhood of x_0 is a number s such that

$$f(x) \geq f(x_0) + (x - x_0)s$$

for all x in the neighborhood of x_0 . In other words, s is the slope of a line which supports the function $f(x)$ from below. Such a line may not be unique.

It can be shown that the possible values of s lie in the interval $[a, b]$, where

$$a = \lim_{\varepsilon \downarrow 0} \frac{f(x_0) - f(x_0 - \varepsilon)}{\varepsilon}$$

is the derivative from the left and

$$b = \lim_{\varepsilon \downarrow 0} \frac{f(x_0 + \varepsilon) - f(x_0)}{\varepsilon}$$

is the derivative from the right.

SUBDIFFERENTIAL

The set $[a, b]$ is called the *subdifferential* of f at x_0 and is denoted as $\partial f(x_0)$. This set is either an interval, or a single point (when $a = b$). It cannot be an empty set.

When f is differentiable at x_0 , then we know that $a = b$ and hence $s = a = b$ is a singleton equal to the classical derivative at x_0 .

Contrariwise, if the subdifferential $\partial f(x_0)$ contains only one element, then that element is the classical derivative at x_0 .

Going back to the function $|x|$, then we can say that its subdifferential at any $x < 0$ is the singleton set $\{-1\}$, while for $x > 0$ it is $\{1\}$. The subdifferential of $|x|$ at $x = 0$ is the set $[-1, 1]$.

In summary, a subderivative g of $|x|$ satisfies

$$g = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \\ \in [-1, 1], & x = 0 \end{cases}.$$

SUBGRADIENT

Similarly, we denote by $\mathbf{g} = [g_1, \dots, g_p]^\top$ a *subgradient* of $\sum_i |b_i|$ with respect to $\mathbf{b}$ and the components of $\mathbf{g}$ satisfy:

$$g_k = \begin{cases} -1, & b_k < 0 \\ 1, & b_k > 0 \\ \in [-1, 1], & b_k = 0 \end{cases}.$$

We are now ready to state the equations satisfied by the lasso estimate.

Normally, with a differentiable convex function we set the gradient to zero to find its optimum. Here we do the same, but replace problematic gradients with subgradients at points that are not differentiable. The lasso solution then satisfies the so-called *Karush–Kuhn–Tucker* (KKT) conditions ($\varsigma := \sqrt{n}\lambda$):

KARUSH–KUHN–TUCKER

$$\mathbf{v}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \|\mathbf{v}_k\| \varsigma g_k, \quad k = 1, \dots, p, \quad (3.9)$$

where $\mathbf{g}$ belongs to the subdifferential of $\sum_i |b_i|$. Unlike the ridge estimator, we cannot solve (3.9) explicitly, because $\mathbf{g}$ depends on the values of $\mathbf{b}$ in a nonlinear way.

However, we can find an explicit solution when solving for the k -th component in $\mathbf{b}$, while keeping every other $b_i, i \neq k$ fixed. If $\hat{\beta}_k$ is the solution to the k -th equation in (3.9), then $\hat{\beta}_k$ satisfies

$$\mathbf{v}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b} - \mathbf{v}_k(\hat{\beta}_k - b_k)) = \|\mathbf{v}_k\| \varsigma g_k.$$

Rearranging yields:

$$\hat{\beta}_k = \frac{\mathbf{v}_k^\top \mathbf{r}_{-k} - \|\mathbf{v}_k\| \varsigma g_k}{\|\mathbf{v}_k\|^2},$$

where $\mathbf{r}_{/k} := \mathbf{y} - \mathbf{X}\mathbf{b} + \mathbf{v}_k b_k$ is the residual that omits the k -th feature. If we define the Lasso *shrinkage operator*:

SHRINKAGE
OPERATOR

$$S_{\lambda}^{\text{Lasso}}(x) := x[1 - \lambda/|x|]_+,$$

then, using the fact that $g_k = \text{sign}(b_k)$ if $b_k \neq 0$ and $g_k \in [-1, 1]$ for $b_k = 0$, we can verify that an explicit solution for $\hat{\beta}_k$ is

$$\hat{\beta}_k = S_{\frac{\varsigma}{\|\mathbf{v}_k\|}}^{\text{Lasso}} \left(\frac{\mathbf{v}_k^{\top} \mathbf{r}_{/k}}{\|\mathbf{v}_k\|^2} \right). \quad (3.10)$$

If we were computing (3.8), where $\|\mathbf{v}_k\|^2 = n$ after standardization, this coordinate descent formula simplifies to

$$\hat{\beta}_k = S_{\lambda}^{\text{Lasso}} \left(\frac{\mathbf{v}_k^{\top} \mathbf{r}_{/k}}{n} \right).$$

This solution now clearly shows how a coefficient can be set to zero, forcing the selection of features.

Next, we apply the formula $b_k \leftarrow S_{\lambda}^{\text{Lasso}}(\mathbf{v}_k^{\top} \mathbf{r}_{/k}/n)$ sequentially for each $k = 1, \dots, p$, and repeat until each b_k does not change by the update $b_k \leftarrow S_{\lambda}^{\text{Lasso}}(\mathbf{v}_k^{\top} \mathbf{r}_{/k}/n)$. It can be shown that this procedure, called *coordinate descent* (CD), converges to the solution of (3.9) and thus delivers the lasso estimator (3.7). The idea is summarized in the following pseudo-algorithm, which assumes no standardization of the predictors.

COORDINATE
DESCENT

Algorithm 3.3.1: Simple CD to minimize $\frac{1}{2n} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \frac{\lambda}{\sqrt{n}} \sum_i \|\mathbf{v}_i\| \times |b_i|$

input: initial guess $\mathbf{b}$ and residual $\mathbf{r} := \mathbf{y} - \mathbf{X}\mathbf{b}$, matrix $\mathbf{X}$, tolerance ε , and λ

output: (global) minimizer $\mathbf{b}$ and corresponding residual $\mathbf{r}$

```

1  $\varsigma \leftarrow \sqrt{n}\lambda$ 
2 repeat // iterate over CD cycles
3    $\mathbf{b}_{\text{old}} \leftarrow \mathbf{b}$ 
4   for  $k = 1, \dots, p$  do // coordinate-descent cycle
5      $b_{\text{new}} \leftarrow S_{\varsigma/\|\mathbf{v}_k\|}^{\text{Lasso}}(b_k + \mathbf{v}_k^{\top} \mathbf{r} / \|\mathbf{v}_k\|^2)$  // this step costs  $O(n)$ 
6      $\mathbf{r} \leftarrow \mathbf{r} + (b_k - b_{\text{new}})\mathbf{v}_k$  // this update costs  $O(n)$ 
7      $b_k \leftarrow b_{\text{new}}$ 
8 until  $\|\mathbf{b} - \mathbf{b}_{\text{old}}\| < \varepsilon$ 
9 return  $\mathbf{b}$  &  $\mathbf{r}$ 

```

If, instead of computing (3.7), we are interested in computing the minimizer of

$$\frac{1}{2n} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \frac{\lambda}{\sqrt{n}} \sum_i w_i \times \|\mathbf{v}_i\| \times |b_i|,$$

where a weight $w_i \geq 0$ is attached to each $|b_i|$ in the lasso penalty, then we need only change line 5 of Algorithm 3.3.1 to

$$b_{\text{new}} \leftarrow S_{w_k \varsigma / \|\mathbf{v}_k\|}^{\text{Lasso}}(b_k + \mathbf{v}_k^\top \mathbf{r} / \|\mathbf{v}_k\|^2). \quad (3.11)$$

Note that one coordinate cycle in Algorithm 3.3.1 costs $O(p \times n)$, which may be too expensive if n is very large.

3.3.2 Picking λ via Cross-Validation

To determine a good value for λ (or, equivalently, $\varsigma = \sqrt{n}\lambda$) we can use cross-validation as in Section 1.2.1. This requires that we compute the lasso estimator $\hat{\boldsymbol{\beta}}_\lambda$ on a grid of m values

$$\lambda_1 < \lambda_2 < \cdots < \lambda_m.$$

Typically, $m = 100$ or $m = 1000$. Larger values of λ correspond to sparser solutions (more zeros in $\hat{\boldsymbol{\beta}}_\lambda$) and smaller values of λ correspond to fewer zeros in the solution vector.

A good way to determine the range of values for this grid is to pick λ_m to be the smallest lambda for which $\hat{\boldsymbol{\beta}}_\lambda = \mathbf{0}$.

What is the value of this λ ? Denote

$$\mathcal{A} := \{k : b_k \neq 0\}$$

to be the set of *working features*, that is, the set of nonzero coefficients in $\mathbf{b}$. Recall the gradient-equals-zero equations (3.9). From these equations we can deduce that

$$\begin{aligned} |\mathbf{v}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b})| &= \|\mathbf{v}_k\| \varsigma, & k \in \mathcal{A} \\ |\mathbf{v}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b})| &\leq \|\mathbf{v}_k\| \varsigma, & k \notin \mathcal{A}. \end{aligned} \quad (3.12)$$

Therefore, if $\mathcal{A}$ is empty, meaning that $\mathbf{b} = \mathbf{0}$, then $|\mathbf{v}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{0})| \leq \|\mathbf{v}_k\| \sqrt{n}\lambda$ for all k . In other words, we smallest λ for which $\mathcal{A}$ is empty satisfies:

$$\lambda > \lambda^* := \max_k \frac{|\mathbf{v}_k^\top \mathbf{y}|}{\|\mathbf{v}_k\| \sqrt{n}}.$$

The idea is then to start with the zero solution $\hat{\boldsymbol{\beta}}_{\lambda_m} = \mathbf{0}$ corresponding to $\lambda_m := \lambda^*$ and then try to compute the solution $\hat{\boldsymbol{\beta}}_{\lambda_{m-1}}$ for a smaller λ_{m-1} using $\hat{\boldsymbol{\beta}}_{\lambda_m}$ as an initial guess for the CD iteration. Once the $\hat{\boldsymbol{\beta}}_{\lambda_{m-1}}$ is computed, then we compute $\hat{\boldsymbol{\beta}}_{\lambda_{m-2}}$ (corresponding to λ_{m-2}) using $\hat{\boldsymbol{\beta}}_{\lambda_{m-1}}$ as an initial guess and so on, until we go through the entire grid of size m .

Typically, λ_1/λ_m equals a small number, say 10^{-5} and the values inbetween the smallest and the largest lambda are determined by creating a log-spaced grid:

$$\lambda_j = \exp(\ln[\lambda_1] + j/m(\ln[\lambda_m] - \ln[\lambda_1])), \quad j = 1, \dots, m.$$

14

WORKING
FEATURES

In other words, the logarithms of the lambdas are equally spaced on a grid with m points. We summarize this procedure in the following algorithm.

Algorithm 3.3.2: Lasso estimate computed on a grid for lambda.

input: training data $\{\mathbf{X}, \mathbf{y}\}$, integer m , error tolerance $\varepsilon > 0$
output: lasso coefficient $\mathbf{b}_j$ corresponding to λ_j for $j = 1, \dots, m$

```

1  $\lambda_m \leftarrow \max_k |\mathbf{v}_k^\top \mathbf{y}| / \|\sqrt{n} \mathbf{v}_k\|$  // the smallest  $\lambda$  forcing  $\mathbf{b} = \mathbf{0}$ 
2  $\lambda_{\min} \leftarrow \lambda_m / 10^5$  // the smallest  $\lambda$  on the grid
3  $\lambda_j \leftarrow \exp(\ln(\lambda_{\min}) + j/m \ln(\lambda_m / \lambda_{\min}))$ ,  $j = 1, \dots, m$  // log-spaced grid
4  $\mathbf{b} \leftarrow \mathbf{0}$ ,  $\mathbf{r} \leftarrow \mathbf{y}$  // initial  $\mathbf{b}$  and residual
5 for  $j = m, m-1, m-2, \dots, 2, 1$  do // decrement the  $\lambda$ 
6    $(\mathbf{b}, \mathbf{r}) \leftarrow \text{lassoCD}(\mathbf{b}, \mathbf{r}, \mathbf{X}, \varepsilon, \lambda_j)$  // implements Algorithm 3.3.1
7    $\mathbf{b}_j \leftarrow \mathbf{b}$ 
```

■ **Example 3.5 (Polynomial Regression)** ■

LOGISTIC REGRESSION

4.1 Logistic Regression

Recall that when $y \in \{0, \dots, c-1\}$, then predicting y is the same as classifying the input $\mathbf{x}$ into c categories, a machine learning problem called *classification*. In this chapter we consider the case where $y \in \{0, 1\}$.

The simplest model for Y is to assume that, conditional on $\mathbf{X} = \mathbf{x}$ (and given fixed $\boldsymbol{\beta}, \beta_0$), the responses are independent Bernoulli random variables with success probability:

$$\mathbb{P}[Y = 1 \mid \mathbf{X} = \mathbf{x}] = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = h(\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}),$$

where $h : \mathbb{R} \mapsto [0, 1]$ is a function mapping the input $\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}$ into a probability value between zero and one. The most common choice for h is the *logistic function*:

$$h(x) := \frac{1}{1 + \exp(-x)}.$$

LOGISTIC
FUNCTION

This choice for h yields the so-called *logistic regression* for the binary response Y . The logistic regression is one of the simplest nonlinear extensions of the linear regression in Chapter 2. In the logistic regression, the log-odds probability (of belonging to one of the two classes) is modeled as a linear function:

LOGISTIC
REGRESSION

$$\ln \frac{\mathbb{P}[Y = 1 \mid \mathbf{X} = \mathbf{x}]}{\mathbb{P}[Y = 0 \mid \mathbf{X} = \mathbf{x}]} = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta}.$$

Assuming that the binary response $\mathbf{Y} = [Y_1, \dots, Y_n]^\top$ consists of n independent Bernoulli random variables, the joint pdf of $\mathbf{Y}$ is:

$$g(\mathbf{y} \mid \boldsymbol{\beta}, \beta_0, \mathbf{X}) = \prod_{i=1}^n [h(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})]^{y_i} [1 - h(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})]^{1-y_i}.$$

Here we keep the intercept β_0 separate from $\boldsymbol{\beta}$ and $\mathbf{X}$ does not contain the constant feature $\mathbf{1}$. For this reason we use the following notation in which we merge β_0 and $\boldsymbol{\beta}$.

Let $\check{\mathbf{b}} := [b_0, \mathbf{b}^\top]^\top$ and $\check{\mathbf{X}} := [\mathbf{1}, \mathbf{X}]$, so that

$$\check{\mathbf{X}}\check{\mathbf{b}} = \mathbf{1}b_0 + \mathbf{X}\mathbf{b}.$$

Since

$$\ln h(\check{\mathbf{x}}_i^\top \check{\mathbf{b}}) = -\ln(1 + \exp(-\check{\mathbf{x}}_i^\top \check{\mathbf{b}}))$$

and

$$\ln[1 - h(\check{\mathbf{x}}_i^\top \check{\mathbf{b}})] = -\check{\mathbf{x}}_i^\top \check{\mathbf{b}} - \ln(1 + \exp(-\check{\mathbf{x}}_i^\top \check{\mathbf{b}})),$$

the negative of the log-likelihood, $-\ln g(\mathbf{y} | \check{\mathbf{b}}, \check{\mathbf{X}})$, is given by:

$$-\ln g(\mathbf{y} | \check{\mathbf{b}}, \check{\mathbf{X}}) = \frac{1}{n} \sum_{i=1}^n (1 - y_i)[b_0 + \mathbf{x}_i^\top \mathbf{b}] + \ln(1 + \exp(-b_0 - \mathbf{x}_i^\top \mathbf{b})).$$

In the machine learning literature we call $-\ln g(\mathbf{y} | \check{\mathbf{b}}, \check{\mathbf{X}})/n$ the *cross-entropy training loss*, denoted as

CROSS-
ENTROPY
TRAINING
LOSS

$$r_\tau(\check{\mathbf{b}}) := \frac{1}{n} \sum_{i=1}^n (1 - y_i)\check{\mathbf{x}}_i^\top \check{\mathbf{b}} + \ln(1 + \exp(-\check{\mathbf{x}}_i^\top \check{\mathbf{b}})). \quad (4.1)$$

The logistic learner which classifies the input $\check{\mathbf{x}} = [1, \mathbf{x}^\top]^\top$ into category 1 or 0 is:

$$\mathbb{1}\{h(\check{\mathbf{x}}^\top \check{\mathbf{b}}) > 1/2\},$$

where $\check{\mathbf{b}}$ is now the minimizer of the cross-entropy training loss (4.1).

The main challenge is the computation of $\operatorname{argmin}_{\check{\mathbf{b}}} r_\tau(\check{\mathbf{b}})$. The problem is nonlinear, but it can be solved by successive and repeated linearizations. Suppose that we have a current best guess $\check{\mathbf{b}}_t$ for the minimizer of (4.1). Given this $\check{\mathbf{b}}_t$, we consider the quadratic *multivariate Taylor approximation* to r_τ at the point $\check{\mathbf{b}}_t$:

MULTIVARI-
ATE TAYLOR
APPROXIMA-
TION

$$r_\tau(\check{\mathbf{b}}_t) + (\check{\mathbf{b}} - \check{\mathbf{b}}_t)^\top \nabla r_\tau(\check{\mathbf{b}}_t) + \frac{1}{2}(\check{\mathbf{b}} - \check{\mathbf{b}}_t)^\top \mathbf{H}(\check{\mathbf{b}}_t)(\check{\mathbf{b}} - \check{\mathbf{b}}_t), \quad (4.2)$$

where the gradient and Hessian are given by the formulas:

$$\begin{aligned} \mu_{t,i} &:= h(b_0 + \mathbf{x}_i^\top \mathbf{b}_t) \\ \boldsymbol{\mu}_t &:= [\mu_{t,1}, \dots, \mu_{t,n}]^\top \\ \nabla r_\tau(\check{\mathbf{b}}_t) &:= \frac{1}{n} \check{\mathbf{X}}^\top (\boldsymbol{\mu}_t - \mathbf{y}) \\ \mathbf{w}_t &:= \sqrt{\boldsymbol{\mu}_t \odot (\mathbf{1} - \boldsymbol{\mu}_t)} \quad (w_{t,i} = \sqrt{\mu_{t,i}(1 - \mu_{t,i})}, \forall i) \\ \mathbf{D}_t &:= \operatorname{diag}(\mathbf{w}_t) \\ \mathbf{H}(\check{\mathbf{b}}_t) &:= \frac{1}{n} \check{\mathbf{X}}^\top \mathbf{D}_t^2 \check{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n w_{t,i}^2 \check{\mathbf{x}}_i \check{\mathbf{x}}_i^\top. \end{aligned}$$

Then, we update $\check{\mathbf{b}}_t$ with $\check{\mathbf{b}}_{t+1}$, which is the minimizer of (4.2), namely:

$$\check{\mathbf{b}}_{t+1} = \check{\mathbf{b}}_t - [\mathbf{H}(\check{\mathbf{b}}_t)]^{-1} \nabla r_\tau(\check{\mathbf{b}}_t).$$

Straightforward algebra shows that this $\check{\mathbf{b}}_{t+1}$ can also be written as the solution to the following *weighed least-squares*:

$$\check{\mathbf{b}}_{t+1} := \underset{\check{\mathbf{b}}}{\operatorname{argmin}} (\check{\mathbf{y}}_t - \check{\mathbf{X}}\check{\mathbf{b}})^\top \mathbf{D}_t^2 (\check{\mathbf{y}}_t - \check{\mathbf{X}}\check{\mathbf{b}}),$$

WEIGHED
LEAST-
SQUARES

where $\check{\mathbf{y}}_t := \check{\mathbf{X}}\check{\mathbf{b}}_t + \mathbf{D}_t^{-2}(\mathbf{y} - \boldsymbol{\mu}_t)$ is the so-called *adjusted response*. If we denote

$$\begin{aligned} \check{\mathbf{X}}_t &:= \mathbf{D}_t \check{\mathbf{X}}, \\ \mathbf{X}_t &:= \mathbf{D}_t \mathbf{X}, \\ \mathbf{y}_t &:= \mathbf{D}_t \check{\mathbf{y}}_t \\ &= \check{\mathbf{X}}_t \check{\mathbf{b}}_t + \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu}_t) \\ &= \mathbf{w}_t b_{t,0} + \mathbf{X}_t \mathbf{b}_t + \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu}_t), \end{aligned}$$

and use the fact that

$$(\check{\mathbf{y}}_t - \check{\mathbf{X}}\check{\mathbf{b}})^\top \mathbf{D}_t^2 (\check{\mathbf{y}}_t - \check{\mathbf{X}}\check{\mathbf{b}}) = \|\mathbf{D}_t \check{\mathbf{y}}_t - \mathbf{D}_t \check{\mathbf{X}}\check{\mathbf{b}}\|^2,$$

we can write the weighed least-squares as

$$(b_{t+1,0}, \mathbf{b}_{t+1}) = \underset{b_{0,t}, \mathbf{b}_t}{\operatorname{argmin}} \|\mathbf{y}_t - \mathbf{w}_t b_{0,t} - \mathbf{X}_t \mathbf{b}_t\|^2.$$

Since the first column of $\check{\mathbf{X}}_t$ is $\mathbf{w}_t$ (for $\check{\mathbf{X}}$ it is $\mathbf{1}$), we can eliminate the intercept term b_0 from the optimization by applying the centering (projection) matrix

$$\mathbf{C}_t = \mathbf{I} - \mathbf{w}_t \mathbf{w}_t^\top / \|\mathbf{w}_t\|^2$$

to both $\mathbf{X}_t$ and $\mathbf{y}_t$. Once the $\mathbf{b}_{t+1}$ is computed, then we recover $b_{0,t+1} = \mathbf{w}_t^\top (\mathbf{y}_t - \mathbf{X}_t \mathbf{b}_{t+1})$. This gives the following formulas for updating $(b_{0,t}, \mathbf{b}_t)$ to $(b_{0,t+1}, \mathbf{b}_{t+1})$:

$$\begin{aligned} \mathbf{b}_{t+1} &:= \underset{\mathbf{b}}{\operatorname{argmin}} \|\mathbf{C}_t \mathbf{y}_t - \mathbf{C}_t \mathbf{X}_t \mathbf{b}\|^2 \\ b_{0,t+1} &:= \frac{\mathbf{w}_t^\top (\mathbf{y}_t - \mathbf{X}_t \mathbf{b}_{t+1})}{\|\mathbf{w}_t\|^2}. \end{aligned}$$

We iterate for $t = 1, 2, \dots$ until a convergence criterion is met. This procedure is usually called *Iterative Reweighed Least Squares* (IRLS).

ITERATIVE
REWEIGHED
LEAST
SQUARES

4.2 Regularized Logistic Regression

One possible pitfall of minimizing (4.1) is that a minimum may no exist in the case of $p > n$. To make the logistic regression applicable even in the frequently occurring case of $p > n$, we consider its lasso version.

The lasso regularized logistic learner is $\mathbb{1}\{h(\check{\mathbf{x}}^\top \check{\mathbf{b}}_\lambda) > 1/2\}$, where:

$$\check{\mathbf{b}}_\lambda := \operatorname{argmin}_{b_0, \mathbf{b}} r_\tau(\check{\mathbf{b}}) + \lambda \sum_{i=1}^p |b_i|.$$

Note that here we choose not to standardize the features. There are two main challenges. First, we need to be able to compute $\check{\mathbf{b}}_\lambda$ for a given λ . Second, we also need to compute $\check{\mathbf{b}}_\lambda$ on a grid of lambda values, which will assist us in using cross-validation to determine a suitable value for λ .

Computation for a given λ . The computation of $\check{\mathbf{b}}_\lambda$ proceeds in exactly the same way as the IRLS, except that we add the penalty $\|\mathbf{b}\|_1$ term to the optimization. In other words, we iterate for $t = 1, 2, \dots$:

$$\begin{aligned} \mathbf{b}_{t+1} &= \operatorname{argmin}_{\mathbf{b}} \frac{\|\mathbf{C}_t \mathbf{y}_t - \mathbf{C}_t \mathbf{X}_t \mathbf{b}\|^2}{2n} + \lambda \sum_{i=1}^p |b_i| \\ b_{0,t+1} &= \frac{\mathbf{w}_t^\top (\mathbf{y}_t - \mathbf{X}_t \mathbf{b}_{t+1})}{\|\mathbf{w}_t\|^2}. \end{aligned} \quad (4.3)$$

Of course, unlike the penalty-free logistic regression, here (4.3) cannot be computed via a simple formula, but has to be computed using the iterative Algorithm 3.3.1 with output/response $\mathbf{C}_t \mathbf{y}_t$, input/feature matrix $\mathbf{C}_t \mathbf{X}_t$, and line 5 changed to (3.11) with $w_k = \sqrt{n}/\|\mathbf{v}_k\|$ for all k .

Grid of λ 's for Cross-Validation. As with the vanilla Lasso for linear regression in Chapter 3, we need to compute the estimate $\check{\mathbf{b}}_\lambda$ on a grid of m values: $\lambda_1 < \lambda_2 < \dots < \lambda_m$. Again, larger values of λ correspond to sparser solutions. To construct the necessary grid, we now repeat the arguments in Section 3.3.2.

First, from the formula for $\nabla r_\tau(\check{\mathbf{b}})$, we find that the analogue of (3.9) here is the KKT condition:

$$\frac{1}{n} \check{\mathbf{X}}^\top (\boldsymbol{\mu} - \mathbf{y}) = \lambda \mathbf{g},$$

where $\mathbf{g}$ is the subgradient of $\sum_i |b_i|$. If $\mathcal{A}$ is the set of working features, then we can rewrite the KKT condition as:

$$\begin{aligned} |\mathbf{v}_k^\top (\mathbf{y} - \boldsymbol{\mu})| &= n\lambda, & k \in \mathcal{A} \\ |\mathbf{v}_k^\top (\mathbf{y} - \boldsymbol{\mu})| &\leq n\lambda, & k \notin \mathcal{A}. \end{aligned}$$

Therefore, the smallest value of λ for which we have $\mathcal{A} = \emptyset$ satisfies:

$$\lambda > \lambda^* := \max_j |\mathbf{v}_j^\top (\mathbf{y} - h(\mathbf{1}b_0 + \mathbf{X}\mathbf{0}))|/n.$$

Since the intercept term b_0 satisfies:

$$\mathbf{1}^\top (h(\mathbf{1}b_0 + \mathbf{X}\mathbf{b}) - \mathbf{y}) = 0,$$

that is, $\bar{y} = \frac{1}{n} \sum_i \frac{1}{1 + \exp(-b_0 - \mathbf{x}_i^\top \mathbf{b})}$, we have for $\mathbf{b} = \mathbf{0}$:

$$b_0 = -\ln(1/\bar{y} - 1)$$

$$h(\mathbf{1}b_0 + \mathbf{X}\mathbf{0}) = \mathbf{1}\bar{y}.$$

Therefore, the smallest λ that will ensure $\mathbf{b} = \mathbf{0}$ satisfies:

$$\lambda > \|\mathbf{X}^\top (\mathbf{y} - \mathbf{1}\bar{y})\|_\infty / n.$$

In summary, the algorithm is as follows.

Algorithm 4.2.1: CD for Lasso-Logistic model (without standardizing $\mathbf{X}$)

input: $\mathbf{y}, \mathbf{X}$, error tolerance $\varepsilon > 0$ and grid size m
output: Regularized solution $(b_{0,j}, \mathbf{b}_j)$ for each λ_j $j = 1, \dots, m$

```

1  $\lambda_m \leftarrow \|\mathbf{X}^\top (\mathbf{y} - \mathbf{1}\bar{y})\|_\infty / n$  // the smallest  $\lambda$  forcing  $\mathbf{b} = \mathbf{0}$ 
2  $\lambda_{\min} \leftarrow \lambda_m / 10^5$  // the smallest  $\lambda$  on the grid
3  $\lambda_j \leftarrow \exp(\ln(\lambda_{\min}) + j/m \ln(\lambda_m / \lambda_{\min}))$ ,  $j = 1, \dots, m$  // the grid
4  $b_0 \leftarrow -\ln(1/\bar{y} - 1)$ ,  $\mathbf{b} \leftarrow \mathbf{0}$  // initializing values for  $\lambda_m$ 
5 for  $j = m, m-1, m-2, \dots, 2, 1$  do // outer loop over  $\{\lambda_j\}$ 
6    $t \leftarrow 0$ 
7   repeat // middle loop is over the Taylor expansion
8      $\mathbf{b}_{\text{old}} \leftarrow \mathbf{b}$ 
9      $\mu \leftarrow h(\mathbf{1}b_0 + \mathbf{X}\mathbf{b})$  // mean response
10     $\mathbf{w} \leftarrow \sqrt{\mu \odot (\mathbf{1} - \mu)}$  // weights
11     $\mathbf{X}_t \leftarrow \text{diag}(\mathbf{w})\mathbf{X}$ 
12     $\mathbf{y}_t \leftarrow b_0\mathbf{w} + \mathbf{X}_t\mathbf{b} + (\mathbf{y} - \mu) \div \mathbf{w}$  // adjusted & weighted response
13     $c_y \leftarrow \mathbf{y}_t^\top \mathbf{w} / \|\mathbf{w}\|^2$ ,  $\mathbf{c}_x \leftarrow \mathbf{X}_t^\top \mathbf{w} / \|\mathbf{w}\|^2$  // adjustments for intercept
14     $\mathbf{b} \leftarrow \text{lassoCD}(\mathbf{b}, \mathbf{y}_t - \mathbf{w}c_y, \mathbf{X}_t - \mathbf{w}\mathbf{c}_x^\top, \varepsilon, \lambda_j)$  // inner loop
15     $b_0 \leftarrow c_y - \mathbf{c}_x^\top \mathbf{b}$  // intercept update given  $\mathbf{b}$ 
16     $t \leftarrow t + 1$ 
17  until  $\|\mathbf{b}_{\text{old}} - \mathbf{b}\| < \varepsilon$ 
18   $(b_{0,j}, \mathbf{b}_j) \leftarrow (b_0, \mathbf{b})$  // store values for each grid point
19 return  $b_{0,j}, \mathbf{b}_j$ ,  $j = 1, \dots, m$ 
```

In line 14 of Algorithm 4.2.1 we call Algorithm 3.3.1 with the modification (3.11) and $w_k = \sqrt{n}/\|\mathbf{v}_k\|$.

■ **Example 4.1 (Affairs Dataset)** ■

LINEAR ALGEBRA

The purpose of this appendix is to review some important topics in linear algebra. We assume that the reader has some familiarity with matrix and vector operations, including matrix multiplication and the computation of determinants.

A.1 Vector Spaces, Bases, and Matrices

Linear algebra is the study of vector spaces and linear mappings. Vectors are, by definition, elements of some *vector space* $\mathcal{V}$ and satisfy the usual rules of addition and scalar multiplication, e.g.,

VECTOR
SPACE

if $\mathbf{x} \in \mathcal{V}$ and $\mathbf{y} \in \mathcal{V}$, then $\alpha\mathbf{x} + \beta\mathbf{y} \in \mathcal{V}$ for all $\alpha, \beta \in \mathbb{R}$ (or $\mathbb{C}$).

We will be dealing mostly with vectors in the Euclidean vector space $\mathbb{R}^n$ for some n . That is, we view the points of $\mathbb{R}^n$ as objects that can be added up and multiplied with a scalar, e.g., $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ for points in $\mathbb{R}^2$. Sometimes it is convenient to work with the complex vector space $\mathbb{C}^n$ instead of $\mathbb{R}^n$; see also Section A.4.

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are called *linearly independent* if none of them can be expressed as a linear combination of the others; that is, if $\alpha_1\mathbf{v}_1 + \dots + \alpha_n\mathbf{v}_n = \mathbf{0}$, then it must hold that $\alpha_i = 0$ for all $i = 1, \dots, n$.

LINEARLY IN-
DEPENDENT

Definition A.1: Basis of a Vector Space

BASIS

A set of vectors $\mathcal{B} = \{v_1, \dots, v_n\}$ is called a *basis* of the vector space $\mathcal{V}$ if every vector $x \in \mathcal{V}$ can be written as a unique linear combination of the vectors in $\mathcal{B}$:

$$x = \alpha_1 v_1 + \dots + \alpha_n v_n.$$

DIMENSION

The (possibly infinite) number n is called the *dimension* of $\mathcal{V}$.

Using a basis $\mathcal{B}$ of $\mathcal{V}$, we can thus represent each vector $x \in \mathcal{V}$ as a row or column of numbers

$$[\alpha_1, \dots, \alpha_n] \quad \text{or} \quad \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}. \quad (\text{A.1})$$

STANDARD BASIS

Typically, vectors in $\mathbb{R}^n$ are represented via the *standard basis*, consisting of unit vectors (points) $e_1 = (1, 0, \dots, 0), \dots, e_n = (0, 0, \dots, 0, 1)$. As a consequence, any point $(x_1, \dots, x_n) \in \mathbb{R}^n$ can be represented, using the standard basis, as a row or column vector of the form (A.1) above, with $\alpha_i = x_i, i = 1, \dots, n$. We will also write

TRANSPOSE

$[x_1, x_2, \dots, x_n]^\top$, for the corresponding column vector, where $^\top$ denotes the *transpose*.



To avoid confusion, we will use the convention from now on that a generic vector x is always represented via the standard basis as a *column* vector. The corresponding row vector is denoted by $x^\top$.

MATRIX LINEAR TRANSFORMATION

A *matrix* can be viewed as an array of m rows and n columns that defines a *linear transformation* from $\mathbb{R}^n$ to $\mathbb{R}^m$ (or for complex matrices, from $\mathbb{C}^n$ to $\mathbb{C}^m$). The matrix is said to be *square* if $m = n$. If $a_1, a_2, \dots, a_n$ are the columns of $\mathbf{A}$, that is, $\mathbf{A} = [a_1, a_2, \dots, a_n]$, and if $x = [x_1, \dots, x_n]^\top$, then $\mathbf{A}x = x_1 a_1 + \dots + x_n a_n$. In particular, the standard basis vector e_k is mapped to the vector $a_k, k = 1, \dots, n$. We sometimes use the notation $\mathbf{A} = [a_{ij}]$, to denote a matrix whose (i, j) -th element is a_{ij} . When we wish to emphasize that a matrix $\mathbf{A}$ is real-valued with m rows and n columns, we write $\mathbf{A} \in \mathbb{R}^{m \times n}$. The *rank* of a matrix is the number of linearly independent rows or, equivalently, the number of linearly independent columns.

RANK

■ **Example A.1 (Linear Transformation)** Take the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -0.5 & -2 \end{bmatrix}.$$

It transforms the two basis vectors $[1, 0]^\top$ and $[0, 1]^\top$, shown in red and blue in the left panel of Figure A.1, to the vectors $[1, -0.5]^\top$ and $[1, -2]^\top$, shown on the right panel. Similarly, the points on the unit circle are transformed to an ellipse.

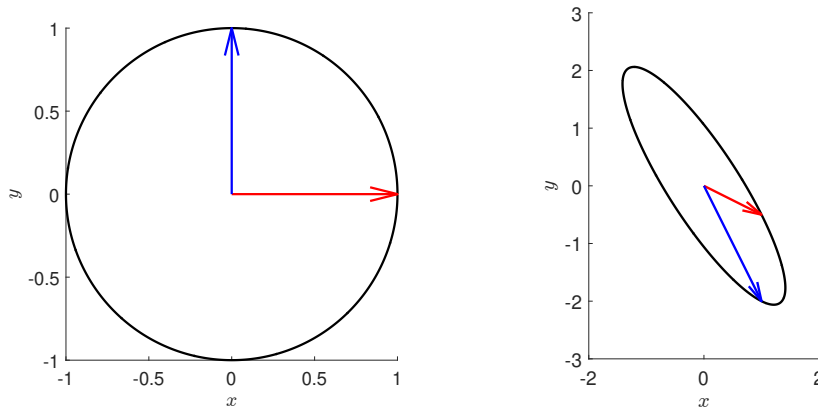


Figure A.1: A linear transformation of the unit circle. ■

Suppose $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$, where the $\mathcal{A} = \{\mathbf{a}_i\}$ form a basis of $\mathbb{R}^n$. Take any vector $\mathbf{x} = [x_1, \dots, x_n]_{\mathcal{E}}^{\top}$ with respect to the standard basis $\mathcal{E}$ (we write subscript $\mathcal{E}$ to stress this). Then the representation of this vector with respect to $\mathcal{A}$ is simply

$$\mathbf{y} = \mathbf{A}^{-1}\mathbf{x},$$

where $\mathbf{A}^{-1}$ is the *inverse* of $\mathbf{A}$; that is, the matrix such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$, where $\mathbf{I}_n$ is the n -dimensional identity matrix. To see this, note that $\mathbf{A}^{-1}\mathbf{a}_i$ gives the i -th unit vector representation, for $i = 1, \dots, n$, and recall that each vector in $\mathbb{R}^n$ is a unique linear combination of these basis vectors.

INVERSE

■ **Example A.2 (Basis Representation)** Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{with inverse} \quad \mathbf{A}^{-1} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}. \quad (\text{A.2})$$

The vector $\mathbf{x} = [1, 1]_{\mathcal{E}}^{\top}$ in the standard basis has representation $\mathbf{y} = \mathbf{A}^{-1}\mathbf{x} = [-1, 1]_{\mathcal{A}}^{\top}$ in the basis consisting of the columns of $\mathbf{A}$. Namely,

$$\mathbf{A}\mathbf{y} = -\begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

■

The *transpose* of a matrix $\mathbf{A} = [a_{ij}]$ is the matrix $\mathbf{A}^{\top} = [a_{ji}]$; that is, the (i, j) -th element of $\mathbf{A}^{\top}$ is the (j, i) -th element of $\mathbf{A}$. The *trace* of a square matrix is the sum of its diagonal elements. A useful result is the following cyclic property.

TRANSPOSE

TRACE

Theorem A.1: Cyclic Property

The trace is invariant under cyclic permutations: $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}) = \text{tr}(\mathbf{CAB})$.

Proof: It suffices to show that $\text{tr}(\mathbf{DE})$ is equal to $\text{tr}(\mathbf{ED})$ for any $m \times n$ matrix $\mathbf{D} = [d_{ij}]$ and $n \times m$ matrix $\mathbf{E} = [e_{ij}]$. The diagonal elements of $\mathbf{DE}$ are $\sum_{j=1}^n d_{ij} e_{ji}, i = 1, \dots, m$ and the diagonal elements of $\mathbf{ED}$ are $\sum_{i=1}^m e_{ji} d_{ij}, j = 1, \dots, n$. They sum up to the same number $\sum_{i=1}^m \sum_{j=1}^n d_{ij} e_{ji}$. $\square$

A square matrix has an inverse if and only if its columns (or rows) are linearly independent. This is the same as the matrix being of *full rank*; that is, its rank is equal to the number of columns. An equivalent statement is that its determinant is not zero. The *determinant* of an $n \times n$ matrix $\mathbf{A} = [a_{i,j}]$ is defined as

DETERMIN-
ANT

$$\det(\mathbf{A}) := \sum_{\pi} (-1)^{\zeta(\pi)} \prod_{i=1}^n a_{\pi_i, i}, \quad (\text{A.3})$$

where the sum is over all permutations $\pi = (\pi_1, \dots, \pi_n)$ of $(1, \dots, n)$, and $\zeta(\pi)$ is the number of pairs (i, j) for which $i < j$ and $\pi_i > \pi_j$. For example, $\zeta(2, 3, 4, 1) = 3$ for the pairs $(1, 4), (2, 4), (3, 4)$. The determinant of a *diagonal matrix* — a matrix with only zero elements off the diagonal — is simply the product of its diagonal elements.

DIAGONAL
MATRIX

Geometrically, the determinant of a square matrix $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ is the (signed) *volume* of the parallelepiped (n -dimensional parallelogram) defined by the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$; that is, the set of points $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{a}_i$, where $0 \leq \alpha_i \leq 1, i = 1, \dots, n$.

The easiest way to compute a determinant of a general matrix is to apply simple operations to the matrix that potentially reduce its complexity (as in the number of non-zero elements, for example), while retaining its determinant:

- Adding a multiple of one column (or row) to another, does not change the determinant.
- Multiplying a column (or row) with a number multiplies the determinant by the same number.
- Swapping two rows changes the sign of the determinant.

By applying these rules repeatedly one can reduce any matrix to a diagonal matrix. It follows then that the determinant of the original matrix is equal to the product of the diagonal elements of the resulting diagonal matrix multiplied by a known constant.

■ **Example A.3 (Determinant and Volume)** Figure A.2 illustrates how the determinant of a matrix can be viewed as a signed volume, which can be computed by repeatedly applying the first rule above. Here, we wish to compute the area of red parallelogram determined by the matrix $\mathbf{A}$ given in (A.2). In particular, the corner points of the parallelogram correspond to the vectors $[0, 0]^T, [1, 3]^T, [2, 4]^T$, and $[3, 7]^T$.

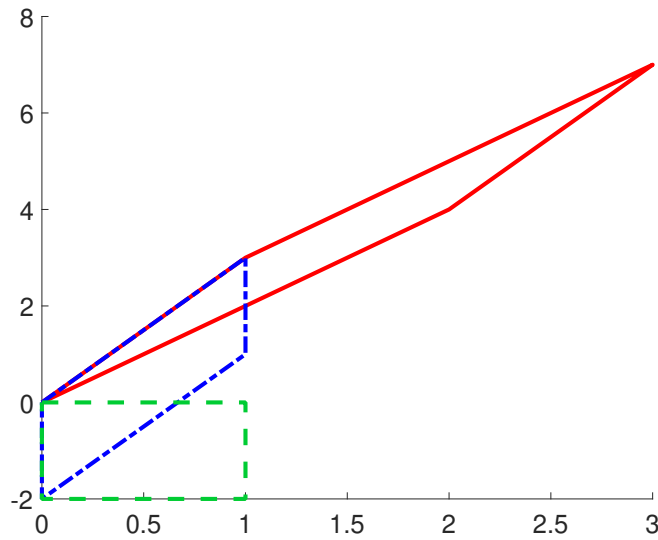


Figure A.2: The volume of the red parallelogram can be obtained by a number of shear operations that do not change the volume.

Adding -2 times the first column of $\mathbf{A}$ to the second column gives the matrix

$$\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix},$$

corresponding to the blue parallelogram. The linear operation that transforms the red to the blue parallelogram can be thought of as a succession of two linear transformations. The first is to transform the coordinates of points on the red parallelogram (in standard basis) to the basis formed by the columns of $\mathbf{A}$. Second, relative to this new basis, we apply the matrix $\mathbf{B}$ above. Note that the input of this matrix is with respect to the new basis, whereas the output is with respect to the standard basis. The matrix for the combined operation is now

$$\mathbf{BA}^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -9 & 4 \end{bmatrix},$$

which maps $[1, 3]^\top$ to $[1, 3]^\top$ (does not change) and $[2, 4]^\top$ to $[0, -2]^\top$. We say that we apply a *shear* in the direction $[1, 3]^\top$. The significance of such an operation is that a shear *does not alter the volume of the parallelogram*. The second (blue) parallelogram has an easier form, because one of the sides is parallel to the y-axis. By applying another shear, in the direction $[0, -2]^\top$, we can obtain a simple (green) rectangle, whose volume is 2. In matrix terms, we add $3/2$ times the second column of $\mathbf{B}$ to the first column of $\mathbf{B}$, to obtain the matrix

$$\mathbf{C} = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix},$$

SHEAR

which is a diagonal matrix, whose determinant is -2 , corresponding to the volume 2 of all the parallelograms. ■

Theorem A.2 summarizes a number of useful matrix rules for the concepts that we have discussed so far. We leave the proofs, which typically involves “writing out” the equations, as an exercise for the reader.

Theorem A.2: Useful Matrix Rules

1. $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$
2. $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$
3. $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1} =: \mathbf{A}^{-\top}$
4. $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$
5. $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top)$
6. $\det(\mathbf{A}) = \prod_i a_{ii}$ if $\mathbf{A} = [a_{ij}]$ is triangular

Next, consider an $n \times p$ matrix $\mathbf{A}$ for which the matrix inverse fails to exist. That is, $\mathbf{A}$ is either non-square ($n \neq p$) or its determinant is 0. Instead of the inverse, we can use its so-called pseudo-inverse, which always exists.

Definition A.2: Moore–Penrose Pseudo-Inverse

The *Moore–Penrose pseudo-inverse* of a real matrix $\mathbf{A} \in \mathbb{R}^{n \times p}$ is defined as the unique matrix $\mathbf{A}^+ \in \mathbb{R}^{p \times n}$ that satisfies the conditions:

1. $\mathbf{A} \mathbf{A}^+ \mathbf{A} = \mathbf{A}$
2. $\mathbf{A}^+ \mathbf{A} \mathbf{A}^+ = \mathbf{A}^+$
3. $(\mathbf{A} \mathbf{A}^+)^\top = \mathbf{A} \mathbf{A}^+$
4. $(\mathbf{A}^+ \mathbf{A})^\top = \mathbf{A}^+ \mathbf{A}$

We can write $\mathbf{A}^+$ explicitly in terms of $\mathbf{A}$ when $\mathbf{A}$ has a full column or row rank. For example, we always have

$$\mathbf{A}^\top \mathbf{A} \mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^+)^\top = ((\mathbf{A} \mathbf{A}^+) \mathbf{A})^\top = (\mathbf{A})^\top = \mathbf{A}^\top. \quad (\text{A.4})$$

If $\mathbf{A}$ has a full column rank p , then $(\mathbf{A}^\top \mathbf{A})^{-1}$ exists, so that from (A.4) it follows that $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$. This is referred to as the *left pseudo-inverse*, as $\mathbf{A}^+ \mathbf{A} = \mathbf{I}_p$.

MOORE–
PENROSE
PSEUDO-
INVERSE

LEFT
PSEUDO-
INVERSE

Similarly, if $\mathbf{A}$ has a full row rank n , that is, $(\mathbf{A}\mathbf{A}^\top)^{-1}$ exists, then it follows from

$$\mathbf{A}^+\mathbf{A}\mathbf{A}^\top = (\mathbf{A}^+\mathbf{A})^\top\mathbf{A}^\top = (\mathbf{A}(\mathbf{A}^+\mathbf{A}))^\top = \mathbf{A}^\top$$

that $\mathbf{A}^+ = \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}$. This is the *right pseudo-inverse*, as $\mathbf{A}\mathbf{A}^+ = \mathbf{I}_n$. Finally, if $\mathbf{A}$ is of full rank and square, then $\mathbf{A}^+ = \mathbf{A}^{-1}$.

RIGHT
PSEUDO-
INVERSE

A.2 Inner Product

The (Euclidean) *inner product* of two real vectors $\mathbf{x} = [x_1, \dots, x_n]^\top$ and $\mathbf{y} = [y_1, \dots, y_n]^\top$ is defined as the number

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i = \mathbf{x}^\top \mathbf{y}.$$

INNER
PRODUCT

Here $\mathbf{x}^\top \mathbf{y}$ is the matrix multiplication of the $(1 \times n)$ matrix $\mathbf{x}^\top$ and the $(n \times 1)$ matrix $\mathbf{y}$. The inner product induces a geometry on the linear space $\mathbb{R}^n$, allowing for the definition of length, angle, and so on. The inner product satisfies the following properties:

1. $\langle \alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$;
2. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
3. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$;
4. $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Vectors $\mathbf{x}$ and $\mathbf{y}$ are called *perpendicular* (or *orthogonal*) if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. The *Euclidean norm* (or length) of a vector $\mathbf{x}$ is defined as

ORTHO-
GONAL
EUCLIDEAN
NORM

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

If $\mathbf{x}$ and $\mathbf{y}$ are perpendicular, then *Pythagoras' theorem* holds:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2. \quad (\text{A.5})$$

Py-
THAGORAS'
THEOREM

A basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of $\mathbb{R}^n$ in which all the vectors are pairwise perpendicular and have norm 1 is called an *orthonormal* (short for orthogonal and normalized) basis. For example, the standard basis is orthonormal.

ORTHONOR-
MAL

Theorem A.3: Orthonormal Basis Representation

If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$, then any vector $\mathbf{x} \in \mathbb{R}^n$ can be expressed as

$$\mathbf{x} = \langle \mathbf{x}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \dots + \langle \mathbf{x}, \mathbf{v}_n \rangle \mathbf{v}_n. \quad (\text{A.6})$$

Proof: Observe that, because the $\{\mathbf{v}_i\}$ form a basis, there exist unique $\alpha_1, \dots, \alpha_n$ such that $\mathbf{x} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$. By the linearity of the inner product and the orthonormality of the $\{\mathbf{v}_i\}$ it follows that $\langle \mathbf{x}, \mathbf{v}_j \rangle = \langle \sum_i \alpha_i \mathbf{v}_i, \mathbf{v}_j \rangle = \alpha_j$. $\square$

An $n \times n$ matrix $\mathbf{V}$ whose columns form an orthonormal basis is called an *orthogonal matrix*.¹ Note that for an orthogonal matrix $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n]$, we have

$$\mathbf{V}^\top \mathbf{V} = \begin{bmatrix} \mathbf{v}_1^\top \\ \mathbf{v}_2^\top \\ \vdots \\ \mathbf{v}_n^\top \end{bmatrix} [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] = \begin{bmatrix} \mathbf{v}_1^\top \mathbf{v}_1 & \mathbf{v}_1^\top \mathbf{v}_2 & \dots & \mathbf{v}_1^\top \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^\top \mathbf{v}_1 & \mathbf{v}_n^\top \mathbf{v}_2 & \dots & \mathbf{v}_n^\top \mathbf{v}_n \end{bmatrix} = \mathbf{I}_n.$$

Hence, $\mathbf{V}^{-1} = \mathbf{V}^\top$. Note also that an orthogonal transformation is *length preserving*; that is, $\mathbf{V}\mathbf{x}$ has the same length as $\mathbf{x}$. This follows from

$$\|\mathbf{V}\mathbf{x}\|^2 = \langle \mathbf{V}\mathbf{x}, \mathbf{V}\mathbf{x} \rangle = \mathbf{x}^\top \mathbf{V}^\top \mathbf{V} \mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|^2.$$

A.3 Kronecker Product

The *vectorization* of a matrix $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m] \in \mathbb{R}^{n \times m}$ is the $n \times m$ column vector

$$\text{vec}(\mathbf{A}) := [\mathbf{a}_1^\top, \dots, \mathbf{a}_m^\top]^\top.$$

If $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$, then the *Kronecker product* of $\mathbf{A}$ and $\mathbf{B}$, denoted as $\mathbf{A} \otimes \mathbf{B}$ is the $np \times mq$ matrix

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{1,1}\mathbf{B} & \dots & a_{1,m}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{n,1}\mathbf{B} & \dots & a_{n,m}\mathbf{B} \end{bmatrix}.$$

The Kronecker product behaves in the same way as multiplication of two scalars, except that it is not commutative, that is, in general $\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$. Some properties of the Kronecker product and vectorization are as follows:

1. Transpose of Kronecker product:

$$(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top.$$

2. Mixed-product property:

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD})$$

for $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ of appropriate dimension.

¹The qualifier “orthogonal” for such matrices has been fixed by history. A better term would have been “orthonormal”.

ORTHO-
GONAL
MATRIX

LENGTH
PRESERVING

VECTORIZA-
TION

KRONECKER
PRODUCT

3. Vectorization in standard basis:

$$\text{vec}(\mathbf{A}) = \sum_{i=1}^n \mathbf{e}_i \otimes (\mathbf{A}\mathbf{e}_i) = \sum_{i=1}^m (\mathbf{u}_i^\top \mathbf{A}) \otimes \mathbf{u}_i,$$

where $\mathbf{e}_1, \dots, \mathbf{e}_n$ is the standard basis for $\mathbb{R}^n$, and $\mathbf{u}_1, \dots, \mathbf{u}_m$ is the standard basis for $\mathbb{R}^m$.

68

■ **Example A.4 (Pseudo-Inverse & Kronecker Product)** A number of other identities follow from the mixed-product identity. For example, if both $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{B} \in \mathbb{R}^{m \times m}$ are square and invertible, then by the mixed-product identity:

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}) = \mathbf{I}_n \otimes \mathbf{I}_m = \mathbf{I}_{m \times n}.$$

Similarly, $(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1})(\mathbf{A} \otimes \mathbf{B}) = \mathbf{I}_{m \times n}$ and hence $(\mathbf{A} \otimes \mathbf{B})^{-1} = (\mathbf{A}^{-1} \otimes \mathbf{B}^{-1})$. More generally, the mixed-product property implies that that pseudo-inverse is:

$$(\mathbf{A} \otimes \mathbf{B})^+ = \mathbf{A}^+ \otimes \mathbf{B}^+.$$

■

■ **Example A.5 (Vectorization & Trace)** Using the standard basis representation of the vectorization, we can find a relationship with the matrix trace, as follows:

$$\begin{aligned} \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B}) &= \left[\sum_i \mathbf{e}_i \otimes (\mathbf{A}\mathbf{e}_i) \right]^\top \sum_j \mathbf{e}_j \otimes (\mathbf{B}\mathbf{e}_j) \\ &= \sum_{i,j} (\mathbf{e}_i^\top \otimes (\mathbf{A}\mathbf{e}_i)^\top) (\mathbf{e}_j \otimes (\mathbf{B}\mathbf{e}_j)) \\ &= \sum_{i,j} (\mathbf{e}_i^\top \mathbf{e}_j) \otimes (\mathbf{e}_i^\top \mathbf{A}^\top \mathbf{B} \mathbf{e}_j) = \sum_{i,j} \mathbb{1}_{i=j} \otimes ([\mathbf{A}^\top \mathbf{B}]_{i,j}) \\ &= \sum_i [\mathbf{A}^\top \mathbf{B}]_{i,i} = \text{tr}(\mathbf{A}^\top \mathbf{B}). \end{aligned}$$

■

■ **Example A.6 (Vectorization & Kronecker Product)** An important relationship between vectorization and the Kronecker product is given by $\text{vec}(\mathbf{u}\mathbf{v}^\top) = \mathbf{v} \otimes \mathbf{u}$, as seen from the following argument:

$$\begin{aligned} \text{vec}(\mathbf{u}\mathbf{v}^\top) &= \sum_i \mathbf{e}_i \otimes (\mathbf{u}\mathbf{v}^\top \mathbf{e}_i) = \sum_i \mathbf{e}_i \otimes (\mathbf{u}v_i) \\ &= \sum_i \mathbf{e}_i v_i \otimes \mathbf{u} = \mathbf{v} \otimes \mathbf{u}. \end{aligned}$$

This relationship can be used to show that

$$\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B}) \quad (\text{A.7})$$

for $\mathbf{A}, \mathbf{B}, \mathbf{C}$ of appropriate dimension. To see this, write $\mathbf{B} = \sum_i \mathbf{p}_i \mathbf{q}_i^\top$ as a sum of rank one matrices. Then,

$$\begin{aligned} \text{vec}(\mathbf{ABC}) &= \sum_i \text{vec}([\mathbf{A}\mathbf{p}_i][\mathbf{q}_i^\top \mathbf{C}]) = \sum_i (\mathbf{C}^\top \mathbf{q}_i) \otimes (\mathbf{A}\mathbf{p}_i) \\ &= \sum_i (\mathbf{C}^\top \otimes \mathbf{A})(\mathbf{q}_i \otimes \mathbf{p}_i) = (\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B}). \end{aligned}$$

■

A.4 Complex Vectors and Matrices

Instead of the vector space $\mathbb{R}^n$ of n -dimensional real vectors, it is sometimes useful to consider the vector space $\mathbb{C}^n$ of n -dimensional complex vectors. In this case the *adjoint* or *conjugate transpose* operation ($*$) replaces the transpose operation ($\top$). This involves the usual transposition of the matrix or vector with the additional step that any complex number $z = x + i y$ is replaced by its complex conjugate $\bar{z} = x - i y$. For example, if

$$\mathbf{x} = \begin{bmatrix} a_1 + i b_1 \\ a_2 + i b_2 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} a_{11} + i b_{11} & a_{12} + i b_{12} \\ a_{21} + i b_{21} & a_{22} + i b_{22} \end{bmatrix},$$

then

$$\mathbf{x}^* = [a_1 - i b_1, a_2 - i b_2] \quad \text{and} \quad \mathbf{A}^* = \begin{bmatrix} a_{11} - i b_{11} & a_{21} - i b_{21} \\ a_{12} - i b_{12} & a_{22} - i b_{22} \end{bmatrix}.$$

The (Euclidean) inner product of $\mathbf{x}$ and $\mathbf{y}$ (viewed as column vectors) is now defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x} = \sum_{i=1}^n x_i \bar{y}_i,$$

which is no longer symmetric: $\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}$. Note that this generalizes the real-valued inner product. The determinant of a complex matrix $\mathbf{A}$ is defined exactly as in (A.3). As a consequence, $\det(\mathbf{A}^*) = \overline{\det(\mathbf{A})}$.

A complex matrix is said to be *Hermitian* or *self-adjoint* if $\mathbf{A}^* = \mathbf{A}$, and *unitary* if $\mathbf{A}^* \mathbf{A} = \mathbf{I}$ (that is, if $\mathbf{A}^* = \mathbf{A}^{-1}$). For real matrices “Hermitian” is the same as “symmetric”, and “unitary” is the same as “orthogonal”.

ADJOINT

HERMITIAN
UNITARY

A.5 Orthogonal Projections

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. The set

$$\mathcal{V} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = \{\alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k, \alpha_1, \dots, \alpha_k \in \mathbb{R}\},$$

is called the *linear subspace spanned by* $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. The *orthogonal complement* of $\mathcal{V}$, denoted by $\mathcal{V}^\perp$, is the set of all vectors $\mathbf{w}$ that are orthogonal to $\mathcal{V}$, in the sense that $\langle \mathbf{w}, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in \mathcal{V}$. The matrix $\mathbf{P}$ such that $\mathbf{P}\mathbf{x} = \mathbf{x}$, for all $\mathbf{x} \in \mathcal{V}$, and $\mathbf{P}\mathbf{x} = \mathbf{0}$, for all $\mathbf{x} \in \mathcal{V}^\perp$ is called the *orthogonal projection matrix* onto $\mathcal{V}$. Suppose that $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_k]$ has full rank, in which case $\mathbf{U}^\top \mathbf{U}$ is an invertible matrix. The orthogonal projection matrix $\mathbf{P}$ onto $\mathcal{V} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is then given by

$$\mathbf{P} = \mathbf{U}(\mathbf{U}^\top \mathbf{U})^{-1} \mathbf{U}^\top.$$

Namely, since $\mathbf{P}\mathbf{U} = \mathbf{U}$, the matrix $\mathbf{P}$ projects any vector in $\mathcal{V}$ onto itself. Moreover, $\mathbf{P}$ projects any vector in $\mathcal{V}^\perp$ onto the zero vector. Using the pseudo-inverse, it is possible to specify the projection matrix also for the case where $\mathbf{U}$ is not of full rank, leading to the following theorem.

LINEAR
SUBSPACE
ORTHOGONAL
COMPLEMENT
ORTHO-
GONAL
PROJECTION
MATRIX

Theorem A.4: Orthogonal Projection

Let $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_k]$. Then, the orthogonal projection matrix $\mathbf{P}$ onto $\mathcal{V} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is given by

$$\mathbf{P} = \mathbf{U} \mathbf{U}^+, \quad (\text{A.8})$$

where $\mathbf{U}^+$ is the (right) pseudo-inverse of $\mathbf{U}$.

Proof: By Property 1 of Definition A.2 we have $\mathbf{P}\mathbf{U} = \mathbf{U}\mathbf{U}^+\mathbf{U} = \mathbf{U}$, so that $\mathbf{P}$ projects any vector in $\mathcal{V}$ onto itself. Moreover, $\mathbf{P}$ projects any vector in $\mathcal{V}^\perp$ onto the zero vector. $\square$

Note that in the special case where $\mathbf{u}_1, \dots, \mathbf{u}_k$ above form an orthonormal basis of $\mathcal{V}$, then the projection onto $\mathcal{V}$ is very simple to describe, namely we have

$$\mathbf{P}\mathbf{x} = \mathbf{U}\mathbf{U}^\top \mathbf{x} = \sum_{i=1}^k \langle \mathbf{x}, \mathbf{u}_i \rangle \mathbf{u}_i. \quad (\text{A.9})$$

For any point $\mathbf{x} \in \mathbb{R}^n$, the point in $\mathcal{V}$ that is closest to $\mathbf{x}$ is its orthogonal projection $\mathbf{P}\mathbf{x}$, as the following theorem shows.

Theorem A.5: Orthogonal Projection and Minimal Distance

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthonormal basis of subspace $\mathcal{V}$ and let $\mathbf{P}$ be the orthogonal projection matrix onto $\mathcal{V}$. The solution to the minimization program

$$\min_{\mathbf{y} \in \mathcal{V}} \|\mathbf{x} - \mathbf{y}\|^2$$

is $\mathbf{y} = \mathbf{P}\mathbf{x}$. That is, $\mathbf{P}\mathbf{x} \in \mathcal{V}$ is closest to $\mathbf{x}$.

Proof: We can write each point $\mathbf{y} \in \mathcal{V}$ as $\mathbf{y} = \sum_{i=1}^k \alpha_i \mathbf{u}_i$. Consequently,

$$\|\mathbf{y} - \mathbf{x}\|^2 = \left\| \mathbf{x} - \sum_{i=1}^k \alpha_i \mathbf{u}_i, \mathbf{x} - \sum_{i=1}^k \alpha_i \mathbf{u}_i \right\|^2 = \|\mathbf{x}\|^2 - 2 \sum_{i=1}^k \alpha_i \langle \mathbf{x}, \mathbf{u}_i \rangle + \sum_{i=1}^k \alpha_i^2.$$

Minimizing this with respect to the $\{\alpha_i\}$ gives $\alpha_i = \langle \mathbf{x}, \mathbf{u}_i \rangle, i = 1, \dots, k$. In view of (A.9), the optimal $\mathbf{y}$ is thus $\mathbf{P}\mathbf{x}$. $\square$

A.6 Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be an $n \times n$ matrix. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for some number λ and non-zero vector $\mathbf{v}$, then λ is called an *eigenvalue* of $\mathbf{A}$ with *eigenvector* $\mathbf{v}$.

If $(\lambda, \mathbf{v})$ is an (eigenvalue, eigenvector) pair, the matrix $\lambda\mathbf{I} - \mathbf{A}$ maps any multiple of $\mathbf{v}$ to the zero vector. Consequently, the columns of $\lambda\mathbf{I} - \mathbf{A}$ are linearly *dependent*, and hence its determinant is 0. This provides a way to identify the eigenvalues, namely as the $r \leq n$ different roots $\lambda_1, \dots, \lambda_r$ of the *characteristic polynomial*

$$\det(\lambda\mathbf{I} - \mathbf{A}) = (\lambda - \lambda_1)^{\alpha_1} \cdots (\lambda - \lambda_r)^{\alpha_r},$$

where $\alpha_1 + \cdots + \alpha_r = n$. The integer α_i is called the *algebraic multiplicity* of λ_i . The eigenvectors that correspond to an eigenvalue λ_i lie in the *kernel* or *null space* of the matrix $\lambda_i\mathbf{I} - \mathbf{A}$; that is, the linear space of vectors $\mathbf{v}$ such that $(\lambda_i\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$. This space is called the *eigenspace* of λ_i . Its dimension, $d_i \in \{1, \dots, n\}$, is called the *geometric multiplicity* of λ_i . It always holds that $d_i \leq \alpha_i$. If $\sum_i d_i = n$, then we can construct a basis for $\mathbb{R}^n$ consisting of eigenvectors, as illustrated next.

■ **Example A.7 (Linear Transformation (cont.))** We revisit the linear transformation in Figure A.1, where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1/2 & -2 \end{bmatrix}.$$

The characteristic polynomial is $(\lambda - 1)(\lambda + 2) + 1/2$, with roots $\lambda_1 = -1/2 - \sqrt{7}/2 \approx -1.8229$ and $\lambda_2 = -1/2 + \sqrt{7}/2 \approx 0.8229$. The corresponding unit eigenvectors are

EIGENVALUE
EIGENVECTOR

CHARACTER-
ISTIC
POLYNOMIAL

ALGEBRAIC
MULTIPLI-
CITY
NULL SPACE

GEOMETRIC
MULTIPLI-
CITY

$\mathbf{v}_1 \approx [0.3339, -0.9426]^\top$ and $\mathbf{v}_2 \approx [0.9847, -0.1744]^\top$. The eigenspace corresponding to λ_1 is $\mathcal{V}_1 = \text{Span}\{\mathbf{v}_1\} = \{\beta \mathbf{v}_1 : \beta \in \mathbb{R}\}$ and the eigenspace corresponding to λ_2 is $\mathcal{V}_2 = \text{Span}\{\mathbf{v}_2\}$. The algebraic and geometric multiplicities are 1 in this case. Any pair of vectors taken from $\mathcal{V}_1$ and $\mathcal{V}_2$ forms a basis for $\mathbb{R}^2$. Figure A.3 shows how $\mathbf{v}_1$ and $\mathbf{v}_2$ are transformed to $\mathbf{A}\mathbf{v}_1 \in \mathcal{V}_1$ and $\mathbf{A}\mathbf{v}_2 \in \mathcal{V}_2$, respectively.

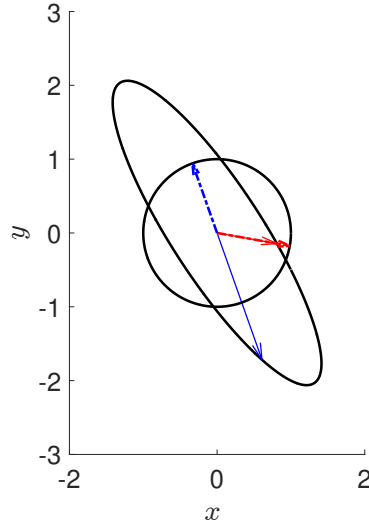


Figure A.3: The dashed arrows are the unit eigenvectors $\mathbf{v}_1$ (blue) and $\mathbf{v}_2$ (red) of matrix $\mathbf{A}$. Their transformed values $\mathbf{A}\mathbf{v}_1$ and $\mathbf{A}\mathbf{v}_2$ are indicated by solid arrows.

A matrix for which the algebraic and geometric multiplicities of all its eigenvalues are the same is called *semi-simple*. This is equivalent to the matrix being *diagonalizable*, meaning that there is a matrix $\mathbf{V}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{VDV}^{-1}.$$

SEMI-SIMPLE
 DIAGONALIZABLE

To see that this so-called *eigen-decomposition* holds, suppose $\mathbf{A}$ is a semi-simple matrix with eigenvalues

$$\underbrace{\lambda_1, \dots, \lambda_1}_{d_1}, \dots, \underbrace{\lambda_r, \dots, \lambda_r}_{d_r}.$$

EIGEN-
 DECOMPOSITION

Let $\mathbf{D}$ be the diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$, and let $\mathbf{V}$ be a matrix whose columns are linearly independent eigenvectors corresponding to these eigenvalues. Then, for each (eigenvalue, eigenvector) pair $(\lambda, \mathbf{v})$, we have $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. Hence, in matrix notation, we have $\mathbf{AV} = \mathbf{VD}$, and so $\mathbf{A} = \mathbf{VDV}^{-1}$.

A.6.1 Left- and Right-Eigenvectors

The eigenvector as defined in the previous section is called a *right*-eigenvector, as it lies on the right of $\mathbf{A}$ in the equation $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.

If $\mathbf{A}$ is a complex matrix with an eigenvalue λ , then the eigenvalue's complex conjugate $\bar{\lambda}$ is an eigenvalue of $\mathbf{A}^*$. To see this, define $\mathbf{B} := \lambda \mathbf{I} - \mathbf{A}$ and $\mathbf{B}^* := \bar{\lambda} \mathbf{I} - \mathbf{A}^*$. Since λ is an eigenvalue, we have $\det(\mathbf{B}) = 0$. Applying the identity $\det(\mathbf{B}) = \overline{\det(\mathbf{B}^*)}$, we see that therefore $\det(\mathbf{B}^*) = 0$, and hence that $\bar{\lambda}$ is an eigenvalue of $\mathbf{A}^*$. Let $\mathbf{w}$ be an eigenvector corresponding to $\bar{\lambda}$. Then, $\mathbf{A}^* \mathbf{w} = \bar{\lambda} \mathbf{w}$ or, equivalently,

$$\mathbf{w}^* \mathbf{A} = \lambda \mathbf{w}^*.$$

LEFT-
EIGENVECTOR

NORMAL
MATRIX

For this reason, we call $\mathbf{w}^*$ the *left-eigenvector* of $\mathbf{A}$ for eigenvalue λ . If $\mathbf{v}$ is a (right-) eigenvector of $\mathbf{A}$, then its adjoint $\mathbf{v}^*$ is usually *not* a left-eigenvector, unless $\mathbf{A}^* \mathbf{A} = \mathbf{A} \mathbf{A}^*$ (such matrices are called *normal*; a real symmetric matrix is normal). However, the important property holds that left- and right-eigenvectors belonging to *different* eigenvalues are *orthogonal*. Namely, if $\mathbf{w}^*$ is a left-eigenvector of λ_1 and $\mathbf{v}$ a right-eigenvector of $\lambda_2 \neq \lambda_1$, then

$$\lambda_1 \mathbf{w}^* \mathbf{v} = \mathbf{w}^* \mathbf{A} \mathbf{v} = \lambda_2 \mathbf{w}^* \mathbf{v},$$

which can only be true if $\mathbf{w}^* \mathbf{v} = 0$.

Theorem A.6: Schur Triangulation

For any complex matrix $\mathbf{A}$, there exists a unitary matrix $\mathbf{U}$ such that $\mathbf{T} = \mathbf{U}^{-1} \mathbf{A} \mathbf{U}$ is upper triangular.

Proof: The proof is by induction on the dimension n of the matrix. Clearly, the statement is true for $n = 1$, as $\mathbf{A}$ is simply a complex number and we can take $\mathbf{U}$ equal to 1. Suppose that the result is true for dimension n . We wish to show that it also holds for dimension $n + 1$. Any matrix $\mathbf{A}$ always has at least one eigenvalue λ with eigenvector $\mathbf{v}$, normalized to have length 1. Let $\mathbf{U}$ be any unitary matrix whose first column is $\mathbf{v}$. Such a matrix can always be constructed². As $\mathbf{U}$ is unitary, the first row of $\mathbf{U}^{-1}$ is $\mathbf{v}^*$, and $\mathbf{U}^{-1} \mathbf{A} \mathbf{U}$ is of the form

$$\begin{bmatrix} \mathbf{v}^* \\ * \end{bmatrix} \mathbf{A} \underbrace{\begin{bmatrix} \mathbf{v} & * \end{bmatrix}}_{\mathbf{U}} = \begin{bmatrix} \lambda & * \\ \mathbf{0} & \mathbf{B} \end{bmatrix},$$

for some matrix $\mathbf{B}$. By the induction hypothesis, there exists a unitary matrix $\mathbf{W}$ and an upper triangular matrix $\mathbf{T}$ such that $\mathbf{W}^{-1} \mathbf{B} \mathbf{W} = \mathbf{T}$. Now, define

$$\mathbf{V} := \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{W} \end{bmatrix}.$$

²After specifying $\mathbf{v}$ we can complete the rest of the unitary matrix via the Gram–Schmidt procedure, for example; see Section A.7.4.

Then,

$$\mathbf{V}^{-1}(\mathbf{U}^{-1}\mathbf{A}\mathbf{U})\mathbf{V} = \left[\begin{array}{c|c} 1 & \mathbf{0}^\top \\ \hline \mathbf{0} & \mathbf{W}^{-1} \end{array} \right] \left[\begin{array}{c|c} \lambda & * \\ \hline \mathbf{0} & \mathbf{B} \end{array} \right] \left[\begin{array}{c|c} 1 & \mathbf{0}^\top \\ \hline \mathbf{0} & \mathbf{W} \end{array} \right] = \left[\begin{array}{c|c} \lambda & * \\ \hline \mathbf{0} & \mathbf{W}^{-1}\mathbf{B}\mathbf{W} \end{array} \right] = \left[\begin{array}{c|c} \lambda & * \\ \hline \mathbf{0} & \mathbf{T} \end{array} \right],$$

which is upper triangular of dimension $n + 1$. Since $\mathbf{UV}$ is unitary, this completes the induction, and hence the result is true for all n . $\square$

The theorem above can be used to prove an important property of *Hermitian* matrices, i.e., matrices for which $\mathbf{A}^* = \mathbf{A}$.

Theorem A.7: Eigenvalues of a Hermitian Matrix

Any $n \times n$ Hermitian matrix has real eigenvalues. The corresponding matrix of normalized eigenvectors is a unitary matrix.

Proof: Let $\mathbf{A}$ be a Hermitian matrix. By Theorem A.6 there exists a unitary matrix $\mathbf{U}$ such that $\mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \mathbf{T}$, where $\mathbf{T}$ is upper triangular. It follows that the adjoint $(\mathbf{U}^{-1}\mathbf{A}\mathbf{U})^* = \mathbf{T}^*$ is lower triangular. However, $(\mathbf{U}^{-1}\mathbf{A}\mathbf{U})^* = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$, since $\mathbf{A}^* = \mathbf{A}$ and $\mathbf{U}^* = \mathbf{U}^{-1}$. Hence, $\mathbf{T}$ and $\mathbf{T}^*$ must be the same, which can only be the case if $\mathbf{T}$ is a *real diagonal* matrix $\mathbf{D}$. Since $\mathbf{A}\mathbf{U} = \mathbf{D}\mathbf{U}$, the diagonal elements are exactly the eigenvalues and the corresponding eigenvectors are the columns of $\mathbf{U}$. $\square$

In particular, the eigenvalues of a real symmetric matrix are real. We can now repeat the proof of Theorem A.6 with real eigenvalues and eigenvectors, so that there exists an orthogonal matrix $\mathbf{Q}$ such that $\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{Q}^\top\mathbf{A}\mathbf{Q} = \mathbf{D}$. The eigenvectors can be chosen as the columns of $\mathbf{Q}$, which form an orthonormal basis. This proves the following theorem.

Theorem A.8: Real Symmetric Matrices are Orthogonally Diagonalizable

Any real symmetric matrix $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top,$$

where $\mathbf{D}$ is the diagonal matrix of (real) eigenvalues and $\mathbf{Q}$ is an orthogonal matrix whose columns are eigenvectors of $\mathbf{A}$.

■ **Example A.8 (Real Symmetric Matrices and Ellipses)** As we have seen, linear transformations map circles into ellipses. We can use the above theory for real symmetric matrices to identify the principal axes. Consider, for example, the transformation with matrix $\mathbf{A} = [1, 1; -1/2, -2]$ in (A.1). A point $\mathbf{x}$ on the unit circle is mapped to a point $\mathbf{y} = \mathbf{A}\mathbf{x}$. Since for such points $\|\mathbf{x}\|^2 = \mathbf{x}^\top\mathbf{x} = 1$, we have that $\mathbf{y}$ satisfies

$\mathbf{y}^\top (\mathbf{A}^{-1})^\top \mathbf{A}^{-1} \mathbf{y} = 1$, which gives the equation for the ellipse

$$\frac{17 y_1^2}{9} + \frac{20 y_1 y_2}{9} + \frac{8 y_2^2}{9} = 1.$$

Let $\mathbf{Q}$ be the orthogonal matrix of eigenvectors of the symmetric matrix $(\mathbf{A}^{-1})^\top \mathbf{A}^{-1} = (\mathbf{A}\mathbf{A}^\top)^{-1}$, so $\mathbf{Q}^\top (\mathbf{A}\mathbf{A}^\top)^{-1} \mathbf{Q} = \mathbf{D}$ for some diagonal matrix $\mathbf{D}$. Taking the inverse on both sides of the previous equation, we have $\mathbf{Q}^\top \mathbf{A}\mathbf{A}^\top \mathbf{Q} = \mathbf{D}^{-1}$, which shows that $\mathbf{Q}$ is also the matrix of eigenvectors of $\mathbf{A}\mathbf{A}^\top$. These eigenvectors point precisely in the direction of the principal axes, as shown in Figure A.4. It turns out, see Section A.7.5, that the square roots of the eigenvalues of $\mathbf{A}\mathbf{A}^\top$, here approximately 2.4221 and 0.6193, correspond to the sizes of the principal axes of the ellipse, as illustrated in Figure A.4.

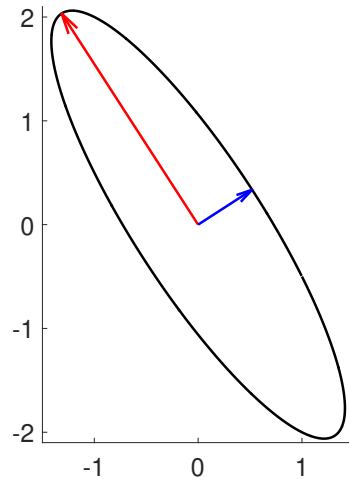


Figure A.4: The eigenvectors and eigenvalues of $\mathbf{A}\mathbf{A}^\top$ determine the principal axes of the ellipse. ■

■ **Example A.9 (Eigenvalues and Kronecker Product)** Suppose that we have the eigenvalue decompositions $\mathbf{A}_1 = \mathbf{Q}_1 \mathbf{D}_1 \mathbf{Q}_1^\top$ and $\mathbf{A}_2 = \mathbf{Q}_2 \mathbf{D}_2 \mathbf{Q}_2^\top$, where $\lambda_{1,j}$, $j = 1, \dots, n_1$ and $\lambda_{2,j}$, $j = 1, \dots, n_2$ are the eigenvalues of $\mathbf{A}_1$ and $\mathbf{A}_2$, respectively. Then, the $n_1 \times n_2$ eigenvalues of the Kronecker product $\mathbf{A}_1 \otimes \mathbf{A}_2$ are $\lambda_{1,i} \times \lambda_{2,j}$ for $i = 1, \dots, n_1$ and $j = 1, \dots, n_2$. To see this note that repeated application of the mixed-product identity yields

$$\begin{aligned} (\mathbf{A}_1 \otimes \mathbf{A}_2)(\mathbf{Q}_1 \otimes \mathbf{Q}_2) &= (\mathbf{A}_1 \mathbf{Q}_1) \otimes (\mathbf{A}_2 \mathbf{Q}_2) \\ &= (\mathbf{Q}_1 \mathbf{D}_1) \otimes (\mathbf{Q}_2 \mathbf{D}_2) \\ &= (\mathbf{Q}_1 \otimes \mathbf{Q}_2)(\mathbf{D}_1 \otimes \mathbf{D}_2). \end{aligned}$$

Hence, the $n_1 \times n_2$ diagonal matrix $\mathbf{D}_1 \otimes \mathbf{D}_2$ contains all the eigenvalues. This implies,

for example, that

$$\begin{aligned}\det(\mathbf{A}_1 \otimes \mathbf{A}_2) &= \det(\mathbf{D}_1 \otimes \mathbf{D}_2) = \prod_i \det(\lambda_{1,i} \mathbf{D}_2) \\ &= \det(\mathbf{D}_1)^{n_2} \det(\mathbf{D}_2)^{n_1} = \det(\mathbf{A}_1)^{n_2} \det(\mathbf{A}_2)^{n_1}.\end{aligned}$$

■

The following definition generalizes the notion of positivity of a real variable to that of a (Hermitian) matrix.

Definition A.3: Positive (Semi)Definite Matrix

A Hermitian matrix $\mathbf{A}$ is called *positive semidefinite* (we write $\mathbf{A} \geq 0$) if $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$ for all $\mathbf{x}$. It is called *positive definite* (we write $\mathbf{A} > 0$) if $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

POSITIVE
SEMIDEFIN-
ITE

The positive (semi)definiteness of a matrix can be directly related to the positivity of its eigenvalues, as follows:

Theorem A.9: Eigenvalues of a Positive Semidefinite Matrix

All eigenvalues of a positive semidefinite matrix are non-negative and all eigenvalues of a positive definite matrix are strictly positive.

Proof: Let $\mathbf{A}$ be a positive semidefinite matrix. By Theorem A.7, the eigenvalues of $\mathbf{A}$ are all *real*. Suppose λ is an eigenvalue with eigenvector $\mathbf{v}$. As $\mathbf{A}$ is positive semidefinite, we have

$$0 \leq \langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle = \lambda \langle \mathbf{v}, \mathbf{v} \rangle = \lambda \|\mathbf{v}\|^2,$$

which can only be true if $\lambda \geq 0$. Similarly, for a positive definite matrix, λ must be strictly greater than 0. $\square$

Corollary A.1 Any real positive semidefinite matrix $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{B}\mathbf{B}^\top$$

for some real matrix $\mathbf{B}$. Conversely, for any real matrix $\mathbf{B}$, the matrix $\mathbf{B}\mathbf{B}^\top$ is positive semidefinite.

Proof: The matrix $\mathbf{A}$ is both Hermitian (by definition) and real (by assumption) and hence it is symmetric. By Theorem A.8, we can write $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$, where $\mathbf{D}$ is the diagonal matrix of (real) eigenvalues of $\mathbf{A}$. By Theorem A.9 all eigenvalues are non-negative, and thus their square root is real-valued. Now, define $\mathbf{B} = \mathbf{Q}\sqrt{\mathbf{D}}$, where

$\sqrt{\mathbf{D}}$ is defined as the diagonal matrix whose diagonal elements are the *square roots* of the eigenvalues of $\mathbf{A}$. Then, $\mathbf{B}\mathbf{B}^\top = \mathbf{Q}\sqrt{\mathbf{D}}(\sqrt{\mathbf{D}})^\top\mathbf{Q}^\top = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top = \mathbf{A}$. The converse statement follows from the fact that $\mathbf{x}^\top\mathbf{B}\mathbf{B}^\top\mathbf{x} = \|\mathbf{B}^\top\mathbf{x}\|^2 \geq 0$ for all $\mathbf{x}$. $\square$

A.7 Matrix Decompositions

Matrix decompositions are frequently used in linear algebra to simplify proofs, avoid numerical instability, and to speed up computations. We mention three important matrix decompositions: (P)LU, QR, and SVD.

A.7.1 (P)LU Decomposition

Every invertible matrix $\mathbf{A}$ can be written as the product of three matrices:

$$\mathbf{A} = \mathbf{P}\mathbf{L}\mathbf{U}, \quad (\text{A.10})$$

where $\mathbf{L}$ is a lower triangular matrix, $\mathbf{U}$ an upper triangular matrix, and $\mathbf{P}$ a *permutation matrix*. A permutation matrix is a square matrix with a single 1 in each row and column, and zeros otherwise. The matrix product $\mathbf{P}\mathbf{B}$ simply permutes the rows of a matrix $\mathbf{B}$ and, likewise, $\mathbf{B}\mathbf{P}$ permutes its columns. A decomposition of the form (A.10) is called a *PLU decomposition*. As a permutation matrix is orthogonal, its transpose is equal to its inverse, and so we can write (A.10) as

$$\mathbf{P}^\top\mathbf{A} = \mathbf{L}\mathbf{U}.$$

The decomposition is not unique, and in many cases $\mathbf{P}$ can be taken to be the identity matrix, in which case we speak of the *LU decomposition* of $\mathbf{A}$, also called the LR for left–right (triangular) decomposition.

A PLU decomposition of an invertible $n \times n$ matrix $\mathbf{A}_0$ can be obtained recursively as follows. The first step is to swap the rows of $\mathbf{A}_0$ such that the element in the first column and first row of the pivoted matrix is as large as possible in absolute value. Write the resulting matrix as

$$\tilde{\mathbf{P}}_0\mathbf{A}_0 = \begin{bmatrix} a_1 & \mathbf{b}_1^\top \\ \mathbf{c}_1 & \mathbf{D}_1 \end{bmatrix},$$

where $\tilde{\mathbf{P}}_0$ is the permutation matrix that swaps the first and k -th row, where k is the row that contains the largest element in the first column. Next, add the matrix $-\mathbf{c}_1[1, \mathbf{b}_1^\top/a_1]$ to the last $n - 1$ rows of $\tilde{\mathbf{P}}_0\mathbf{A}_0$, to obtain the matrix

$$\begin{bmatrix} a_1 & \mathbf{b}_1^\top \\ \mathbf{0} & \mathbf{D}_1 - \mathbf{c}_1\mathbf{b}_1^\top/a_1 \end{bmatrix} =: \begin{bmatrix} a_1 & \mathbf{b}_1^\top \\ \mathbf{0} & \mathbf{A}_1 \end{bmatrix}.$$

PERMUTA-
TION MATRIX

PLU DECOM-
POSITION

In effect, we add some multiple of the first row to each of the remaining rows in order to obtain zeros in the first column, except for the first element.

We now apply the same procedure to $\mathbf{A}_1$ as we did to $\mathbf{A}_0$ and then to subsequent smaller matrices $\mathbf{A}_2, \dots, \mathbf{A}_{n-1}$:

1. Swap the first row with the row having the maximal absolute value element in the first column.
2. Make every other element in the first column equal to 0 by adding appropriate multiples of the first row to the other rows.

Suppose that $\mathbf{A}_t$ has a PLU decomposition $\mathbf{P}_t \mathbf{L}_t \mathbf{U}_t$. Then it is easy to check that

$$\underbrace{\widetilde{\mathbf{P}}_{t-1}^\top \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{P}_t \end{bmatrix}}_{\mathbf{P}_{t-1}} \underbrace{\begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{P}_t^\top \mathbf{c}_t / a_t & \mathbf{L}_t \end{bmatrix}}_{\mathbf{L}_{t-1}} \underbrace{\begin{bmatrix} a_t & \mathbf{b}_t^\top \\ \mathbf{0} & \mathbf{U}_t \end{bmatrix}}_{\mathbf{U}_{t-1}} \quad (\text{A.11})$$

is a PLU decomposition of $\mathbf{A}_{t-1}$. Since the PLU decomposition for the scalar $\mathbf{A}_{n-1}$ is trivial, by working backwards we obtain a PLU decomposition $\mathbf{P}_0 \mathbf{L}_0 \mathbf{U}_0$ of $\mathbf{A}$.

■ **Example A.10 (PLU Decomposition)** Take

$$\mathbf{A} = \begin{bmatrix} 0 & -1 & 7 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

Our goal is to modify $\mathbf{A}$ via Steps 1 and 2 above so as to obtain an upper triangular matrix with maximal elements on the diagonal. We first swap the first and second row. Next, we add $-1/3$ times the first row to the third row and $1/3$ times the second row to the third row:

$$\begin{bmatrix} 0 & -1 & 7 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 2 & 0 \\ 0 & -1 & 7 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 2 & 0 \\ 0 & -1 & 7 \\ 0 & 1/3 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 2 & 0 \\ 0 & -1 & 7 \\ 0 & 0 & 10/3 \end{bmatrix}.$$

The final matrix is $\mathbf{U}_0$, and in the process we have applied the permutation matrices

$$\widetilde{\mathbf{P}}_0 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \widetilde{\mathbf{P}}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Using the recursion (A.11) we can now recover $\mathbf{P}_0$ and $\mathbf{L}_0$. Namely, at the final iteration we have $\mathbf{P}_2 = \mathbf{I}$, $\mathbf{L}_2 = \mathbf{I}$, and $\mathbf{U}_2 = 10/3$. And subsequently,

$$\mathbf{P}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{L}_1 = \begin{bmatrix} 1 & 0 \\ -1/3 & 1 \end{bmatrix}, \quad \mathbf{P}_0 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{L}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/3 & -1/3 & 1 \end{bmatrix},$$

observing that $a_1 = 3$, $\mathbf{c}_1 = [0, 1]^\top$, $a_2 = -1$, and $\mathbf{c}_2 = 1/3$. ■

PLU decompositions can be used to solve large systems of linear equations of the form $\mathbf{Ax} = \mathbf{b}$ efficiently, especially when such an equation has to be solved for many different $\mathbf{b}$. This is done by first decomposing $\mathbf{A}$ into $\mathbf{PLU}$, and then solving two triangular systems:

1. $\mathbf{Ly} = \mathbf{P}^\top \mathbf{b}$.

2. $\mathbf{Ux} = \mathbf{y}$.

The first equation can be solved efficiently via *forward substitution*, and the second via *backward substitution*, as illustrated in the following example.

■ **Example A.11 (Solving Linear Equations with an LU Decomposition)** Let $\mathbf{A} = \mathbf{PLU}$ be the same as in Example A.10. We wish to solve $\mathbf{Ax} = [1, 2, 3]^\top$. First, solving

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/3 & -1/3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

gives, $y_1 = 2$, $y_2 = 1$ and $y_3 = 3 - 2/3 + 1/3 = 8/3$, by forward substitution. Next,

$$\begin{bmatrix} 3 & 2 & 0 \\ 0 & -1 & 7 \\ 0 & 0 & 10/3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 8/3 \end{bmatrix}$$

gives $x_3 = 4/5$, $x_2 = -1 + 28/5 = 23/5$, and $x_1 = 2(1 - 23/5)/3 = -12/5$, so $\mathbf{x} = [-12, 23, 4]^\top/5$. ■

A.7.2 Woodbury Identity

LU (or more generally PLU) decompositions can also be applied to block matrices. A starting point is the following LU decomposition for a general 2×2 matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & d - bc/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix},$$

which holds as long as $a \neq 0$; this can be seen by simply writing out the matrix product. The block matrix generalization for matrices $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times k}$, $\mathbf{C} \in \mathbb{R}^{k \times n}$, $\mathbf{D} \in \mathbb{R}^{k \times k}$ is

$$\Sigma := \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{O}_{n \times k} \\ \mathbf{C} & \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O}_{k \times n} & \mathbf{I}_k \end{bmatrix}, \quad (\text{A.12})$$

provided that $\mathbf{A}$ is invertible (again, write out the block matrix product). Here, we use the notation $\mathbf{O}_{p \times q}$ to denote the $p \times q$ matrix of zeros. We can further rewrite this as:

$$\Sigma = \begin{bmatrix} \mathbf{I}_n & \mathbf{O}_{n \times k} \\ \mathbf{CA}^{-1} & \mathbf{I}_k \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{O}_{n \times k} \\ \mathbf{O}_{k \times n} & \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O}_{k \times n} & \mathbf{I}_k \end{bmatrix}.$$

FORWARD
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BACKWARD
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Thus, inverting both sides, we obtain

$$\Sigma^{-1} = \begin{bmatrix} \mathbf{I}_n & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O}_{k \times n} & \mathbf{I}_k \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{A} & \mathbf{O}_{n \times k} \\ \mathbf{O}_{k \times n} & \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{I}_n & \mathbf{O}_{n \times k} \\ \mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_k \end{bmatrix}^{-1}.$$

Inversion of the above block matrices gives (again write out)

$$\Sigma^{-1} = \begin{bmatrix} \mathbf{I}_n & -\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O}_{k \times n} & \mathbf{I}_k \end{bmatrix} \begin{bmatrix} \mathbf{A}^{-1} & \mathbf{O}_{n \times k} \\ \mathbf{O}_{k \times n} & (\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{O}_{n \times k} \\ -\mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_k \end{bmatrix}. \quad (\text{A.13})$$

Assuming that $\mathbf{D}$ is invertible, we could also perform a block UL (as opposed to LU) decomposition:

$$\Sigma = \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{B} \\ \mathbf{O}_{k \times n} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{O}_{n \times k} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_k \end{bmatrix}, \quad (\text{A.14})$$

which, after a similar calculation as the one above, yields

$$\Sigma^{-1} = \begin{bmatrix} \mathbf{I}_n & \mathbf{O}_{n \times k} \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_k \end{bmatrix} \begin{bmatrix} (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} & \mathbf{O}_{n \times k} \\ \mathbf{O}_{k \times n} & \mathbf{D}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & -\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O}_{k \times n} & \mathbf{I}_k \end{bmatrix}. \quad (\text{A.15})$$

The upper-left block of Σ^{-1} from (A.15) must be the same as the upper-left block of Σ^{-1} from (A.13), leading to the *Woodbury identity*:

$$(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} = \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{C}\mathbf{A}^{-1}. \quad (\text{A.16})$$

From (A.12) and the fact that the determinant of a product is the product of the determinants, we see that $\det(\Sigma) = \det(\mathbf{A}) \det(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})$. Similarly, from (A.14) we have $\det(\Sigma) = \det(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}) \det(\mathbf{D})$, leading to the identity

$$\det(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}) \det(\mathbf{D}) = \det(\mathbf{A}) \det(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}). \quad (\text{A.17})$$

The following special cases of (A.17) and (A.16) are of particular importance.

Theorem A.10: Sherman–Morrison Formula

Suppose that $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then,

$$\det(\mathbf{A} + \mathbf{x}\mathbf{y}^\top) = \det(\mathbf{A})(1 + \mathbf{y}^\top \mathbf{A}^{-1} \mathbf{x}).$$

If in addition $\mathbf{y}^\top \mathbf{A}^{-1} \mathbf{x} \neq -1$, then the *Sherman–Morrison formula* holds:

$$(\mathbf{A} + \mathbf{x}\mathbf{y}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top \mathbf{A}^{-1} \mathbf{x}}.$$

WOODBURY
IDENTITY

SHERMAN–
MORRISON
FORMULA

Proof: Take $\mathbf{B} = \mathbf{x}$, $\mathbf{C} = -\mathbf{y}^\top$, and $\mathbf{D} = 1$ in (A.17) and (A.16). $\square$

A.7.3 Cholesky Decomposition

If $\mathbf{A}$ is a real-valued positive definite matrix (and therefore symmetric), e.g., a covariance matrix, then an LU decomposition can be achieved with matrices $\mathbf{L}$ and $\mathbf{U} = \mathbf{L}^\top$.

Theorem A.11: Cholesky Decomposition

A real-valued positive definite matrix $\mathbf{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$ can be decomposed as

$$\mathbf{A} = \mathbf{L}\mathbf{L}^\top,$$

where the real $n \times n$ lower triangular matrix $\mathbf{L} = [l_{kj}]$ satisfies the recursive formula

$$l_{kj} = \frac{a_{kj} - \sum_{i=1}^{j-1} l_{ji} l_{ki}}{\sqrt{a_{jj} - \sum_{i=1}^{j-1} l_{ji}^2}}, \quad \text{where} \quad \sum_{i=1}^0 l_{ji} l_{ki} := 0 \quad (\text{A.18})$$

for $k = 1, \dots, n$ and $j = 1, \dots, k$.

Proof: The proof is by inductive construction. For $k = 1, \dots, n$, let $\mathbf{A}_k$ be the left-upper $k \times k$ submatrix of $\mathbf{A} = \mathbf{A}_n$. With $\mathbf{e}_1 := [1, 0, \dots, 0]^\top$, we have $\mathbf{A}_1 = a_{11} = \mathbf{e}_1^\top \mathbf{A} \mathbf{e}_1 > 0$ by the positive-definiteness of $\mathbf{A}$. It follows that $l_{11} = \sqrt{a_{11}}$. Suppose that $\mathbf{A}_{k-1}$ has a Cholesky factorization $\mathbf{L}_{k-1} \mathbf{L}_{k-1}^\top$ with $\mathbf{L}_{k-1}$ having strictly positive diagonal elements, we can construct a Cholesky factorization of $\mathbf{A}_k$ as follows. First write

$$\mathbf{A}_k = \begin{bmatrix} \mathbf{L}_{k-1} \mathbf{L}_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix}$$

and propose $\mathbf{L}_k$ to be of the form

$$\mathbf{L}_k = \begin{bmatrix} \mathbf{L}_{k-1} & \mathbf{0} \\ \mathbf{l}_{k-1}^\top & l_{kk} \end{bmatrix}$$

for some vector $\mathbf{l}_{k-1} \in \mathbb{R}^{k-1}$ and scalar l_{kk} , for which it must hold that

$$\begin{bmatrix} \mathbf{L}_{k-1} \mathbf{L}_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_{k-1} & \mathbf{0} \\ \mathbf{l}_{k-1}^\top & l_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{L}_{k-1}^\top & \mathbf{l}_{k-1} \\ \mathbf{0}^\top & l_{kk} \end{bmatrix}.$$

To establish that such an $\mathbf{l}_{k-1}$ and l_{kk} exist, we must verify that the set of equations

$$\begin{aligned} \mathbf{L}_{k-1} \mathbf{l}_{k-1} &= \mathbf{a}_{k-1} \\ \mathbf{l}_{k-1}^\top \mathbf{l}_{k-1} + l_{kk}^2 &= a_{kk} \end{aligned} \quad (\text{A.19})$$

has a solution. The system $\mathbf{L}_{k-1} \mathbf{l}_{k-1} = \mathbf{a}_{k-1}$ has a unique solution, because (by assumption) $\mathbf{L}_{k-1}$ is lower diagonal with strictly positive entries down the main diagonal and we can solve for $\mathbf{l}_{k-1}$ using forward substitution: $\mathbf{l}_{k-1} = \mathbf{L}_{k-1}^{-1} \mathbf{a}_{k-1}$. We can solve

the second equation as $l_{kk} = \sqrt{a_{kk} - \|\mathbf{l}_{k-1}\|^2}$, provided that the term within the square root is positive. We demonstrate this using the fact that $\mathbf{A}$ is a positive definite matrix. In particular, for $\mathbf{x} \in \mathbb{R}^n$ of the form $[\mathbf{x}_1^\top, x_2, \mathbf{0}^\top]^\top$, where $\mathbf{x}_1$ is a non-zero $(k-1)$ -dimensional vector and x_2 a non-zero number, we have

$$0 < \mathbf{x}^\top \mathbf{A} \mathbf{x} = [\mathbf{x}_1^\top, x_2] \begin{bmatrix} \mathbf{L}_{k-1} \mathbf{L}_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} [\mathbf{x}_1, x_2] = \|\mathbf{L}_{k-1}^\top \mathbf{x}_1\|^2 + 2\mathbf{x}_1^\top \mathbf{a}_{k-1} x_2 + a_{kk} x_2^2.$$

Now take $\mathbf{x}_1 = -x_2 \mathbf{L}_{k-1}^{-\top} \mathbf{l}_{k-1}$ to obtain $0 < \mathbf{x}^\top \mathbf{A} \mathbf{x} = x_2^2 (a_{kk} - \|\mathbf{l}_{k-1}\|^2)$. Therefore, (A.19) can be uniquely solved. As we have already solved it for $k = 1$, we can solve it for any $k = 1, \dots, n$, leading to the recursive formula (A.18) and Algorithm A.7.1 below. $\square$

An implementation of Cholesky's decomposition that uses the notation in the proof of Theorem A.7.3 is the following algorithm, whose running cost is $O(n^3)$.

Algorithm A.7.1: Cholesky Decomposition

input: Positive-definite $n \times n$ matrix $\mathbf{A}_n$ with entries $\{a_{ij}\}$.

output Lower triangular $\mathbf{L}_n$ such that $\mathbf{L}_n \mathbf{L}_n^\top = \mathbf{A}_n$.

```

1   $\mathbf{L}_1 \leftarrow \sqrt{a_{11}}$ 
2  for  $k = 2, \dots, n$  do
3     $\mathbf{a}_{k-1} \leftarrow [a_{1k}, \dots, a_{k-1,k}]^\top$ 
4     $\mathbf{l}_{k-1} \leftarrow \mathbf{L}_{k-1}^{-1} \mathbf{a}_{k-1}$  (computed in  $O(k^2)$  time via forward substitution)
5     $l_{kk} \leftarrow \sqrt{a_{kk} - \mathbf{l}_{k-1}^\top \mathbf{l}_{k-1}}$ 
6     $\mathbf{L}_k \leftarrow \begin{bmatrix} \mathbf{L}_{k-1} & \mathbf{0} \\ \mathbf{l}_{k-1}^\top & l_{kk} \end{bmatrix}$ 
7  return  $\mathbf{L}_n$ 

```

A.7.4 QR Decomposition and the Gram–Schmidt Procedure

Let $\mathbf{A}$ be an $n \times p$ matrix, where $p \leq n$. Then, there exists a matrix $\mathbf{Q} \in \mathbb{R}^{n \times p}$ satisfying $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$, and an upper triangular matrix $\mathbf{R} \in \mathbb{R}^{p \times p}$, such that

$$\mathbf{A} = \mathbf{Q}\mathbf{R}.$$

This is the *QR decomposition* for real-valued matrices. When $\mathbf{A}$ has full column rank, such a decomposition can be obtained via the *Gram–Schmidt* procedure, which constructs an *orthonormal basis* $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ of the column space of $\mathbf{A}$ spanned by $\{\mathbf{a}_1, \dots, \mathbf{a}_p\}$, in the following way (see also Figure A.5):

1. Take $\mathbf{u}_1 = \mathbf{a}_1 / \|\mathbf{a}_1\|$.

GRAM–
SCHMIDT

2. Let $\mathbf{p}_1$ be the projection of $\mathbf{a}_2$ onto $\text{Span}\{\mathbf{u}_1\}$. That is, $\mathbf{p}_1 = \langle \mathbf{u}_1, \mathbf{a}_2 \rangle \mathbf{u}_1$. Now take $\mathbf{u}_2 = (\mathbf{a}_2 - \mathbf{p}_1) / \|\mathbf{a}_2 - \mathbf{p}_1\|$. This vector is perpendicular to $\mathbf{u}_1$ and has unit length.
3. Let $\mathbf{p}_2$ be the projection of $\mathbf{a}_3$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. That is, $\mathbf{p}_2 = \langle \mathbf{u}_1, \mathbf{a}_3 \rangle \mathbf{u}_1 + \langle \mathbf{u}_2, \mathbf{a}_3 \rangle \mathbf{u}_2$. Now take $\mathbf{u}_3 = (\mathbf{a}_3 - \mathbf{p}_2) / \|\mathbf{a}_3 - \mathbf{p}_2\|$. This vector is perpendicular to both $\mathbf{u}_1$ and $\mathbf{u}_2$ and has unit length.
4. Continue this process to obtain $\mathbf{u}_4, \dots, \mathbf{u}_p$.

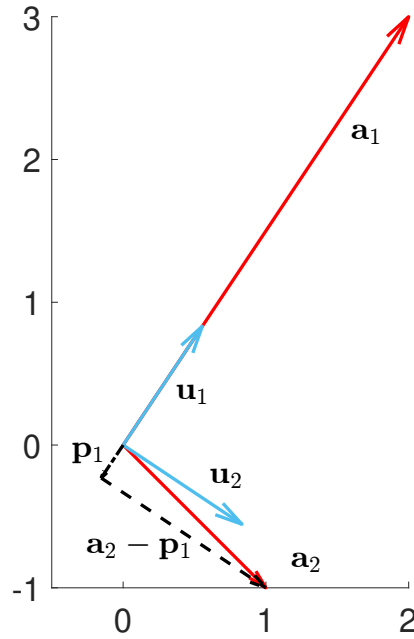


Figure A.5: Illustration of the Gram–Schmidt procedure.

At the end of the procedure, a set $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ of p orthonormal vectors are obtained. Consequently, as a result of Theorem A.3,

$$\mathbf{a}_j = \sum_{i=1}^j \underbrace{\langle \mathbf{a}_j, \mathbf{u}_i \rangle}_{r_{ij}} \mathbf{u}_i, \quad j = 1, \dots, p,$$

for some numbers r_{ij} , $j = 1, \dots, i$, $i = 1, \dots, p$. Denoting the corresponding upper triangular matrix $[r_{ij}]$ by $\mathbf{R}$, we have in matrix notation:

$$\mathbf{QR} = [\mathbf{u}_1, \dots, \mathbf{u}_p] \begin{bmatrix} r_{11} & r_{12} & r_{13} & \dots & r_{1p} \\ 0 & r_{22} & r_{23} & \dots & r_{2p} \\ \vdots & 0 & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & r_{pp} \end{bmatrix} = [\mathbf{a}_1, \dots, \mathbf{a}_p] = \mathbf{A},$$

which yields a QR decomposition. The QR decomposition can be used to efficiently solve least-squares problems; this will be shown shortly. It can also be used to calculate the determinant of the matrix $\mathbf{A}$, whenever $\mathbf{A}$ is square. Namely, $\det(\mathbf{A}) = \det(\mathbf{Q})\det(\mathbf{R}) = \det(\mathbf{R})$; and since $\mathbf{R}$ is triangular, its determinant is the product of its diagonal elements.

An important application of the QR decomposition is found in solving the least-squares problem in $O(p^2 n)$ time:

$$\min_{\beta \in \mathbb{R}^p} \|\mathbf{X}\beta - \mathbf{y}\|^2$$

for some $\mathbf{X} \in \mathbb{R}^{n \times p}$ (model) matrix. Using the defining properties of the pseudo-inverse in Definition A.2, one can show that $\|\mathbf{X}\mathbf{X}^+\mathbf{y} - \mathbf{y}\|^2 \leq \|\mathbf{X}\beta - \mathbf{y}\|^2$ for any β . In other words, $\hat{\beta} := \mathbf{X}^+\mathbf{y}$ minimizes $\|\mathbf{X}\beta - \mathbf{y}\|$. If we have the QR decomposition $\mathbf{X} = \mathbf{Q}\mathbf{R}$, then a numerically stable way to calculate $\hat{\beta}$ with an $O(p^2 n)$ cost is via

$$\hat{\beta} = (\mathbf{Q}\mathbf{R})^+\mathbf{y} = \mathbf{R}^+\mathbf{Q}^+\mathbf{y} = \mathbf{R}^+\mathbf{Q}^\top\mathbf{y}.$$

If $\mathbf{X}$ has full column rank, then $\mathbf{R}^+ = \mathbf{R}^{-1}$.

Note that while the QR decomposition is the method of choice for solving the ordinary least-squares regression problem.

A.7.5 Singular Value Decomposition

One of the most useful matrix decompositions is the *singular value decomposition* (SVD).

SINGULAR
VALUE DE-
COMPOSITION

Theorem A.12: Singular Value Decomposition

Any (complex) matrix $m \times n$ matrix $\mathbf{A}$ admits a unique decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*,$$

where $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices of dimension m and n , respectively, and $\mathbf{\Sigma}$ is a real $m \times n$ diagonal matrix. If $\mathbf{A}$ is real, then $\mathbf{U}$ and $\mathbf{V}$ are both orthogonal matrices.

Proof: Without loss of generality we can assume that $m \geq n$ (otherwise consider the transpose of $\mathbf{A}$). Then $\mathbf{A}^*\mathbf{A}$ is a positive semidefinite Hermitian matrix, because $\langle \mathbf{A}^*\mathbf{A}\mathbf{v}, \mathbf{v} \rangle = \mathbf{v}^*\mathbf{A}^*\mathbf{A}\mathbf{v} = \|\mathbf{A}\mathbf{v}\|^2 \geq 0$ for all $\mathbf{v}$. Hence, $\mathbf{A}^*\mathbf{A}$ has non-negative real eigenvalues, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$. By Theorem A.7 the matrix $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ of right-eigenvectors is a unitary matrix. Define the i -th *singular value* as $\sigma_i = \sqrt{\lambda_i}$, $i = 1, \dots, n$ and suppose $\lambda_1, \dots, \lambda_r$ are all greater than 0, and $\lambda_{r+1}, \dots, \lambda_n = 0$. In particular, $\mathbf{A}\mathbf{v}_i = \mathbf{0}$ for $i = r+1, \dots, n$. Let $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$, $i = 1, \dots, r$. Then, for

SINGULAR
VALUE

72

$i, j \leq r$,

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \mathbf{u}_j^* \mathbf{u}_i = \frac{\mathbf{v}_j^* \mathbf{A}^* \mathbf{A} \mathbf{v}_i}{\sigma_i \sigma_j} = \frac{\lambda_i \mathbb{1}\{i=j\}}{\sigma_i \sigma_j} = \mathbb{1}\{i=j\}.$$

We can extend $\mathbf{u}_1, \dots, \mathbf{u}_r$ to an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ of $\mathbb{C}^n$ (e.g., using the Gram–Schmidt procedure). Let $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_n]$ be the corresponding unitary matrix. Defining $\mathbf{\Sigma}$ to be the $m \times n$ diagonal matrix with diagonal $(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$, we have,

$$\mathbf{U}\mathbf{\Sigma} = [\mathbf{A}\mathbf{v}_1, \dots, \mathbf{A}\mathbf{v}_r, \mathbf{0}, \dots, \mathbf{0}] = \mathbf{A}\mathbf{V},$$

and hence $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$. $\square$

Note that

$$\mathbf{A}\mathbf{A}^* = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*\mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^* = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^*\mathbf{U}^* \quad \text{and} \quad \mathbf{A}^*\mathbf{A} = \mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^*\mathbf{U}\mathbf{\Sigma}\mathbf{V}^* = \mathbf{V}\mathbf{\Sigma}^*\mathbf{\Sigma}\mathbf{V}^*.$$

So, $\mathbf{U}$ is a unitary matrix whose columns are eigenvectors of $\mathbf{A}\mathbf{A}^*$ and $\mathbf{V}$ is a unitary matrix whose columns are eigenvectors of $\mathbf{A}^*\mathbf{A}$.

The SVD makes it possible to write the matrix $\mathbf{A}$ as a sum of rank-1 matrices, weighted by the singular values $\{\sigma_i\}$:

$$\mathbf{A} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \begin{bmatrix} \sigma_1 & 0 & \dots & \dots & 0 \\ 0 & \ddots & 0 & \dots & 0 \\ 0 & \dots & \sigma_r & \dots & 0 \\ 0 & \dots & \dots & 0 & \dots & 0 \\ 0 & \dots & \dots & \dots & \ddots & 0 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \vdots \\ \mathbf{v}_n^* \end{bmatrix} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^*, \quad (\text{A.20})$$

which is called the *dyade* or *spectral representation* of $\mathbf{A}$.

Since $\text{rank}(\mathbf{A}) = r \leq \min\{m, n\}$, the representation (A.20) makes it possible to formally distinguish three different forms of the SVD.

Full SVD. This is simply the representation (A.20) where $\mathbf{U}$ and $\mathbf{V}$ are both square matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$.

Thin SVD. This is the case when $m > n = r$ so that $\mathbf{A}$ can be written as

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^* = [\mathbf{u}_1, \dots, \mathbf{u}_n] \text{diag}(\sigma_1, \dots, \sigma_n) \begin{bmatrix} \mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \vdots \\ \mathbf{v}_n^* \end{bmatrix} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*,$$

where both $\mathbf{V} \in \mathbb{R}^{n \times n}$ and $\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$ are square, but $\mathbf{U} \in \mathbb{R}^{m \times n}$ is rectangular.

Compact SVD. This is the case when $r < \min\{m, n\}$ so that $\mathbf{A}$ can be written as

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^* = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*,$$

where $\mathbf{\Sigma} \in \mathbb{R}^{r \times r}$ is square, but $\mathbf{U} \in \mathbb{R}^{m \times r}$ and $\mathbf{V} \in \mathbb{R}^{n \times r}$ are rectangular.

For real-valued matrices, the SVD has a nice geometric interpretation, illustrated in Figure A.6. The linear mapping defined by matrix $\mathbf{A}$ can be thought of as a succession of three linear operations: (1) an orthogonal transformation (i.e., a rotation with a possible flipping of some axes), corresponding to matrix $\mathbf{V}^\top$, followed by (2) a simple scaling of the unit vectors, corresponding to $\mathbf{\Sigma}$, followed by (3) another orthogonal transformation, corresponding to $\mathbf{U}$.

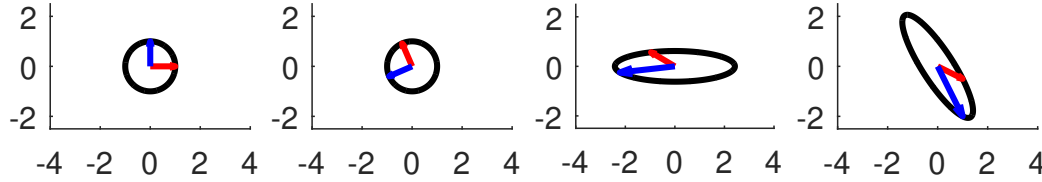


Figure A.6: The figure shows how the unit circle and unit vectors (first panel) are first rotated (second panel), then scaled (third panel), and finally rotated and flipped.

■ **Example A.12 (Ellipses)** We continue Example A.8. We have the following SVD matrices for matrix $\mathbf{A}$:

$$\mathbf{U} = \begin{bmatrix} -0.5430 & 0.8398 \\ 0.8398 & 0.5430 \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} 2.4221 & 0 \\ 0 & 0.6193 \end{bmatrix}, \quad \text{and} \quad \mathbf{V}^\top = \begin{bmatrix} -0.3975 & 0.9176 \\ -0.9176 & -0.3975 \end{bmatrix}.$$

Figure A.4 shows the columns of the matrix $\mathbf{U}\mathbf{\Sigma}$ as the two principal axes of the ellipse that is obtained by applying matrix $\mathbf{A}$ to the points of the unit circle. ■

82

A practical method to compute the pseudo-inverse of a real-valued matrix $\mathbf{A}$ is via the singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, where $\mathbf{\Sigma}$ is the diagonal matrix collecting all the positive singular values, say $\sigma_1, \dots, \sigma_r$, as in Theorem A.12. In this case, $\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top$, where $\mathbf{\Sigma}^+$ is the $n \times m$ diagonal (pseudo-inverse) matrix:

$$\mathbf{\Sigma}^+ = \begin{bmatrix} \sigma_1^{-1} & 0 & \dots & \dots & 0 \\ 0 & \ddots & 0 & \dots & 0 \\ 0 & \dots & \sigma_r^{-1} & \dots & 0 \\ 0 & \dots & \dots & 0 & \dots & 0 \\ 0 & \dots & \dots & \dots & \ddots & 0 \end{bmatrix}.$$

PROBABILITY AND STATISTICS

B.1 Random Experiments and Probability Spaces

The basic notion in probability theory is that of a *random experiment*: an experiment whose outcome cannot be determined in advance. Mathematically, a random experiment is modeled via a triplet $(\Omega, \mathcal{H}, \mathbb{P})$, where:

RANDOM
EXPERIMENT

- Ω is the set of all possible outcomes of the experiment, called the *sample space*.
- $\mathcal{H}$ is the collection of all subsets of Ω to which a probability can be assigned; such subsets are called *events*.
- $\mathbb{P}$ is a *probability measure*, which assigns to each event A a number $\mathbb{P}[A]$ between 0 and 1, indicating the likelihood that the outcome of the random experiment lies in A .

SAMPLE
SPACE

EVENTS

PROBABILITY
MEASURE

Any probability measure $\mathbb{P}$ must satisfy the following *Kolmogorov axioms*:

KOLMOGOROV
AXIOMS

1. $\mathbb{P}[A] \geq 0$ for every event A .
2. $\mathbb{P}[\Omega] = 1$.
3. For any sequence $A_1, A_2, \dots$ of events,

$$\mathbb{P}\left[\bigcup_i A_i\right] \leq \sum_i \mathbb{P}[A_i], \quad (\text{B.1})$$

with strict *equality* whenever the events are *disjoint* (that is, non-overlapping).

When (B.1) holds as an equality, it is often referred to as the *sum rule* of probability. It simply states that if an event can happen in a number of different but not simultaneous ways, the probability of that event is the sum of the probabilities of the comprising events. If the events are allowed to overlap, then the inequality (B.1) is called the *union bound*.

SUM RULE

UNION
BOUND

ELEMENTARY
EVENT

In many applications the sample space is *countable*; that is, $\Omega = \{a_1, a_2, \dots\}$. In this case the easiest way to specify a probability measure $\mathbb{P}$ is to first assign a number p_i to each *elementary event* $\{a_i\}$, with $\sum_i p_i = 1$, and then to define

$$\mathbb{P}[A] = \sum_{i: a_i \in A} p_i \quad \text{for all } A \subseteq \Omega.$$

DISCRETE
PROBABILITY
SPACE

Here the collection of events $\mathcal{H}$ can be taken to be equal to the collection of *all* subsets of Ω . The triple $(\Omega, \mathcal{H}, \mathbb{P})$ is called a *discrete probability space*. This idea is graphically represented in Figure B.1. Each element a_i , represented by a dot, is assigned a weight (that is, probability) p_i , indicated by the size of the dot. The probability of the event A is simply the sum of the weights of all the outcomes in A .

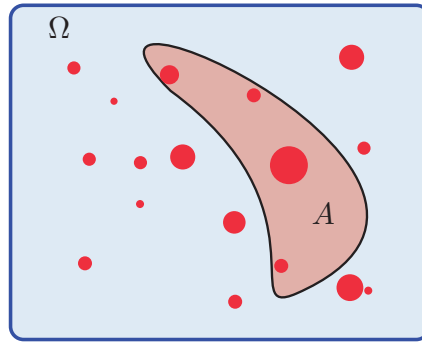


Figure B.1: A discrete probability space.

■ **Remark B.1 (Equilikely Principle)** A special case of a discrete probability space occurs when a random experiment has finitely many outcomes that are all *equally likely*. In this case the probability measure is given by

$$\mathbb{P}[A] = \frac{|A|}{|\Omega|}, \quad (\text{B.2})$$

where $|A|$ denotes the number of outcomes in A and $|\Omega|$ is the total number of outcomes. Thus, the calculation of probabilities reduces to counting numbers of outcomes in events. This is called the *equilikely principle*. ■

EQUILIKELY
PRINCIPLE

B.2 Random Variables and Probability Distributions

It is often convenient to describe a random experiment via “random variables”, representing numerical measurements of the experiment. Random variables are usually denoted by capital letters from the last part of the alphabet. From a mathematical point of view, a *random variable* X is a function from Ω to $\mathbb{R}$ such that sets of the form $\{a < X \leq b\} := \{\omega \in \Omega : a < X(\omega) \leq b\}$ are events (and so can be assigned a probability).

RANDOM
VARIABLE

All probabilities involving a random variable X can be computed, in principle, from its *cumulative distribution function* (cdf), defined by

$$F(x) = \mathbb{P}[X \leq x], \quad x \in \mathbb{R}.$$

CUMULATIVE
DISTRIBUTION
FUNCTION

For example $\mathbb{P}[a < X \leq b] = \mathbb{P}[X \leq b] - \mathbb{P}[X \leq a] = F(b) - F(a)$. Figure B.2 shows a generic cdf. Note that any cdf is right-continuous, increasing, and lies between 0 and 1.

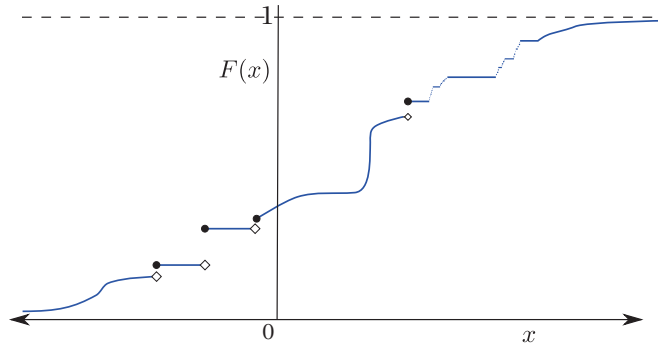


Figure B.2: A cumulative distribution function (cdf).

A cdf F_d is called *discrete* if there exist numbers $x_1, x_2, \dots$ and probabilities $0 < f(x_i) \leq 1$ summing up to 1, such that for all x

DISCRETE
CDF

$$F_d(x) = \sum_{x_i \leq x} f(x_i). \quad (\text{B.3})$$

Such a cdf is piecewise constant and has jumps of sizes $f(x_1), f(x_2), \dots$ at points $x_1, x_2, \dots$, respectively. The function $f(x)$ is called a *probability mass function* or *discrete probability density function* (pdf). It is often easier to use the pdf rather than the cdf, since probabilities can simply be calculated from it via summation:

DISCRETE
PDF

$$\mathbb{P}[X \in B] = \sum_{x \in B} f(x),$$

as illustrated in Figure B.3.

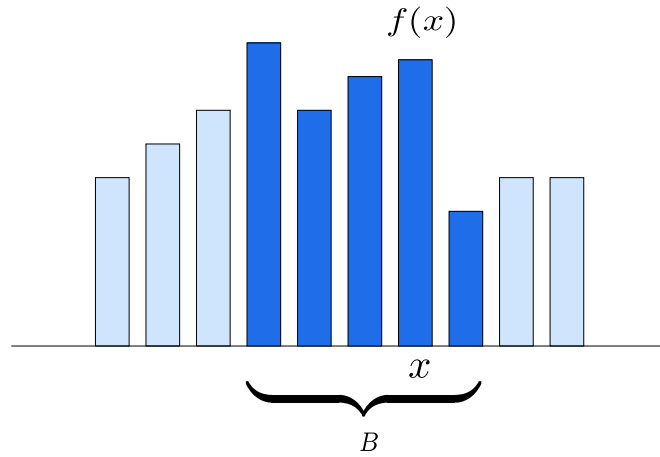


Figure B.3: Discrete probability density function (pdf). The darker area corresponds to the probability $\mathbb{P}[X \in B]$.

CONTINUOUS
CDF

A cdf F_c is called *continuous*¹, if there exists a positive function f such that for all x

$$F_c(x) = \int_{-\infty}^x f(u) du. \quad (\text{B.4})$$

PDF

Note that such an F_c is differentiable (and hence continuous) with derivative f . The function f is called the *probability density function (continuous pdf)*. By the fundamental theorem of integration, we have

$$\mathbb{P}[a < X \leq b] = F(b) - F(a) = \int_a^b f(x) dx.$$

Thus, calculating probabilities reduces to integration, as illustrated in Figure B.4.

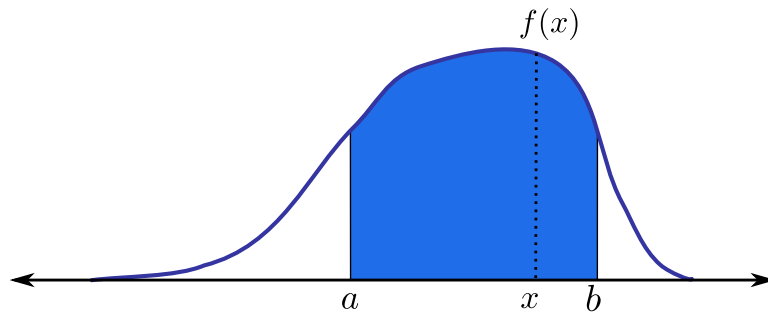


Figure B.4: Continuous probability density function (pdf). The shaded area corresponds to the probability $\mathbb{P}[X \in B]$, with B being here the interval $(a, b]$.

¹In advanced probability, we would say “*absolutely continuous with respect to the Lebesgue measure*”.

■ **Remark B.2 (Probability Density and Probability Mass)** It is important to note that we deliberately use the *same* name, “pdf”, and symbol, f , in both the discrete and the continuous case, rather than distinguish between a probability mass function (pmf) and probability density function (pdf). From a theoretical point of view the pdf plays exactly the same role in the discrete and continuous cases. We use the notation $X \sim \text{Dist}$, $X \sim f$, and $X \sim F$ to indicate that X has distribution Dist, pdf f , and cdf F .

Tables B.1 and B.2 list a number of important continuous and discrete distributions. Note that in Table B.1, Γ is the gamma function: $\Gamma(\alpha) = \int_0^\infty e^{-x} x^{\alpha-1} dx$, $\alpha > 0$.

Table B.1: Commonly used continuous distributions.

Name	Notation	$f(x)$	$x \in$	Parameters
Uniform	$\mathcal{U}[\alpha, \beta]$	$\frac{1}{\beta - \alpha}$	$[\alpha, \beta]$	$\alpha < \beta$
Normal	$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\mathbb{R}$	$\sigma > 0, \mu \in \mathbb{R}$
Gamma	$\text{Gamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Inverse Gamma	$\text{InvGamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{-\alpha-1} e^{-\lambda x^{-1}}}{\Gamma(\alpha)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$\mathbb{R}_+$	$\lambda > 0$
Beta	$\text{Beta}(\alpha, \beta)$	$\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$[0, 1]$	$\alpha, \beta > 0$
Weibull	$\text{Weib}(\alpha, \lambda)$	$\alpha \lambda (\lambda x)^{\alpha-1} e^{-(\lambda x)^\alpha}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Pareto	$\text{Pareto}(\alpha, \lambda)$	$\alpha \lambda (1 + \lambda x)^{-(\alpha+1)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Student	t_ν	$\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$	$\mathbb{R}$	$\nu > 0$
F	$F(m, n)$	$\frac{\Gamma(\frac{m+n}{2}) (m/n)^{m/2} x^{(m-2)/2}}{\Gamma(\frac{m}{2}) \Gamma(\frac{n}{2}) [1 + (m/n)x]^{(m+n)/2}}$	$\mathbb{R}_+$	$m, n \in \mathbb{N}_+$

The $\text{Gamma}(n/2, 1/2)$ distribution is called the *chi-squared distribution* with n degrees of freedom, denoted χ_n^2 . The t_1 distribution is also called the *Cauchy distribution*.

χ_n^2 DISTRIBUTION

Table B.2: Commonly used discrete distributions.

Name	Notation	$f(x)$	$x \in$	Parameters
Bernoulli	$\text{Ber}(p)$	$p^x(1-p)^{1-x}$	$\{0, 1\}$	$0 \leq p \leq 1$
Binomial	$\text{Bin}(n, p)$	$\binom{n}{x} p^x(1-p)^{n-x}$	$\{0, 1, \dots, n\}$	$0 \leq p \leq 1,$ $n \in \mathbb{N}$
Discrete uniform	$\mathcal{U}\{1, \dots, n\}$	$\frac{1}{n}$	$\{1, \dots, n\}$	$n \in \{1, 2, \dots\}$
Geometric	$\text{Geom}(p)$	$p(1-p)^{x-1}$	$\{1, 2, \dots\}$	$0 \leq p \leq 1$
Poisson	$\text{Poi}(\lambda)$	$e^{-\lambda} \frac{\lambda^x}{x!}$	$\mathbb{N}$	$\lambda > 0$

B.3 Expectation

It is often useful to consider different kinds of numerical characteristics of a random variable. One such quantity is the expectation, which measures the “average” value of the distribution.

EXPECTATION

The *expectation* (or expected value or mean) of a random variable X with pdf f , denoted by $\mathbb{E}X$ or² $\mathbb{E}[X]$ (and sometimes μ), is defined by

$$\mathbb{E}X = \begin{cases} \sum_x x f(x) & \text{discrete case,} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{continuous case.} \end{cases}$$

If X is a random variable, then a function of X , such as X^2 or $\sin(X)$, is again a random variable. Moreover, the expected value of a function of X is simply a weighted average of the possible values that this function can take. That is, for any real function h

$$\mathbb{E}h(X) = \begin{cases} \sum_x h(x) f(x) & \text{discrete case,} \\ \int_{-\infty}^{\infty} h(x) f(x) dx & \text{continuous case,} \end{cases}$$

provided that the sum or integral are well-defined.

VARIANCE

The *variance* of a random variable X , denoted by $\mathbb{V}\text{ar } X$ (and sometimes σ^2), is defined by

$$\mathbb{V}\text{ar } X = \mathbb{E}(X - \mathbb{E}[X])^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2.$$

STANDARD
DEVIATION

The square root of the variance is called the *standard deviation*. Table B.3 lists the expectations and variances for some well-known distributions. Both variance and

²We only use brackets in an expectation if it is unclear with respect to which random variable the expectation is taken.

standard deviation measure the spread or dispersion of the distribution. Note, however, that the standard deviation measures the dispersion in the same units as the random variable, unlike the variance, which uses squared units.

Table B.3: Expectations and variances for some well-known distributions.

Dist.	$\mathbb{E}X$	$\mathbb{V}\text{ar } X$
$\text{Bin}(n, p)$	np	$np(1 - p)$
$\text{Geom}(p)$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
$\text{Poi}(\lambda)$	λ	λ
$\mathcal{U}[\alpha, \beta]$	$\frac{\alpha + \beta}{2}$	$\frac{(\beta - \alpha)^2}{12}$
$\text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
t_ν	0 ($\nu > 1$)	$\frac{\nu}{\nu - 2}$ ($\nu > 2$)
Dist.	$\mathbb{E}X$	$\mathbb{V}\text{ar } X$
$\text{Gamma}(\alpha, \lambda)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2
$\text{Beta}(\alpha, \beta)$	$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha\beta}{(\alpha + \beta)^2(1 + \alpha + \beta)}$
$\text{Weib}(\alpha, \lambda)$	$\frac{\Gamma(1/\alpha)}{\alpha\lambda}$	$\frac{2\Gamma(2/\alpha)}{\alpha} - \left(\frac{\Gamma(1/\alpha)}{\alpha\lambda}\right)^2$
$F(m, n)$	$\frac{n}{n-2}$ ($n > 2$)	$\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$ ($n > 4$)

It is sometimes useful to consider the *moment generating function* of a random variable X . This is the function M defined by

$$M(s) = \mathbb{E} e^{sX}, \quad s \in \mathbb{R}. \quad (\text{B.5})$$

**MOMENT
GENERATING
FUNCTION**

The moment generating functions of two random variables coincide if and only if the random variables have the same distribution.

■ **Example B.1 (Moment Generation Function of the $\text{Gamma}(\alpha, \lambda)$ Distribution)**

Let $X \sim \text{Gamma}(\alpha, \lambda)$. For $s < \lambda$, the moment generating function of X at s is given

by

$$\begin{aligned}
 M(s) = \mathbb{E}e^{sX} &= \int_0^\infty e^{sx} \frac{e^{-\lambda x} \lambda^\alpha x^{\alpha-1}}{\Gamma(\alpha)} dx \\
 &= \left(\frac{\lambda}{\lambda-s}\right)^\alpha \underbrace{\int_0^\infty \frac{e^{-(\lambda-s)x} (\lambda-s)^\alpha x^{\alpha-1}}{\Gamma(\alpha)} dx}_{\text{pdf of Gamma}(\alpha, \lambda-s)} = \left(\frac{\lambda}{\lambda-s}\right)^\alpha.
 \end{aligned}$$

For $s \geq \lambda$, $M(s) = \infty$. Interestingly, the moment generating function has a much simpler formula than the pdf. ■

B.4 Joint Distributions

Distributions for random vectors and stochastic processes can be specified in much the same way as for random variables. In particular, the distribution of a random vector $\mathbf{X} = [X_1, \dots, X_n]^\top$ is completely determined by specifying the *joint cdf* F , defined by

$$F(x_1, \dots, x_n) = \mathbb{P}[X_1 \leq x_1, \dots, X_n \leq x_n], \quad x_i \in \mathbb{R}, i = 1, \dots, n.$$

Similarly, the distribution of a *stochastic process*, that is, a collection of random variables $\{X_t, t \in \mathcal{T}\}$, for some index set $\mathcal{T}$, is completely determined by its finite-dimensional distributions; specifically, the distributions of the random vectors $[X_{t_1}, \dots, X_{t_n}]^\top$ for every choice of n and $t_1, \dots, t_n$.

By analogy to the one-dimensional case, a random vector $\mathbf{X} = [X_1, \dots, X_n]^\top$ taking values in $\mathbb{R}^n$ is said to have a pdf f if, in the continuous case,

$$\mathbb{P}[\mathbf{X} \in B] = \int_B f(\mathbf{x}) d\mathbf{x}, \quad (\text{B.6})$$

for all n -dimensional rectangles B . Replace the integral with a sum for the discrete case. The pdf is also called the *joint pdf* of $X_1, \dots, X_n$. The pdfs of the individual components — called *marginal pdfs* — can be recovered from the joint pdf by “integrating out the other variables”. For example, for a continuous random vector $[X, Y]^\top$ with pdf f , the pdf f_X of X is given by

$$f_X(x) = \int f(x, y) dy.$$

B.5 Conditioning and Independence

Conditional probabilities and conditional distributions are used to model additional information on a random experiment. Independence is used to model *lack* of such information.

B.5.1 Conditional Probability

Suppose some event $B \subseteq \Omega$ occurs. Given this fact, event A will occur if and only if $A \cap B$ occurs, and the relative chance of A occurring is therefore $\mathbb{P}[A \cap B]/\mathbb{P}[B]$, provided $\mathbb{P}[B] > 0$. This leads to the definition of the *conditional probability* of A given B :

$$\mathbb{P}[A | B] = \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}, \quad \text{if } \mathbb{P}[B] > 0. \quad (\text{B.7})$$

CONDI-
TIONAL
PROBABILITY

The above definition breaks down if $\mathbb{P}[B] = 0$. Such conditional probabilities must be treated with more care.

Three important consequences of the definition of conditional probability are:

1. *Product rule*: For any sequence of events $A_1, A_2, \dots, A_n$,

PRODUCT
RULE

$$\mathbb{P}[A_1 \cdots A_n] = \mathbb{P}[A_1] \mathbb{P}[A_2 | A_1] \mathbb{P}[A_3 | A_1 A_2] \cdots \mathbb{P}[A_n | A_1 \cdots A_{n-1}], \quad (\text{B.8})$$

using the abbreviation $A_1 A_2 \cdots A_k := A_1 \cap A_2 \cap \cdots \cap A_k$.

2. *Law of total probability*: If $\{B_i\}$ forms a *partition* of Ω (that is, $B_i \cap B_j = \emptyset, i \neq j$ and $\cup_i B_i = \Omega$), then for any event A

LAW OF
TOTAL
PROBABILITY

$$\mathbb{P}[A] = \sum_i \mathbb{P}[A | B_i] \mathbb{P}[B_i]. \quad (\text{B.9})$$

3. *Bayes' rule*: Let $\{B_i\}$ form a partition of Ω . Then, for any event A with $\mathbb{P}[A] > 0$,

BAYES' RULE

$$\mathbb{P}[B_j | A] = \frac{\mathbb{P}[A | B_j] \mathbb{P}[B_j]}{\sum_i \mathbb{P}[A | B_i] \mathbb{P}[B_i]}. \quad (\text{B.10})$$

B.5.2 Independence

Two events A and B are said to be *independent* if the knowledge that B has occurred does not change the probability that A occurs. That is, A, B independent $\Leftrightarrow \mathbb{P}[A | B] = \mathbb{P}[A]$. Since $\mathbb{P}[A | B] \mathbb{P}[B] = \mathbb{P}[A \cap B]$, an alternative definition of independence is

INDEPEND-
ENT EVENTS

$$A, B \text{ independent} \Leftrightarrow \mathbb{P}[A \cap B] = \mathbb{P}[A] \mathbb{P}[B].$$

This definition covers the case where $\mathbb{P}[B] = 0$ and can be extended to arbitrarily many events: events $A_1, A_2, \dots$ are said to be (mutually) independent if for any k and any choice of distinct indices $i_1, \dots, i_k$,

$$\mathbb{P}[A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}] = \mathbb{P}[A_{i_1}] \mathbb{P}[A_{i_2}] \cdots \mathbb{P}[A_{i_k}].$$

The concept of independence can also be formulated for random variables. Random variables $X_1, X_2, \dots$ are said to be *independent* if the events $\{X_{i_1} \leq x_{i_1}\}, \dots, \{X_{i_n} \leq$

INDEPEND-
ENT RANDOM
VARIABLES

$x_{i_n}\}$ are independent for all finite choices of n distinct indices $i_1, \dots, i_n$ and values $x_{i_1}, \dots, x_{i_n}$.

An important characterization of independent random variables is the following.

Theorem B.1: Independence Characterization

Random variables $X_1, \dots, X_n$ with marginal pdfs $f_{X_1}, \dots, f_{X_n}$ and joint pdf f are independent if and only if

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n) \quad \text{for all } x_1, \dots, x_n. \quad (\text{B.11})$$

Many probabilistic models involve random variables $X_1, X_2, \dots$ that are *independent and identically distributed*, abbreviated as *iid*. We use this abbreviation throughout this book.

B.5.3 Expectation and Covariance

Similar to the univariate case, the expected value of a real-valued function h of a random vector $\mathbf{X} \sim f$ is a weighted average of all values that $h(\mathbf{X})$ can take. Specifically, in the continuous case, $\mathbb{E}h(\mathbf{X}) = \int h(\mathbf{x})f(\mathbf{x})d\mathbf{x}$. In the discrete case replace this multidimensional integral with a sum. Using this result, it is not difficult to show that for any collection of dependent or independent random variables $X_1, \dots, X_n$,

$$\mathbb{E}[a + b_1X_1 + b_2X_2 + \cdots + b_nX_n] = a + b_1\mathbb{E}X_1 + \cdots + b_n\mathbb{E}X_n \quad (\text{B.12})$$

for all constants $a, b_1, \dots, b_n$. Moreover, for *independent* random variables,

$$\mathbb{E}[X_1X_2 \cdots X_n] = \mathbb{E}X_1 \mathbb{E}X_2 \cdots \mathbb{E}X_n. \quad (\text{B.13})$$

We leave the proofs as an exercise.

The *covariance* of two random variables X and Y with expectations μ_X and μ_Y , respectively, is defined as

$$\mathbb{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

This is a measure of the amount of linear dependency between the variables. Let $\sigma_X^2 = \mathbb{V}\text{ar } X$ and $\sigma_Y^2 = \mathbb{V}\text{ar } Y$. A scaled version of the covariance is given by the *correlation coefficient*,

$$\varrho(X, Y) = \frac{\mathbb{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

The following properties follow directly from the definitions of variance and covariance.

1. $\mathbb{V}\text{ar } X = \mathbb{E}X^2 - \mu_X^2$.
2. $\mathbb{V}\text{ar}[aX + b] = a^2 \sigma_X^2$.

3. $\mathbb{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X \mu_Y$.
4. $\mathbb{Cov}(X, Y) = \mathbb{Cov}(Y, X)$.
5. $-\sigma_X \sigma_Y \leq \mathbb{Cov}(X, Y) \leq \sigma_X \sigma_Y$.
6. $\mathbb{Cov}(aX + bY, Z) = a \mathbb{Cov}(X, Z) + b \mathbb{Cov}(Y, Z)$.
7. $\mathbb{Cov}(X, X) = \sigma_X^2$.
8. $\mathbb{Var}[X + Y] = \sigma_X^2 + \sigma_Y^2 + 2 \mathbb{Cov}(X, Y)$.
9. If X and Y are independent, then $\mathbb{Cov}(X, Y) = 0$.

As a consequence of Properties 2 and 8 we have that for any sequence of *independent* random variables $X_1, \dots, X_n$ with variances $\sigma_1^2, \dots, \sigma_n^2$,

$$\mathbb{Var}[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \dots + a_n^2 \sigma_n^2, \quad (\text{B.14})$$

for any choice of constants $a_1, \dots, a_n$.

For random column vectors, such as $\mathbf{X} = [X_1, \dots, X_n]^\top$, it is convenient to write the expectations and covariances in vector and matrix notation. For a random vector $\mathbf{X}$ we define its *expectation vector* as the vector of expectations

$$\boldsymbol{\mu} = [\mu_1, \dots, \mu_n]^\top = [\mathbb{E}X_1, \dots, \mathbb{E}X_n]^\top.$$

EXPECTATION
VECTOR

Similarly, if the expectation of a matrix is the matrix of expectations, then given two random vectors $\mathbf{X} \in \mathbb{R}^n$ and $\mathbf{Y} \in \mathbb{R}^m$, the $n \times m$ matrix

$$\mathbb{Cov}(\mathbf{X}, \mathbf{Y}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}\mathbf{X})(\mathbf{Y} - \mathbb{E}\mathbf{Y})^\top] \quad (\text{B.15})$$

has (i, j) -th element $\mathbb{Cov}(X_i, Y_j) = \mathbb{E}[(X_i - \mathbb{E}X_i)(Y_j - \mathbb{E}Y_j)]$. A consequence of this definition is that

$$\mathbb{Cov}(\mathbf{A}\mathbf{X}, \mathbf{B}\mathbf{Y}) = \mathbf{A} \mathbb{Cov}(\mathbf{X}, \mathbf{Y}) \mathbf{B}^\top,$$

where $\mathbf{A}$ and $\mathbf{B}$ are two matrices with n and m columns, respectively.

The *covariance matrix* of the vector $\mathbf{X}$ is defined as the $n \times n$ matrix $\mathbb{Cov}(\mathbf{X}, \mathbf{X})$. The covariance matrix is also denoted as $\mathbb{Var}(\mathbf{X}) = \mathbb{Cov}(\mathbf{X}, \mathbf{X})$, in analogy with the scalar identity $\mathbb{Var}(X) = \mathbb{Cov}(X, X)$.

COVARIANCE
MATRIX

A useful application of the cyclic property of the trace of a matrix (see Theorem A.1) is the following.



69

Theorem B.2: Expectation of a Quadratic Form

Let $\mathbf{A}$ be an $n \times n$ matrix and $\mathbf{X}$ an n -dimensional random vector with expectation vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. The random variable $Y := \mathbf{X}^\top \mathbf{A} \mathbf{X}$ has expectation $\text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

Proof: Since Y is a scalar, it is equal to its trace. Now, using the cyclic property: $\mathbb{E}Y = \mathbb{E} \text{tr}(Y) = \mathbb{E} \text{tr}(\mathbf{X}^\top \mathbf{A} \mathbf{X}) = \mathbb{E} \text{tr}(\mathbf{A} \mathbf{X} \mathbf{X}^\top) = \text{tr}(\mathbf{A} \mathbb{E}[\mathbf{X} \mathbf{X}^\top]) = \text{tr}(\mathbf{A}(\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top)) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \text{tr}(\mathbf{A} \boldsymbol{\mu} \boldsymbol{\mu}^\top) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$. $\square$

B.5.4 Conditional Density and Conditional Expectation

Suppose X and Y are both discrete or both continuous, with joint pdf f , and suppose $f_X(x) > 0$. Then, the *conditional pdf* of Y given $X = x$ is given by

CONDI-
TIONAL PDF

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} \quad \text{for all } y. \quad (\text{B.16})$$

In the discrete case, the formula is a direct translation of (B.7), with $f_{Y|X}(y|x) = \mathbb{P}[Y = y | X = x]$. In the continuous case, a similar interpretation in terms of densities can be used. The corresponding distribution is called the *conditional distribution* of Y given $X = x$. Note that (B.16) implies that

CONDITIONAL
DISTRIBUTION

$$f(x, y) = f_X(x) f_{Y|X}(y|x).$$

This is useful when the marginal and conditional pdfs are given, rather than the joint one. More generally, for the n -dimensional case we have

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) f_{X_2|X_1}(x_2|x_1) \cdots f_{X_n|X_1, \dots, X_{n-1}}(x_n|x_1, \dots, x_{n-1}), \quad (\text{B.17})$$

which is in essence a rephrasing of the product rule (B.8) in terms of probability densities.

103

As a conditional pdf has all the properties of an ordinary pdf, we may define expectations with respect to it. The *conditional expectation* of a random variable Y given $X = x$ is defined as

CONDITIONAL
EXPECTATION

$$\mathbb{E}[Y | X = x] = \begin{cases} \sum_y y f_{Y|X}(y|x) & \text{discrete case,} \\ \int y f_{Y|X}(y|x) dy & \text{continuous case.} \end{cases} \quad (\text{B.18})$$

Note that $\mathbb{E}[Y | X = x]$ is a function of x . The corresponding random variable is written as $\mathbb{E}[Y | X]$. A similar formalism can be used when conditioning on a sequence of random variables $X_1, \dots, X_n$. The conditional expectation has similar properties to the ordinary expectation. Other useful properties are:

1. *Tower property*: If $\mathbb{E}Y$ exists, then

$$\mathbb{E} \mathbb{E}[Y | X] = \mathbb{E}Y. \quad (\text{B.19})$$

2. *Taking out what is known*: If $\mathbb{E}Y$ exists, then

$$\mathbb{E}[XY | X] = X \mathbb{E}[Y | X].$$

B.6 Functions of Random Variables

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ be a column vector in $\mathbb{R}^n$ and $\mathbf{A}$ an $m \times n$ matrix. The mapping $\mathbf{x} \mapsto \mathbf{z}$, with $\mathbf{z} = \mathbf{A}\mathbf{x}$, is a linear transformation, as discussed in Section A.1. Now consider a random vector $\mathbf{X} = [X_1, \dots, X_n]^\top$ and let $\mathbf{Z} := \mathbf{A}\mathbf{X}$. Then $\mathbf{Z}$ is a random vector in $\mathbb{R}^m$. The following theorem details how the distribution of $\mathbf{Z}$ is related to that of $\mathbf{X}$.



67

Theorem B.3: Linear Transformation

If $\mathbf{X}$ has an expectation vector $\boldsymbol{\mu}_X$ and covariance matrix $\boldsymbol{\Sigma}_X$, then the expectation vector of $\mathbf{Z}$ is

$$\boldsymbol{\mu}_Z = \mathbf{A} \boldsymbol{\mu}_X \quad (\text{B.20})$$

and the covariance matrix of $\mathbf{Z}$ is

$$\boldsymbol{\Sigma}_Z = \mathbf{A} \boldsymbol{\Sigma}_X \mathbf{A}^\top. \quad (\text{B.21})$$

If, in addition, $\mathbf{A}$ is an invertible $n \times n$ matrix and $\mathbf{X}$ is a continuous random vector with pdf f_X , then the pdf of the continuous random vector $\mathbf{Z} = \mathbf{A}\mathbf{X}$ is given by

$$f_Z(\mathbf{z}) = \frac{f_X(\mathbf{A}^{-1}\mathbf{z})}{|\det(\mathbf{A})|}, \quad \mathbf{z} \in \mathbb{R}^n, \quad (\text{B.22})$$

where $|\det(\mathbf{A})|$ denotes the absolute value of the determinant of $\mathbf{A}$.

Proof: We have $\boldsymbol{\mu}_Z = \mathbb{E}\mathbf{Z} = \mathbb{E}[\mathbf{A}\mathbf{X}] = \mathbf{A} \mathbb{E}\mathbf{X} = \mathbf{A}\boldsymbol{\mu}_X$ and

$$\begin{aligned} \boldsymbol{\Sigma}_Z &= \mathbb{E}[(\mathbf{Z} - \boldsymbol{\mu}_Z)(\mathbf{Z} - \boldsymbol{\mu}_Z)^\top] = \mathbb{E}[\mathbf{A}(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{A}(\mathbf{X} - \boldsymbol{\mu}_X))^\top] \\ &= \mathbf{A} \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top] \mathbf{A}^\top \\ &= \mathbf{A} \boldsymbol{\Sigma}_X \mathbf{A}^\top. \end{aligned}$$

For $\mathbf{A}$ invertible and $\mathbf{X}$ continuous (as opposed to discrete), let $\mathbf{z} = \mathbf{A}\mathbf{x}$ and $\mathbf{x} = \mathbf{A}^{-1}\mathbf{z}$. Consider the n -dimensional cube $C = [z_1, z_1 + h] \times \dots \times [z_n, z_n + h]$. Then,

$$\mathbb{P}[\mathbf{Z} \in C] \approx h^n f_Z(\mathbf{z}),$$

by definition of the joint density of $\mathbf{Z}$. Let D be the image of C under $\mathbf{A}^{-1}$ — that is, all points $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} \in C$. Recall from Section A.1 that any matrix B linearly transforms an n -dimensional rectangle with volume V into an n -dimensional parallelepiped with volume $V |\det(B)|$. Thus, in addition to the above expression for $\mathbb{P}[\mathbf{Z} \in C]$, we also have

$$\mathbb{P}[\mathbf{Z} \in C] = \mathbb{P}[\mathbf{X} \in D] \approx h^n |\det(\mathbf{A}^{-1})| f_X(\mathbf{x}) = h^n |\det(\mathbf{A})|^{-1} f_X(\mathbf{x}).$$

Equating these two expressions for $\mathbb{P}[\mathbf{Z} \in C]$, dividing both sides by h^n , and letting h go to 0, we obtain (B.22). $\square$



67

For a generalization of the linear transformation rule (B.22), consider an arbitrary mapping $\mathbf{x} \mapsto \mathbf{g}(\mathbf{x})$, written out:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_n(\mathbf{x}) \end{bmatrix}.$$

Theorem B.4: Transformation Rule

Let $\mathbf{X}$ be an n -dimensional vector of continuous random variables with pdf f_X . Let $\mathbf{Z} = \mathbf{g}(\mathbf{X})$, where $\mathbf{g}$ is an invertible mapping with inverse $\mathbf{g}^{-1}$ and *matrix of Jacobi* $\mathbf{J}_g$; that is, the matrix of partial derivatives of $\mathbf{g}$. Then, at $\mathbf{z} = \mathbf{g}(\mathbf{x})$ the random vector $\mathbf{Z}$ has pdf

$$f_Z(\mathbf{z}) = \frac{f_X(\mathbf{x})}{|\det(\mathbf{J}_g(\mathbf{x}))|} = f_X(\mathbf{g}^{-1}(\mathbf{z})) |\det(\mathbf{J}_{g^{-1}}(\mathbf{z}))|, \quad \mathbf{z} \in \mathbb{R}^n. \quad (\text{B.23})$$

Proof: For a fixed $\mathbf{x}$, let $\mathbf{z} = \mathbf{g}(\mathbf{x})$; and thus $\mathbf{x} = \mathbf{g}^{-1}(\mathbf{z})$. In the neighborhood of $\mathbf{x}$, the function $\mathbf{g}$ behaves like a linear function, in the sense that $\mathbf{g}(\mathbf{x} + \delta) \approx \mathbf{g}(\mathbf{x}) + \mathbf{J}_g(\mathbf{x}) \delta$ for small vectors δ .

Consequently, an infinitesimally small n -dimensional rectangle at $\mathbf{x}$ with volume V is transformed into an infinitesimally small n -dimensional parallelepiped at $\mathbf{z}$ with volume $V |\det(\mathbf{J}_g(\mathbf{x}))|$. Now, as in the proof of the linear case, let C be a small cube around $\mathbf{z} = \mathbf{g}(\mathbf{x})$ with volume h^n . Let D be the image of C under $\mathbf{g}^{-1}$. Then,

$$h^n f_Z(\mathbf{z}) \approx \mathbb{P}[\mathbf{Z} \in C] \approx h^n |\mathbf{J}_{g^{-1}}(\mathbf{z})| f_X(\mathbf{x}),$$

and since $|\det(\mathbf{J}_{g^{-1}}(\mathbf{z}))| = 1/|\det(\mathbf{J}_g(\mathbf{x}))|$, (B.23) follows as h goes to 0. $\square$



Typically, in coordinate transformations it is $\mathbf{g}^{-1}$ that is given — that is, an expression for $\mathbf{x}$ as a function of $\mathbf{z}$.

■ **Example B.2 (Polar Transform)** Suppose X, Y are independent and have standard normal distribution. The joint pdf is

$$f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}, \quad (x, y) \in \mathbb{R}^2.$$

In polar coordinates we have

$$X = R \cos \Theta \quad \text{and} \quad Y = R \sin \Theta, \quad (\text{B.24})$$

where $R \geq 0$ is the radius and $\Theta \in [0, 2\pi)$ the angle of the point (X, Y) . What is the joint pdf of R and Θ ? By the radial symmetry of the bivariate normal distribution, we would expect Θ to be uniform on $(0, 2\pi)$. But what is the pdf of R ? To work out the joint pdf, consider the inverse transformation $\mathbf{g}^{-1}$, defined by

$$\begin{bmatrix} r \\ \theta \end{bmatrix} \xrightarrow{\mathbf{g}^{-1}} \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The corresponding matrix of Jacobi is

$$\mathbf{J}_{\mathbf{g}^{-1}}(r, \theta) = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix},$$

which has determinant r . Since $x^2 + y^2 = r^2(\cos^2 \theta + \sin^2 \theta) = r^2$, it follows by the transformation rule (B.23) that the joint pdf of R and Θ is given by

$$f_{R,\Theta}(r, \theta) = f_{X,Y}(x, y) r = \frac{1}{2\pi} e^{-\frac{1}{2}r^2} r, \quad \theta \in (0, 2\pi), \quad r \geq 0.$$

By integrating out θ and r , respectively, we find $f_R(r) = r e^{-r^2/2}$ and $f_\Theta(\theta) = 1/(2\pi)$. Since $f_{R,\Theta}$ is the product of f_R and f_Θ , the random variables R and Θ are independent. ■

B.7 Multivariate Normal Distribution

The normal (or Gaussian) distribution — especially its multidimensional version — plays a central role in data science and machine learning. Recall from Table B.1 that a random variable X is said to have a *normal* distribution with parameters μ and σ^2 if its pdf is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R}. \quad (\text{B.25})$$

NORMAL DIS-
TRIBUTION

We write $X \sim \mathcal{N}(\mu, \sigma^2)$. The parameters μ and σ^2 are the expectation and variance of the distribution, respectively. If $\mu = 0$ and $\sigma = 1$ then

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

and the distribution is known as the *standard normal* distribution. The cdf of the standard normal distribution is often denoted by Φ and its pdf by φ . In Figure B.5 the pdf of the $\mathcal{N}(\mu, \sigma^2)$ distribution for various μ and σ^2 is plotted.

STANDARD
NORMAL

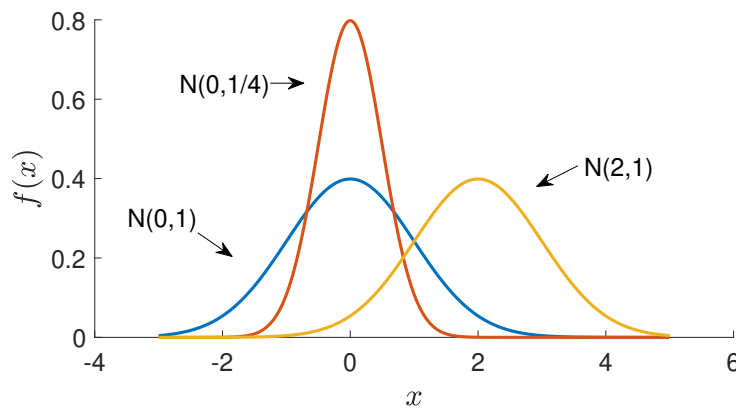


Figure B.5: The pdf of the $\mathcal{N}(\mu, \sigma^2)$ distribution for various μ and σ^2 .

We next consider some important properties of the normal distribution.

Theorem B.5: Standardization

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and define $Z = (X - \mu)/\sigma$. Then Z has a standard normal distribution.

Proof: The cdf of Z is given by

$$\begin{aligned} \mathbb{P}[Z \leq z] &= \mathbb{P}[(X - \mu)/\sigma \leq z] = \mathbb{P}[X \leq \mu + \sigma z] \\ &= \int_{-\infty}^{\mu + \sigma z} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy = \Phi(z), \end{aligned}$$

where we make a change of variable $y = (x - \mu)/\sigma$ in the fourth equation. Hence, $Z \sim \mathcal{N}(0, 1)$. $\square$

STANDARDIZATION

The rescaling procedure in Theorem B.5 is called *standardization*. It follows from Theorem B.5 that any $X \sim \mathcal{N}(\mu, \sigma^2)$ can be written as

$$X = \mu + \sigma Z, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

AFFINE TRANSFORMATION

In other words, any normal random variable can be viewed as an *affine transformation* — that is, a linear transformation plus a constant — of a standard normal random variable.

We now generalize this to n dimensions. Let $Z_1, \dots, Z_n$ be independent and standard normal random variables. The joint pdf of $\mathbf{Z} = [Z_1, \dots, Z_n]^\top$ is given by

$$f_{\mathbf{Z}}(\mathbf{z}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z_i^2} = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\mathbf{z}^\top \mathbf{z}}, \quad \mathbf{z} \in \mathbb{R}^n. \quad (\text{B.26})$$

We write $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\mathbf{I}$ is the identity matrix. Consider the affine transformation

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{B} \mathbf{Z} \quad (\text{B.27})$$

for some $m \times n$ matrix $\mathbf{B}$ and m -dimensional vector $\boldsymbol{\mu}$. Note that, by (B.20) and (B.21), $\mathbf{X}$ has expectation vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{B}^\top$. We say that $\mathbf{X}$ has a *multivariate normal* or *multivariate Gaussian* distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. We write $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

The following theorem states that any affine combination of independent multivariate normal random variables is again multivariate normal.

Theorem B.6: Affine Transformation of Normal Random Vectors

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_r$ be independent m_i -dimensional normal random vectors, with $\mathbf{X}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, $i = 1, \dots, r$. Then, for any $n \times 1$ vector $\mathbf{a}$ and $n \times m_i$ matrices $\mathbf{B}_1, \dots, \mathbf{B}_r$,

$$\mathbf{a} + \sum_{i=1}^r \mathbf{B}_i \mathbf{X}_i \sim \mathcal{N}\left(\mathbf{a} + \sum_{i=1}^r \mathbf{B}_i \boldsymbol{\mu}_i, \sum_{i=1}^r \mathbf{B}_i \boldsymbol{\Sigma}_i \mathbf{B}_i^\top\right). \quad (\text{B.28})$$

Proof: Denote the n -dimensional random vector in the left-hand side of (B.28) by $\mathbf{Y}$. By definition, each $\mathbf{X}_i$ can be written as $\boldsymbol{\mu}_i + \mathbf{A}_i \mathbf{Z}_i$, where the $\{\mathbf{Z}_i\}$ are independent (because the $\{\mathbf{X}_i\}$ are independent), so that

$$\mathbf{Y} = \mathbf{a} + \sum_{i=1}^r \mathbf{B}_i (\boldsymbol{\mu}_i + \mathbf{A}_i \mathbf{Z}_i) = \mathbf{a} + \sum_{i=1}^r \mathbf{B}_i \boldsymbol{\mu}_i + \sum_{i=1}^r \mathbf{B}_i \mathbf{A}_i \mathbf{Z}_i,$$

which is an affine combination of independent standard normal random vectors. Hence, $\mathbf{Y}$ is multivariate normal. Its expectation vector and covariance matrix can be found easily from Theorem B.3. $\square$

■ **Example B.3 (Matrix Normal Distribution)** Let $\mathbf{B}_1 \in \mathbb{R}^{m \times m}$, $\mathbf{B}_2 \in \mathbb{R}^{n \times n}$, $\mathbf{M} \in \mathbb{R}^{m \times n}$ be real matrices, and let $\mathbf{Z} \in \mathbb{R}^{m \times n}$ be a random matrix whose entries are iid $\mathcal{N}(0, 1)$. By Theorem B.6 the entries of the $m \times n$ matrix

$$\mathbf{X} = \mathbf{M} + \mathbf{B}_1 \mathbf{Z} \mathbf{B}_2^\top \quad (\text{B.29})$$

are also normally distributed such that the vectorization $\mathbf{Y} = \text{vec}(\mathbf{X})$ has mean $\mathbb{E}\mathbf{Y} = \text{vec}(\mathbf{M})$ and covariance matrix:

$$\text{Var}(\mathbf{Y}) = \boldsymbol{\Sigma}_2 \otimes \boldsymbol{\Sigma}_1,$$

where $\boldsymbol{\Sigma}_1 = \mathbf{B}_1 \mathbf{B}_1^\top$ and $\boldsymbol{\Sigma}_2 = \mathbf{B}_2 \mathbf{B}_2^\top$. To see this, use the identity (A.7) to write $\mathbf{Y} - \mathbb{E}\mathbf{Y} = \text{vec}(\mathbf{B}_1 \mathbf{Z} \mathbf{B}_2^\top) = (\mathbf{B}_2 \otimes \mathbf{B}_1) \text{vec}(\mathbf{Z})$. Hence, by the mixed-product Kronecker identity,

$$\begin{aligned} \text{Var}(\mathbf{Y}) &= (\mathbf{B}_2 \otimes \mathbf{B}_1) \text{Var}(\text{vec}(\mathbf{Z})) (\mathbf{B}_2 \otimes \mathbf{B}_1)^\top \\ &= (\mathbf{B}_2 \otimes \mathbf{B}_1) (\mathbf{B}_2^\top \otimes \mathbf{B}_1^\top) \\ &= (\mathbf{B}_2 \mathbf{B}_2^\top) \otimes (\mathbf{B}_1 \mathbf{B}_1^\top). \end{aligned}$$



MULTIVARIATE
NORMAL



107



74



76

MATRIX
NORMAL

The matrix $\mathbf{X}$ in (B.29) is said to have *matrix normal* distribution with mean $\mathbf{M}$ and covariance $\Sigma_2 \otimes \Sigma_1$, or simply

$$\text{vec}(\mathbf{X}) \sim \mathcal{N}(\text{vec}(\mathbf{M}), \Sigma_2 \otimes \Sigma_1).$$

■

The next theorem shows that the distribution of a subvector of a multivariate normal random vector is again normal.

Theorem B.7: Marginal Distributions of Normal Random Vectors

Let $X \sim \mathcal{N}(\mu, \Sigma)$ be an n -dimensional normal random vector. Decompose X , μ , and Σ as

$$X = \begin{bmatrix} X_p \\ X_q \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_p \\ \mu_q \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_p & \Sigma_r \\ \Sigma_r^\top & \Sigma_q \end{bmatrix}, \quad (\text{B.30})$$

where Σ_p is the upper left $p \times p$ corner of Σ and Σ_q is the lower right $q \times q$ corner of Σ . Then, $X_p \sim \mathcal{N}(\mu_p, \Sigma_p)$.

Proof: We give a proof assuming that Σ is positive definite. Let $\mathbf{B}\mathbf{B}^\top$ be the (lower) Cholesky decomposition of Σ . We can write

88

$$\begin{bmatrix} X_p \\ X_q \end{bmatrix} = \begin{bmatrix} \mu_p \\ \mu_q \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{B}_p & \mathbf{O} \\ \mathbf{C}_r & \mathbf{C}_q \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} Z_p \\ Z_q \end{bmatrix}, \quad (\text{B.31})$$

where Z_p and Z_q are independent p - and q -dimensional standard normal random vectors. In particular, $X_p = \mu_p + \mathbf{B}_p Z_p$, which means that $X_p \sim \mathcal{N}(\mu_p, \Sigma_p)$, since $\mathbf{B}_p \mathbf{B}_p^\top = \Sigma_p$. □

By relabeling the elements of X we see that Theorem B.7 implies that *any* subvector of X has a multivariate normal distribution. For example, $X_q \sim \mathcal{N}(\mu_q, \Sigma_q)$.

The following theorem shows that not only the marginal distributions of a normal random vector are normal, but also its *conditional distributions*.

Theorem B.8: Conditional Distributions of Normal Random Vectors

Let $X \sim \mathcal{N}(\mu, \Sigma)$ be an n -dimensional normal random vector with $\det(\Sigma) > 0$. If X is decomposed as in (B.30), then

$$(X_q | X_p = x_p) \sim \mathcal{N}(\mu_q + \Sigma_r^\top \Sigma_p^{-1} (x_p - \mu_p), \Sigma_q - \Sigma_r^\top \Sigma_p^{-1} \Sigma_r). \quad (\text{B.32})$$

As a consequence, X_p and X_q are *independent* if and only if they are *uncorrelated*; that is, if $\Sigma_r = \mathbf{O}$ (zero matrix).

Proof: From (B.31) we see that $X_p = \mu_p + \mathbf{B}_p \mathbf{Z}_p$ and $X_q = \mu_q + \mathbf{C}_r \mathbf{Z}_p + \mathbf{C}_q \mathbf{Z}_q$. Consequently,

$$(X_q | X_p = \mathbf{x}_p) = \mu_q + \mathbf{C}_r \mathbf{B}_p^{-1}(\mathbf{x}_p - \mu_p) + \mathbf{C}_q \mathbf{Z}_q,$$

where $\mathbf{Z}_q$ is a q -dimensional multivariate standard normal random vector. It follows that X_q conditional on $X_p = \mathbf{x}_p$ has a $\mathcal{N}(\mu_q + \mathbf{C}_r \mathbf{B}_p^{-1}(\mathbf{x}_p - \mu_p), \mathbf{C}_q \mathbf{C}_q^\top)$ distribution. The proof of (B.32) is completed by observing that $\Sigma_r^\top \Sigma_p^{-1} = \mathbf{C}_r \mathbf{B}_p^\top (\mathbf{B}_p^\top)^{-1} \mathbf{B}_p^{-1} = \mathbf{C}_r \mathbf{B}_p^{-1}$, and

$$\Sigma_q - \Sigma_r^\top \Sigma_p^{-1} \Sigma_r = \mathbf{C}_r \mathbf{C}_r^\top + \mathbf{C}_q \mathbf{C}_q^\top - \underbrace{\mathbf{C}_r \mathbf{B}_p^{-1} \Sigma_r}_{\mathbf{B}_p \mathbf{C}_r^\top} = \mathbf{C}_q \mathbf{C}_q^\top.$$

If X_p and X_q are independent, then they are obviously uncorrelated, as $\Sigma_r = \mathbb{E}[(X_p - \mu_p)(X_q - \mu_q)^\top] = \mathbb{E}(X_p - \mu_p) \mathbb{E}(X_q - \mu_q)^\top = \mathbf{O}$. Conversely, if $\Sigma_r = \mathbf{O}$, then by (B.32) the conditional distribution of X_q given X_p is the same as the unconditional distribution of X_q ; that is, $\mathcal{N}(\mu_q, \Sigma_q)$. In other words, X_q is independent of X_p . $\square$

The next few results are about the relationships between the normal, chi-squared, Student, and F distributions, defined in Table B.1. Recall that the chi-squared family of distributions, denoted by χ_n^2 , are simply Gamma($n/2, 1/2$) distributions, where the parameter $n \in \{1, 2, 3, \dots\}$ is called the *degrees of freedom*.

χ^2 DISTRIBUTION

Theorem B.9: Relationship Between Normal and χ^2 Distributions

If $X \sim \mathcal{N}(\mu, \Sigma)$ is an n -dimensional normal random vector with $\det(\Sigma) > 0$, then

$$(X - \mu)^\top \Sigma^{-1} (X - \mu) \sim \chi_n^2. \quad (\text{B.33})$$

Proof: Let $\mathbf{B}\mathbf{B}^\top$ be the Cholesky decomposition of Σ , where $\mathbf{B}$ is invertible. Since X can be written as $\mu + \mathbf{B}\mathbf{Z}$, where $\mathbf{Z} = [Z_1, \dots, Z_n]^\top$ is a vector of independent standard normal random variables, we have

$$(X - \mu)^\top \Sigma^{-1} (X - \mu) = (X - \mu)^\top (\mathbf{B}\mathbf{B}^\top)^{-1} (X - \mu) = \mathbf{Z}^\top \mathbf{Z} = \sum_{i=1}^n Z_i^2.$$

Using the independence of $Z_1, \dots, Z_n$, the moment generating function of $Y = \sum_{i=1}^n Z_i^2$ is given by

$$\mathbb{E} e^{sY} = \mathbb{E} e^{s(Z_1^2 + \dots + Z_n^2)} = \mathbb{E} [e^{sZ_1^2} \dots e^{sZ_n^2}] = \left(\mathbb{E} e^{sZ^2} \right)^n,$$

where $Z \sim \mathcal{N}(0, 1)$. The moment generating function of Z^2 is

$$\mathbb{E} e^{sZ^2} = \int_{-\infty}^{\infty} e^{sz^2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(1-2s)z^2} dz = \frac{1}{\sqrt{1-2s}},$$



so that $\mathbb{E} e^{sY} = \left(\frac{1}{2} / \left(\frac{1}{2} - s \right) \right)^{\frac{n}{2}}$, $s < \frac{1}{2}$, which is the moment generating function of the Gamma($n/2, 1/2$) distribution; that is, the χ_n^2 distribution — see Example B.1. The result now follows from the uniqueness of the moment generating function. $\square$

101

NONCENTRAL
 χ_n^2 DISTRIBUTION

A consequence of Theorem B.9 is that if $\mathbf{X} = [X_1, \dots, X_n]^\top$ is n -dimensional standard normal, then the squared length $\|\mathbf{X}\|^2 = X_1^2 + \dots + X_n^2$ has a χ_n^2 distribution. If instead $X_i \sim \mathcal{N}(\mu_i, 1)$, $i = 1, \dots, n$, then $\|\mathbf{X}\|^2$ is said to have a *noncentral χ_n^2 distribution*. This distribution depends on the $\{\mu_i\}$ only through the norm $\|\boldsymbol{\mu}\|$. We write $\|\mathbf{X}\|^2 \sim \chi_n^2(\theta)$, where $\theta = \|\boldsymbol{\mu}\|^2$ is the *noncentrality parameter*.

Such distributions frequently occur when considering *projections* of multivariate normal random variables, as summarized in the following theorem.

Theorem B.10: Normal and Noncentral χ^2 Distributions

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{I}_n)$ be an n -dimensional normal random vector and let $\mathcal{V}_k \subset \mathcal{V}_m$ be linear subspaces of dimensions k and m , respectively, with $k < m \leq n$. Let $\mathbf{X}_k$ and $\mathbf{X}_m$ be orthogonal projections of $\mathbf{X}$ onto $\mathcal{V}_k$ and $\mathcal{V}_m$, and let $\boldsymbol{\mu}_k$ and $\boldsymbol{\mu}_m$ be the corresponding projections of $\boldsymbol{\mu}$. Then, the following holds.

1. The random vectors $\mathbf{X}_k$, $\mathbf{X}_m - \mathbf{X}_k$, and $\mathbf{X} - \mathbf{X}_m$ are independent.
2. $\|\mathbf{X}_k\|^2 \sim \chi_k^2(\|\boldsymbol{\mu}_k\|^2)$, $\|\mathbf{X}_m - \mathbf{X}_k\|^2 \sim \chi_{m-k}^2(\|\boldsymbol{\mu}_m - \boldsymbol{\mu}_k\|^2)$, and $\|\mathbf{X} - \mathbf{X}_m\|^2 \sim \chi_{n-m}^2(\|\boldsymbol{\mu} - \boldsymbol{\mu}_m\|^2)$.

77

Proof: Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be an orthonormal basis of $\mathbb{R}^n$ such that $\mathbf{v}_1, \dots, \mathbf{v}_k$ spans $\mathcal{V}_k$ and $\mathbf{v}_1, \dots, \mathbf{v}_m$ spans $\mathcal{V}_m$. By (A.9) we can write the orthogonal projection matrices onto $\mathcal{V}_j$, as $\mathbf{P}_j = \sum_{i=1}^j \mathbf{v}_i \mathbf{v}_i^\top$, $j = k, m, n$, where $\mathcal{V}_n$ is defined as $\mathbb{R}^n$. Note that $\mathbf{P}_n$ is simply the identity matrix. Let $\mathbf{V} := [\mathbf{v}_1, \dots, \mathbf{v}_n]$ and define $\mathbf{Z} := [Z_1, \dots, Z_n]^\top = \mathbf{V}^\top \mathbf{X}$. Recall from Section A.2 that any orthogonal transformation such as $\mathbf{z} = \mathbf{V}^\top \mathbf{x}$ is *length preserving*; that is, $\|\mathbf{z}\| = \|\mathbf{x}\|$.

74

To prove the first statement of the theorem, note that $\mathbf{V}^\top \mathbf{X}_j = \mathbf{V}^\top \mathbf{P}_j \mathbf{X} = [Z_1, \dots, Z_j, 0, \dots, 0]^\top$, $j = k, m$. It follows that $\mathbf{V}^\top (\mathbf{X}_m - \mathbf{X}_k) = [0, \dots, 0, Z_{k+1}, \dots, Z_m, 0, \dots, 0]^\top$ and $\mathbf{V}^\top (\mathbf{X} - \mathbf{X}_m) = [0, \dots, 0, Z_{m+1}, \dots, Z_n]^\top$. Moreover, being a linear transformation of a normal random vector, $\mathbf{Z}$ is also normal, with covariance matrix $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_n$. In particular, the $\{Z_i\}$ are *independent*. This shows that $\mathbf{X}_k$, $\mathbf{X}_m - \mathbf{X}_k$ and $\mathbf{X} - \mathbf{X}_m$ are independent as well.

Next, observe that $\|\mathbf{X}_k\| = \|\mathbf{V}^\top \mathbf{X}_k\| = \|\mathbf{Z}_k\|$, where $\mathbf{Z}_k := [Z_1, \dots, Z_k]^\top$. The latter vector has independent components with variances 1, and its squared norm has therefore (by definition) a $\chi_k^2(\theta)$ distribution. The noncentrality parameter is $\theta = \|\mathbb{E} \mathbf{Z}_k\|^2 = \|\mathbb{E} \mathbf{X}_k\|^2 = \|\boldsymbol{\mu}_k\|^2$, again by the length-preserving property of orthogonal transformations. This shows that $\|\mathbf{X}_k\|^2 \sim \chi_k^2(\|\boldsymbol{\mu}_k\|^2)$. The distributions of $\|\mathbf{X}_m - \mathbf{X}_k\|^2$ and $\|\mathbf{X} - \mathbf{X}_m\|^2$ follow by analogy. $\square$

Theorem B.10 is frequently used in the statistical analysis of *normal linear models*; see Section 2.4. In typical situations μ lies in the subspace $\mathcal{V}_m$ or even $\mathcal{V}_k$ — in which case $\|X_m - X_k\|^2 \sim \chi_{m-k}^2$ and $\|X - X_m\|^2 \sim \chi_{n-m}^2$, independently. The (scaled) quotient then turns out to have an F distribution — a consequence of the following theorem.

39

Theorem B.11: Relationship Between χ^2 and F Distributions

Let $U \sim \chi_m^2$ and $V \sim \chi_n^2$ be independent. Then,

$$\frac{U/m}{V/n} \sim F(m, n).$$

Proof: For notational simplicity, let $c = m/2$ and $d = n/2$. The pdf of $W = U/V$ is given by $f_W(w) = \int_0^\infty f_U(wv) v f_V(v) dv$. Substituting the pdfs of the corresponding Gamma distributions, we have

$$\begin{aligned} f_W(w) &= \int_0^\infty \frac{(wv)^{c-1} e^{-wv/2}}{\Gamma(c) 2^c} v \frac{v^{d-1} e^{-v/2}}{\Gamma(d) 2^d} dv = \frac{w^{c-1}}{\Gamma(c) \Gamma(d) 2^{c+d}} \int_0^\infty v^{c+d-1} e^{-(1+w)v/2} dv \\ &= \frac{\Gamma(c+d)}{\Gamma(c) \Gamma(d)} \frac{w^{c-1}}{(1+w)^{c+d}}, \end{aligned}$$

where the last equality follows from the fact that the integrand is equal to $\Gamma(\alpha)\lambda^\alpha$ times the density of the $\text{Gamma}(\alpha, \lambda)$ distribution with $\alpha = c + d$ and $\lambda = (1 + w)/2$. The density of $Z = \frac{n}{m} \frac{U}{V}$ is given by

$$f_Z(z) = f_W(z m/n) m/n.$$

The proof is completed by comparing the resulting expression with the pdf of the F distribution given in Table B.1. $\square$

99

Corollary B.1 (Relationship Between Normal, χ^2 , and t Distributions) Let $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_n^2$ be independent. Then,

$$\frac{Z}{\sqrt{V/n}} \sim t_n.$$

Proof: Let $T = Z/\sqrt{V/n}$. Because $Z^2 \sim \chi_1^2$, we have by Theorem B.11 that $T^2 \sim F(1, n)$. The result follows now from the symmetry around 0 of the pdf of T and the fact that the square of a t_n random variable has an $F(1, n)$ distribution. $\square$

Finally, the next result shows that computing expectations under the multivariate normal distribution may be significantly simplified.

Theorem B.12: Covariance & Normality

Suppose that $Y \sim \mathcal{N}(\mu, \Sigma)$, where $\Sigma \in \mathbb{R}^{n \times n}$ is an invertible covariance matrix, and $g : \mathbb{R}^n \mapsto \mathbb{R}^n$ is a continuously differentiable function. Then,

$$\text{tr} [\text{Cov}(g(Y), Y)] = \mathbb{E} \text{tr} \left[\frac{\partial g}{\partial Y}^\top \Sigma \right].$$

Proof: Let $f(y)$ be the density of Y . Computing the gradient

$$\frac{\partial}{\partial y} (y - \mu)^\top \Sigma^{-1} (y - \mu) = 2\Sigma^{-1} (y - \mu),$$

we deduce that $\frac{\nabla f(y)}{f(y)} = \nabla \ln f(y) = -\Sigma^{-1} (y - \mu)$. Hence,

$$\begin{aligned} \mathbb{E}[(Y - \mu)^\top g(Y)] &= \int_{\mathbb{R}^n} f(y) (y - \mu)^\top g(y) dy \\ &= \int_{\mathbb{R}^n} f(y) [\Sigma^{-1} (y - \mu)]^\top \Sigma g(y) dy \\ &= - \int_{\mathbb{R}^n} [\nabla f(y)]^\top \Sigma g(y) dy = - \sum_{i,j} \int_{\mathbb{R}^n} \frac{\partial f}{\partial y_i} \Sigma_{i,j} g_j(y) dy. \end{aligned}$$

By integration by parts, $- \int_{\mathbb{R}^n} \frac{\partial f}{\partial y_i} \Sigma_{i,j} g_j(y) dy = \int_{\mathbb{R}^n} f(y) \Sigma_{i,j} \frac{\partial g_j}{\partial y_i} dy$. Therefore,

$$\begin{aligned} \text{tr} [\text{Cov}(g(Y), Y)] &= \mathbb{E}[(Y - \mu)^\top g(Y)] \\ &= \mathbb{E} \sum_{i,j} \Sigma_{i,j} \frac{\partial g_j}{\partial y_i} = \mathbb{E} \text{tr} \left[\frac{\partial g}{\partial y}^\top \Sigma \right]. \end{aligned}$$

□

B.8 Statistics

Statistics deals with the gathering, summarization, analysis, and interpretation of data. The two main branches of statistics are:

FREQUENTIST STATISTICS

1. *Classical or frequentist statistics:* Here the observed data τ is viewed as the outcome of random data $\mathcal{T}$ described by a probabilistic model — usually the model is specified up to a (multidimensional) parameter; that is, $\mathcal{T} \sim g(\cdot | \theta)$ for some θ . The statistical inference is then purely concerned with the model and in particular with the parameter θ . For example, on the basis of the data one may wish to

- (a) estimate the parameter,

- (b) perform statistical tests on the parameter, or
- (c) validate the model.

2. *Bayesian statistics*: In this approach we average over all possible values of the parameter θ using a user-specified weight function $g(\theta)$ and obtain the model $\mathcal{T} \sim \int g(\cdot | \theta) g(\theta) d\theta$. For practical computations, this means that we can treat θ as a random variable with pdf $g(\theta)$. Bayes' formula $g(\theta | \tau) \propto g(\tau | \theta) g(\theta)$ is used to learn θ based on the observed data τ .

BAYESIAN
STATISTICS

■ **Example B.4 (Iid Sample)** The most fundamental statistical model is where the data $\mathcal{T} = X_1, \dots, X_n$ is such that the random variables $X_1, \dots, X_n$ are assumed to be independent and identically distributed:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Dist},$$

according to some known or unknown distribution Dist . An iid sample is often called a *random sample* in the statistics literature. Note that the word “sample” can refer to both a collection of random variables and to a single random variable. It should be clear from the context which meaning is being used.

RANDOM
SAMPLE

Often our guess or model for the true distribution is specified up to an unknown parameter θ , with $\theta \in \Theta$. The most common model is:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2),$$

in which case $\theta = (\mu, \sigma^2)$ and $\Theta = \mathbb{R} \times \mathbb{R}_+$. ■

B.9 Estimation

Suppose the model $g(\cdot | \theta)$ for the data $\mathcal{T}$ is completely specified up to an unknown parameter vector θ . The aim is to estimate θ on the basis of the observed data τ only (an alternative goal could be to estimate $\eta = \psi(\theta)$ for some vector-valued function ψ). Specifically, the goal is to find an *estimator* $T = T(\mathcal{T})$ that is close to the unknown θ . The corresponding outcome $t = T(\tau)$ is the *estimate* of θ . The *bias* of an estimator T of θ is defined as $\mathbb{E}T - \theta$. An estimator T of θ is said to be *unbiased* if $\mathbb{E}_\theta T = \theta$. We often write $\hat{\theta}$ for both an estimator and estimate of θ . The *mean squared error* (MSE) of a real-valued estimator T is defined as

ESTIMATOR
ESTIMATE
BIAS
MEAN
SQUARED
ERROR

$$\text{MSE} = \mathbb{E}_\theta (T - \theta)^2.$$

An estimator T_1 is said to be more *efficient* than an estimator T_2 if the MSE of T_1 is smaller than the MSE of T_2 . The MSE can be written as the sum

EFFICIENT

$$\text{MSE} = (\mathbb{E}_\theta T - \theta)^2 + \text{Var}_\theta T.$$

The first term measures the unbiasedness and the second is the variance of the estimator. In particular, for an unbiased estimator the MSE of an estimator is simply equal to its variance.

For simulation purposes it is often important to include the running time of the estimator in efficiency comparisons. One way to compare two unbiased estimators T_1 and T_2 is to compare their *relative time variance products*,

$$\frac{r_i \text{Var } T_i}{(\mathbb{E} T_i)^2}, \quad i = 1, 2, \quad (\text{B.34})$$

where r_1 and r_2 are the times required to calculate the estimators T_1 and T_2 , respectively. In this scheme, T_1 is considered more efficient than T_2 if its relative time variance product is smaller. We discuss next two systematic approaches for constructing sound estimators.

B.9.1 Method of Moments

Suppose $x_1, \dots, x_n$ are outcomes from an iid sample $X_1, \dots, X_n \sim_{\text{iid}} g(x | \theta)$, where $\theta = [\theta_1, \dots, \theta_k]^\top$ is unknown. The moments of the sampling distribution can be easily estimated. Namely, if $X \sim g(x | \theta)$, then the r -th moment of X , that is $\mu_r(\theta) = \mathbb{E}_\theta X^r$ (assuming it exists), can be estimated through the *sample r -th moment*: $\frac{1}{n} \sum_{i=1}^n x_i^r$. The *method of moments* involves choosing the estimate $\hat{\theta}$ of θ such that each of the first k sample and true moments are matched:

$$\frac{1}{n} \sum_{i=1}^n x_i^r = \mu_r(\hat{\theta}), \quad r = 1, 2, \dots, k.$$

In general, this set of equations is nonlinear and so its solution often has to be found numerically.

■ **Example B.5 (Sample Mean and Sample Variance)** Suppose the data is given by $\mathcal{T} = \{X_1, \dots, X_n\}$, where the $\{X_i\}$ form an iid sample from a general distribution with mean μ and variance $\sigma^2 < \infty$. Matching the first two moments gives the set of equations

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n x_i &= \mu, \\ \frac{1}{n} \sum_{i=1}^n x_i^2 &= \mu^2 + \sigma^2. \end{aligned}$$

The method of moments estimates for μ and σ^2 are therefore the *sample mean*

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad (\text{B.35})$$

and

$$\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n x_i^2 - (\bar{x})^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (\text{B.36})$$

The corresponding estimator for μ , $\bar{X}$, is unbiased. However, the estimator for σ^2 is biased: $\mathbb{E} \widehat{\sigma^2} = \sigma^2(n-1)/n$. An unbiased estimator is the *sample variance*

SAMPLE
VARIANCE

$$S^2 = \widehat{\sigma^2} \frac{n}{n-1} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Its square root, $S = \sqrt{S^2}$, is called the *sample standard deviation*. ■

SAMPLE
STANDARD
DEVIATION

■ **Example B.6 (Sample Covariance Matrix)** The method of moments can also be used to estimate the covariance matrix of a random vector. In particular, let the $X_1, \dots, X_n$ be iid copies of a d -dimensional random vector $\mathbf{X}$ with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. We assume $n \geq d$. The moment estimator for $\boldsymbol{\mu}$ is, as in the $d = 1$ case, $\bar{\mathbf{X}} = (\mathbf{X}_1 + \dots + \mathbf{X}_n)/n$. As the covariance matrix can be written (see (B.15)) as

$$\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top,$$

the method of moments yields the estimator

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top. \quad (\text{B.37})$$

Similar to the one-dimensional case ($d = 1$), replacing the factor $1/n$ with $1/(n-1)$ gives an unbiased estimator, called the *sample covariance matrix*. ■

SAMPLE
COVARIANCE
MATRIX

B.9.2 Maximum Likelihood Method

The concept of *likelihood* is central in statistics. It describes in a precise way the information about model parameters that is contained in the observed data.

Let $\mathcal{T}$ be a (random) data object that is modeled as a draw from the pdf $g(\tau | \boldsymbol{\theta})$ (discrete or continuous) with parameter vector $\boldsymbol{\theta} \in \Theta$. Let τ be an outcome of $\mathcal{T}$. The function $L(\boldsymbol{\theta} | \tau) := g(\tau | \boldsymbol{\theta})$, $\boldsymbol{\theta} \in \Theta$, is called the *likelihood function* of $\boldsymbol{\theta}$, based on τ . The (natural) logarithm of the likelihood function is called the *log-likelihood function* and is often denoted by a lower case l .

LIKELIHOOD
FUNCTION
LOG-
LIKELIHOOD
FUNCTION



Note that $L(\boldsymbol{\theta} | \tau)$ and $g(\tau | \boldsymbol{\theta})$ have the same formula, but the first is viewed as a function of $\boldsymbol{\theta}$ for fixed τ , where the second is viewed as a function of τ for fixed $\boldsymbol{\theta}$.

The concept of likelihood is particularly useful when $\mathcal{T}$ is modeled as an iid sample $\{X_1, \dots, X_n\}$ from some pdf $\hat{g}$. In that case, the likelihood of the data $\tau = \{x_1, \dots, x_n\}$, as a function of θ , is given by the product

$$L(\theta | \tau) = \prod_{i=1}^n \hat{g}(x_i | \theta). \quad (\text{B.38})$$

Let τ be an observation from $\mathcal{T} \sim g(\tau | \theta)$, and suppose that $g(\tau | \theta)$ takes its largest value at $\theta = \hat{\theta}$. In a way this $\hat{\theta}$ is our best estimate for θ , as it maximizes the probability (density) for the observation τ . It is called the *maximum likelihood estimate* (MLE) of θ . Note that $\hat{\theta} = \hat{\theta}(\tau)$ is a function of τ . The corresponding random variable, also denoted $\hat{\theta}$ is the *maximum likelihood estimator* (also abbreviated as MLE).

MAXIMUM
LIKELIHOOD
ESTIMATOR

Maximization of $L(\theta | \tau)$ as a function of θ is equivalent (when searching for the maximizer) to maximizing the log-likelihood $l(\theta | \tau)$, as the natural logarithm is an increasing function. This is often easier, especially when $\mathcal{T}$ is an iid sample from some sampling distribution. For example, for L of the form (B.38), we have

$$l(\theta | \tau) = \sum_{i=1}^n \ln \hat{g}(x_i | \theta).$$

If $l(\theta | \tau)$ is a differentiable function with respect to θ and the maximum is attained in the interior of Θ , and there exists a unique maximum point, then we can find the MLE of θ by solving the equations

$$\frac{\partial}{\partial \theta_i} l(\theta | \tau) = 0, \quad i = 1, \dots, d.$$

■ **Example B.7 (Bernoulli Random Sample)** Suppose we have data $\tau_n = \{x_1, \dots, x_n\}$ and assume the model $X_1, \dots, X_n \sim_{\text{iid}} \text{Ber}(\theta)$. Then, the likelihood function is given by

$$L(\theta | \tau) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^s (1 - \theta)^{n-s}, \quad 0 < \theta < 1, \quad (\text{B.39})$$

where $s := x_1 + \dots + x_n =: n\bar{x}$. The log-likelihood is $l(\theta) = s \ln \theta + (n - s) \ln(1 - \theta)$. Through differentiation with respect to θ , we find the derivative

$$\frac{s}{\theta} - \frac{n-s}{1-\theta} = \frac{s}{\theta(1-\theta)} - \frac{n}{1-\theta}. \quad (\text{B.40})$$

Solving $g'(\theta) = 0$ gives the ML estimate $\hat{\theta} = \bar{x}$ and ML estimator $\hat{\theta} = \bar{X}$. ■

B.10 Confidence Intervals

An essential part in any estimation procedure is to provide an assessment of the *accuracy* of the estimate. Indeed, without information on its accuracy the estimate itself would be meaningless. Confidence intervals (also called *interval estimates*) provide a precise way of describing the uncertainty in the estimate.

INTERVAL
ESTIMATES

Let $X_1, \dots, X_n$ be random variables with a joint distribution depending on a parameter $\theta \in \Theta$. Let $T_1 < T_2$ be statistics; that is, $T_i = T_i(X_1, \dots, X_n)$, $i = 1, 2$ are functions of the data, but not of θ .

1. The random interval (T_1, T_2) is called a *stochastic confidence interval* for θ with confidence $1 - \alpha$ if

STOCHASTIC
CONFIDENCE
INTERVAL

$$\mathbb{P}_\theta[T_1 < \theta < T_2] \geq 1 - \alpha \quad \text{for all } \theta \in \Theta. \quad (\text{B.41})$$

2. If t_1 and t_2 are the observed values of T_1 and T_2 , then the interval (t_1, t_2) is called the (*numerical*) *confidence interval* for θ with confidence $1 - \alpha$ for every $\theta \in \Theta$.
3. If the right-hand side of (B.41) is merely a heuristic estimate or approximation of the true probability, then the resulting interval is called an *approximate confidence interval*.
4. The probability $\mathbb{P}_\theta[T_1 < \theta < T_2]$ is called the *coverage probability*. For a $1 - \alpha$ confidence interval, it must be at least $1 - \alpha$.

(NUMERICAL)
CONFIDENCE
INTERVAL

COVERAGE
PROBABILITY

For multidimensional parameters $\theta \in \mathbb{R}^d$ the stochastic confidence interval is replaced with a stochastic *confidence region* $C \subset \mathbb{R}^d$ such that $\mathbb{P}_\theta[\theta \in C] \geq 1 - \alpha$ for all θ .

CONFIDENCE
REGION

■ **Example B.8 (Approximate Confidence Interval for the Mean)** Let $X_1, X_2, \dots, X_n$ be an iid sample from a distribution with mean μ and variance $\sigma^2 < \infty$ (both assumed to be unknown). By the central limit theorem and the law of large numbers,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \underset{\text{approx.}}{\sim} \mathcal{N}(0, 1),$$

for large n , where S is the sample standard deviation. Rearranging the approximate equality $\mathbb{P}[|T| \leq z_{1-\alpha/2}] \approx 1 - \alpha$, where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ quantile of the standard normal distribution, yields

$$\mathbb{P}\left[\bar{X} - z_{1-\alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\alpha/2} \frac{S}{\sqrt{n}}\right] \approx 1 - \alpha,$$

so that

$$\left(\bar{X} - z_{1-\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{S}{\sqrt{n}}\right), \text{ abbreviated as } \bar{X} \pm z_{1-\alpha/2} \frac{S}{\sqrt{n}}, \quad (\text{B.42})$$

is an approximate stochastic $(1 - \alpha)$ confidence interval for μ . ■

Since (B.42) is an asymptotic result only, care should be taken when applying it to cases where the sample size is small or moderate and the sampling distribution is heavily skewed.

B.11 Hypothesis Testing

Suppose the model for the data $\mathcal{T}$ is described by a family of probability distributions that depend on a parameter $\theta \in \Theta$. The aim of *hypothesis testing* is to decide, on the basis of the observed data τ , which of two competing hypotheses holds true; these being the *null hypothesis*, $H_0 : \theta \in \Theta_0$, and the *alternative hypothesis*, $H_1 : \theta \in \Theta_1$.

In classical statistics the null hypothesis and alternative hypothesis do not play equivalent roles. H_0 contains the “status quo” statement and is only rejected if the observed data are very unlikely to have happened under H_0 .

The decision whether to accept or reject H_0 is dependent on the outcome of a *test statistic* $T = T(\mathcal{T})$. For simplicity, we discuss only the one-dimensional case $T \equiv T$. Two (related) types of decision rules are generally used:

1. **Decision rule 1:** *Reject H_0 if T falls in the critical region.*

Here the *critical region* is any appropriately chosen region in $\mathbb{R}$. In practice a critical region is one of the following:

- *left one-sided:* $(-\infty, c]$,
- *right one-sided:* $[c, \infty)$,
- *two-sided:* $(-\infty, c_1] \cup [c_2, \infty)$.

For example, for a right one-sided test, H_0 is rejected if the outcome of the test statistic is too large. The endpoints c , c_1 , and c_2 of the critical regions are called *critical values*.

2. **Decision rule 2:** *Reject H_0 if the P-value is smaller than some significance level α .*

The *P-value* is the probability that, under H_0 , the (random) test statistic takes a value as extreme as or more extreme than the one observed. In particular, if t is the observed outcome of the test statistic T , then

- *left one-sided test:* $P := \mathbb{P}_{H_0}[T \leq t]$,
- *right one-sided:* $P := \mathbb{P}_{H_0}[T \geq t]$,
- *two-sided:* $P := \min\{2\mathbb{P}_{H_0}[T \leq t], 2\mathbb{P}_{H_0}[T \geq t]\}$.

The smaller the P-value, the greater the strength of the evidence against H_0 provided by the data. As a rule of thumb:

HYPOTHESIS
TESTING

NULL
HYPOTHESIS
ALTERNATIVE
HYPOTHESIS

TEST
STATISTIC

CRITICAL
REGION

CRITICAL
VALUES

P-VALUE

$P < 0.10$ suggestive evidence,
 $P < 0.05$ reasonable evidence,
 $P < 0.01$ strong evidence.

Whether the first or the second decision rule is used, one can make two types of errors, as depicted in Table B.4.

Table B.4: Type I and II errors in hypothesis testing.

Decision	True statement	
	H_0 is true	H_1 is true
Accept H_0	Correct	Type II Error
Reject H_0	Type I Error	Correct

The choice of the test statistic and the corresponding critical region involves a multiobjective optimization criterion, whereby both the probabilities of a type I and type II error should, ideally, be chosen as small as possible. Unfortunately, these probabilities compete with each other. For example, if the critical region is made larger (smaller), the probability of a type II error is reduced (increased), but at the same time the probability of a type I error is increased (reduced).

Since the type I error is considered more serious, Neyman and Pearson suggested the following approach: choose the critical region such that the probability of a type II error is as small as possible, while keeping the probability of a type I error below a predetermined small *significance level* α .

SIGNIFIC-
ANCE LEVEL

■ **Remark B.3 (Equivalence of Decision Rules)** Note that decision rule 1 and 2 are equivalent in the following sense:

Reject H_0 if T falls in the critical region, at significance level α .

$\Leftrightarrow$

Reject H_0 if the P-value is $\leq$ significance level α .

In other words, the P-value of the test is the smallest level of significance that would lead to the rejection of H_0 . ■



In general, a statistical test involves the following steps:

1. Formulate an appropriate statistical model for the data.
2. Give the null (H_0) and alternative (H_1) hypotheses in terms of the para-

meters of the model.

3. Determine the test statistic (a function of the data only).
4. Determine the (approximate) distribution of the test statistic under H_0 .
5. Calculate the outcome of the test statistic.
6. Calculate the P-value *or* the critical region, given a preselected significance level α .
7. Accept or reject H_0 .

The actual choice of an appropriate test statistic is akin to selecting a good estimator for the unknown parameter θ . The test statistic should summarize the information about θ and make it possible to distinguish between the alternative hypotheses.

■ **Example B.9 (Hypothesis Testing)** We are given outcomes $x_1, \dots, x_m$ and $y_1, \dots, y_n$ of two simulation studies obtained via independent runs, with $m = 100$ and $n = 50$. The sample means and standard deviations are $\bar{x} = 1.3$, $s_X = 0.1$ and $\bar{y} = 1.5$, $s_Y = 0.3$. Thus, the $\{x_i\}$ are outcomes of iid random variables $\{X_i\}$, the $\{y_i\}$ are outcomes of iid random variables $\{Y_i\}$, and the $\{X_i\}$ and $\{Y_i\}$ are independent. We wish to assess whether the expectations $\mu_X = \mathbb{E}X_i$ and $\mu_Y = \mathbb{E}Y_i$ are the same or not. Going through the 7 steps above, we have:

1. The model is already specified above.
2. $H_0 : \mu_X - \mu_Y = 0$ versus $H_1 : \mu_X - \mu_Y \neq 0$.
3. For similar reasons as in Example B.8, take

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{S_X^2/m + S_Y^2/n}}.$$

4. By the central limit theorem, the statistic T has, under H_0 , approximately a standard normal distribution (assuming the variances are finite).
5. The outcome of T is $t = (\bar{x} - \bar{y})/\sqrt{s_X^2/m + s_Y^2/n} \approx -4.59$.
6. As this is a two-sided test, the P-value is $2\mathbb{P}_{H_0}[T \leq -4.59] \approx 4 \cdot 10^{-6}$.
7. Because the P-value is extremely small, there is overwhelming evidence that the two expectations are not the same.

